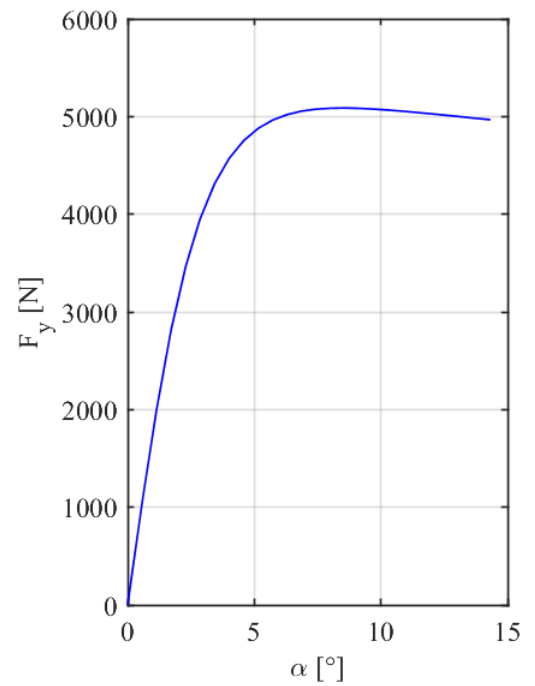
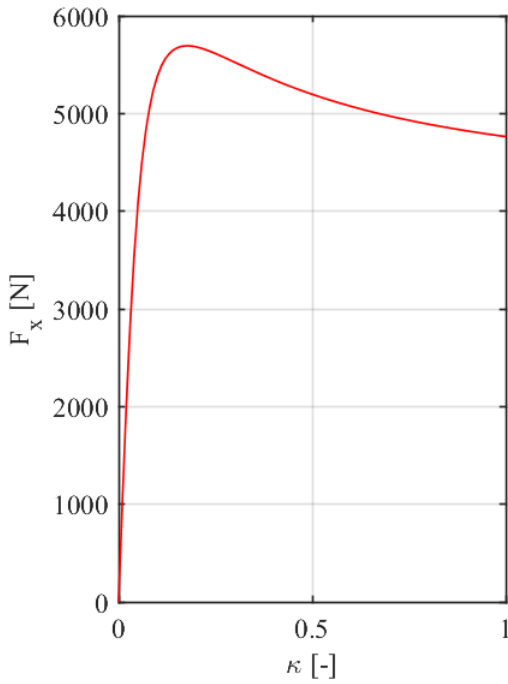


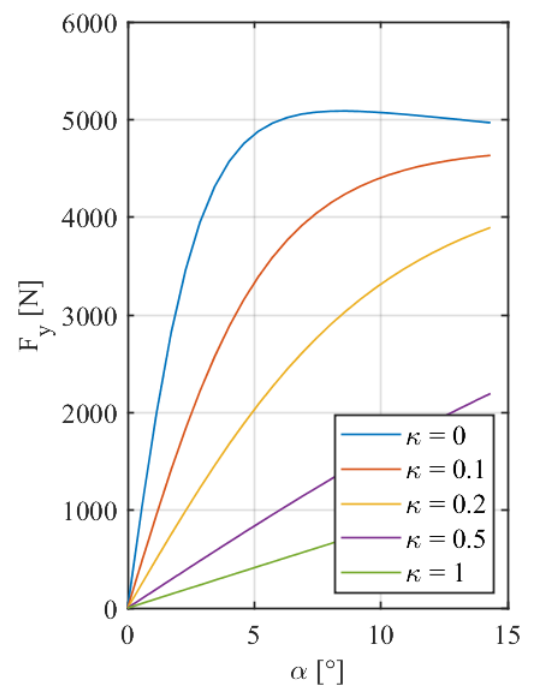
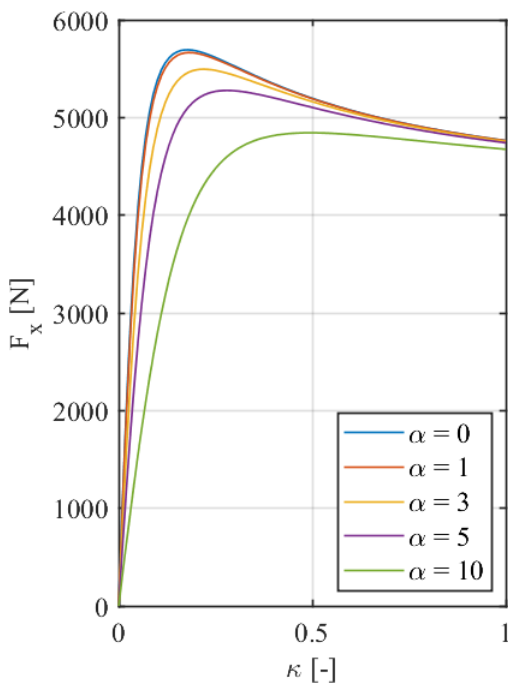
Vehicle Dynamics and Control: Tire Lab

Car segment: **Pick-up**

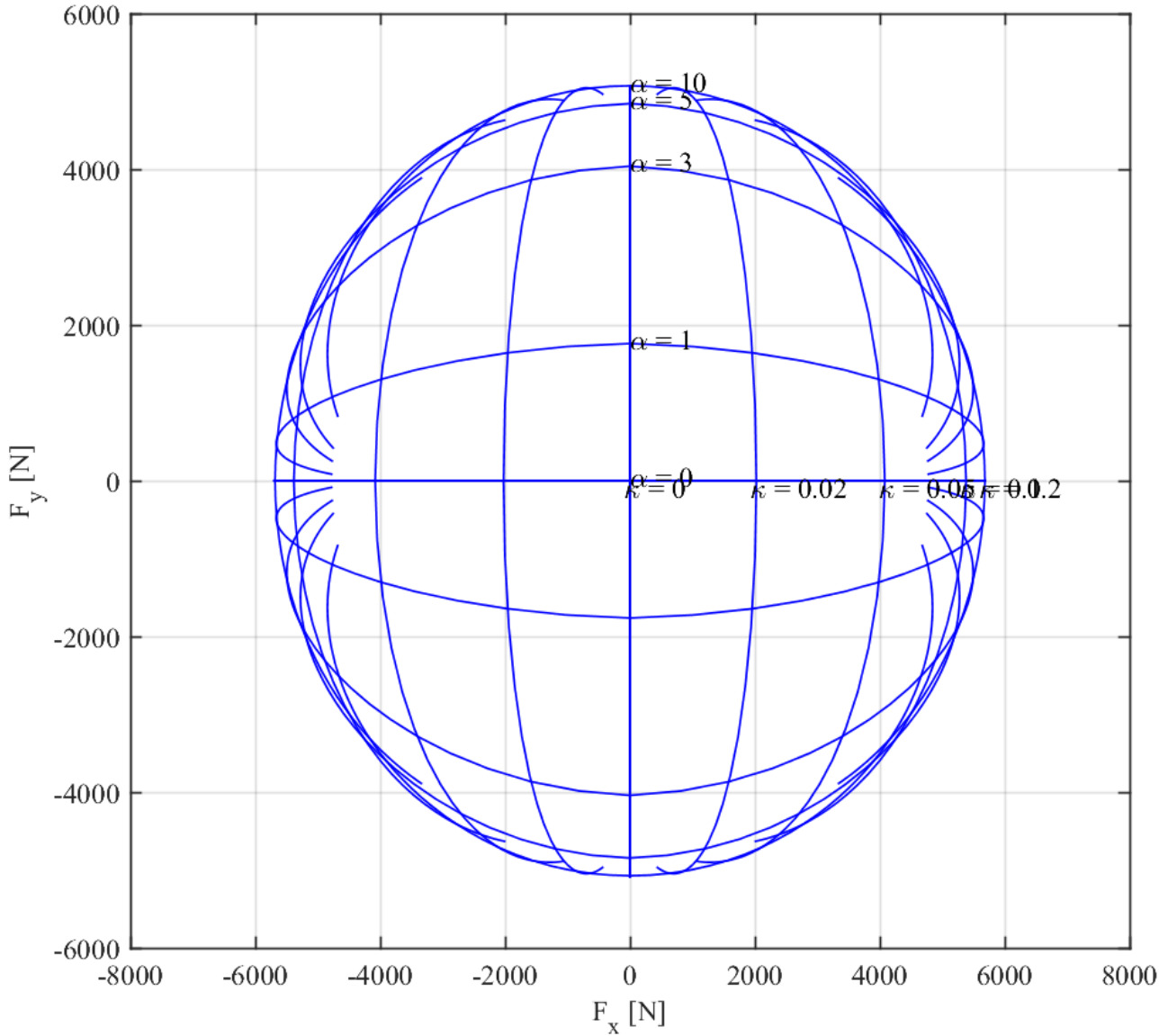
Longitudinal and lateral forces



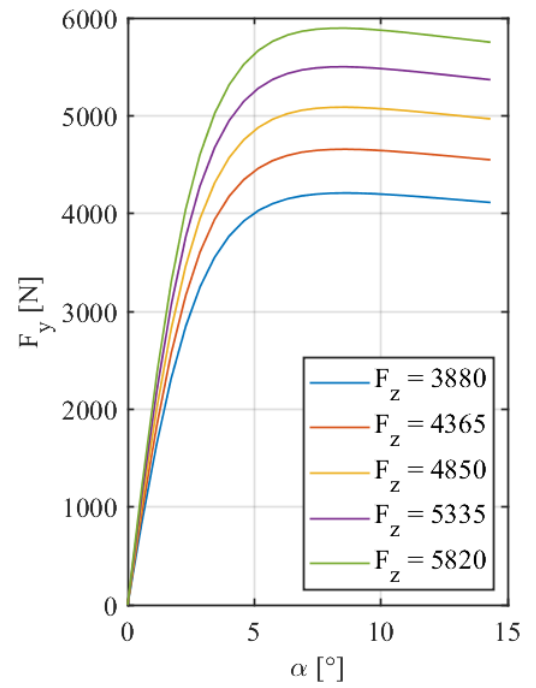
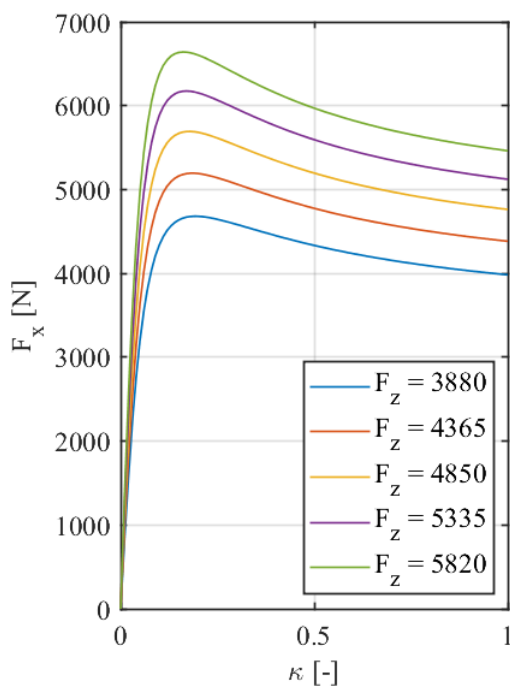
Combined slip effect



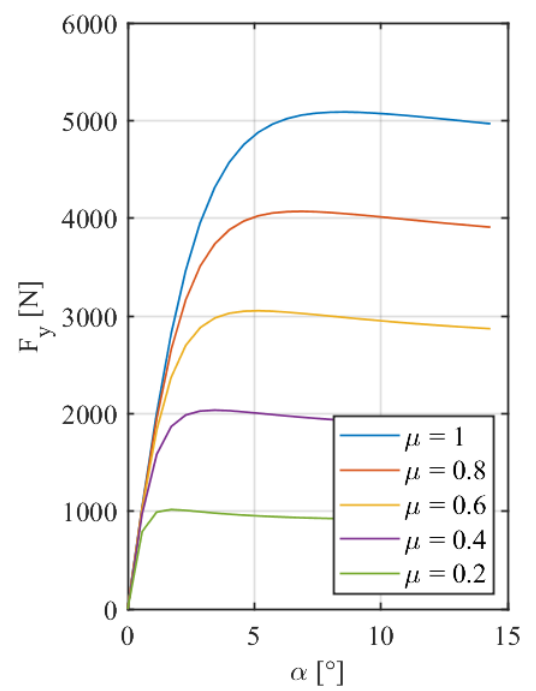
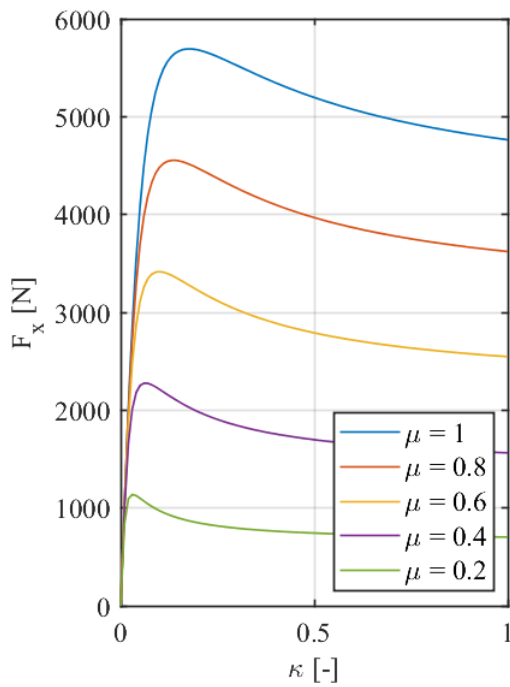
Combined friction forces



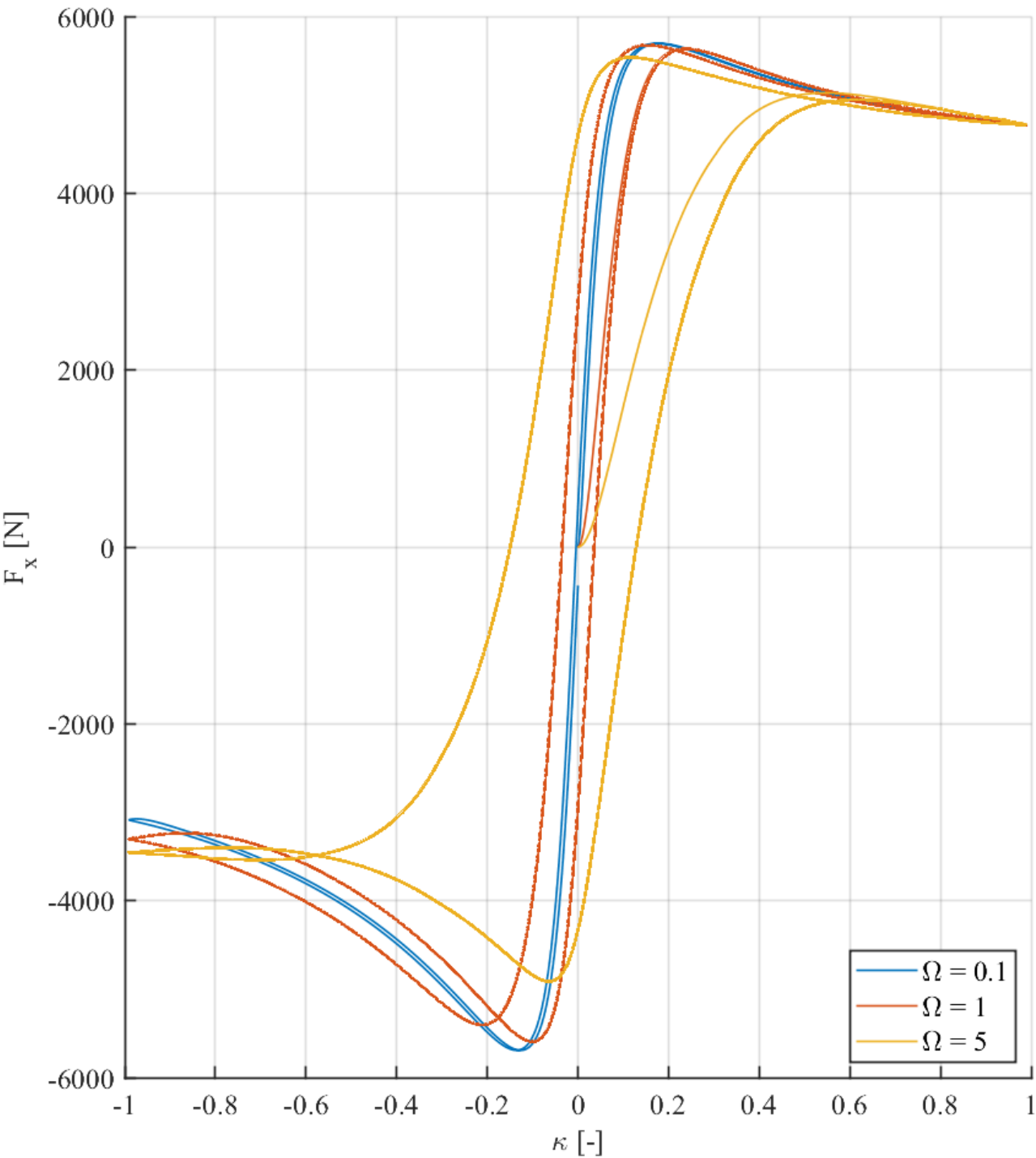
Vertical load effect

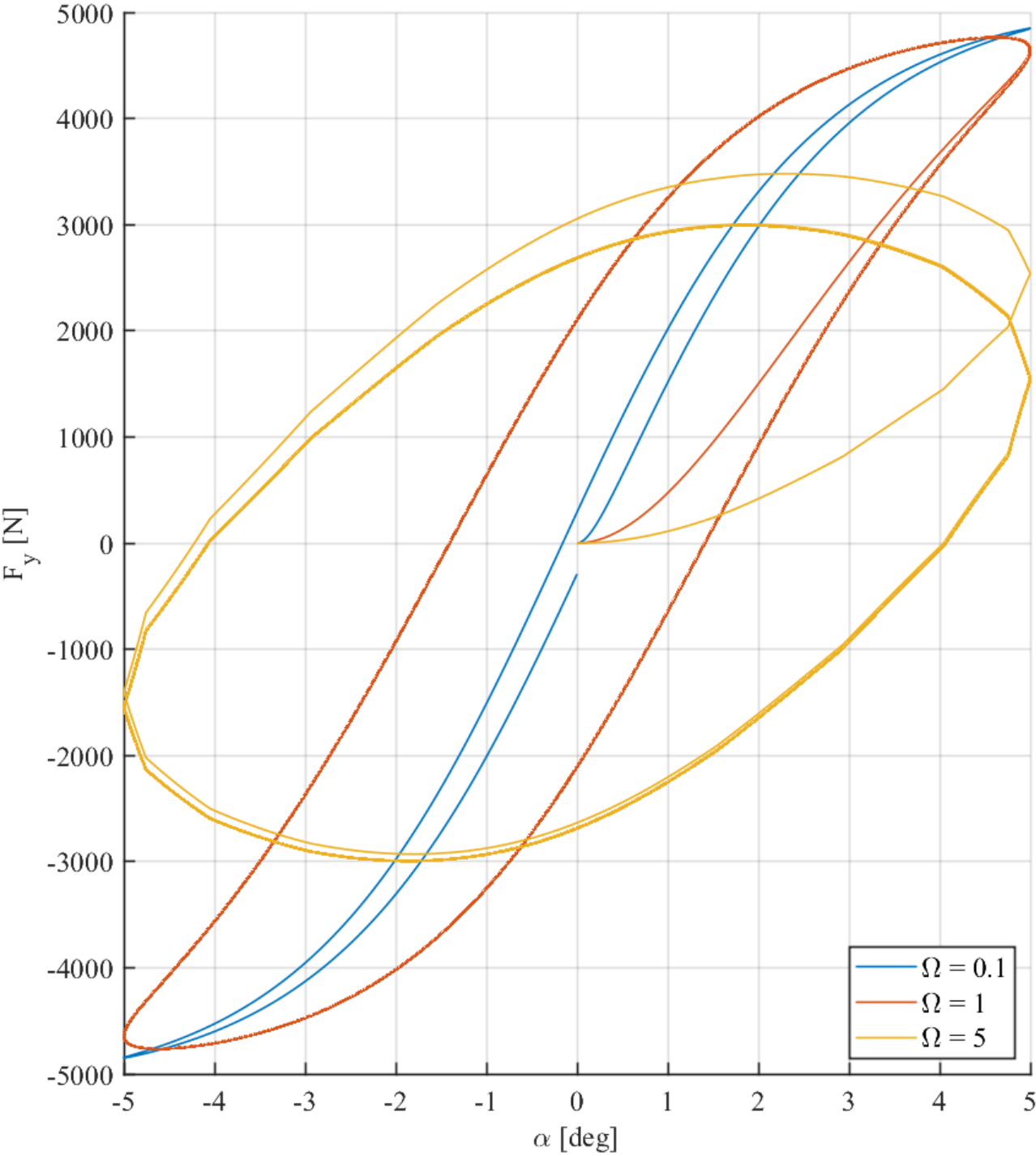


Friction coefficient effect



Transient behaviour

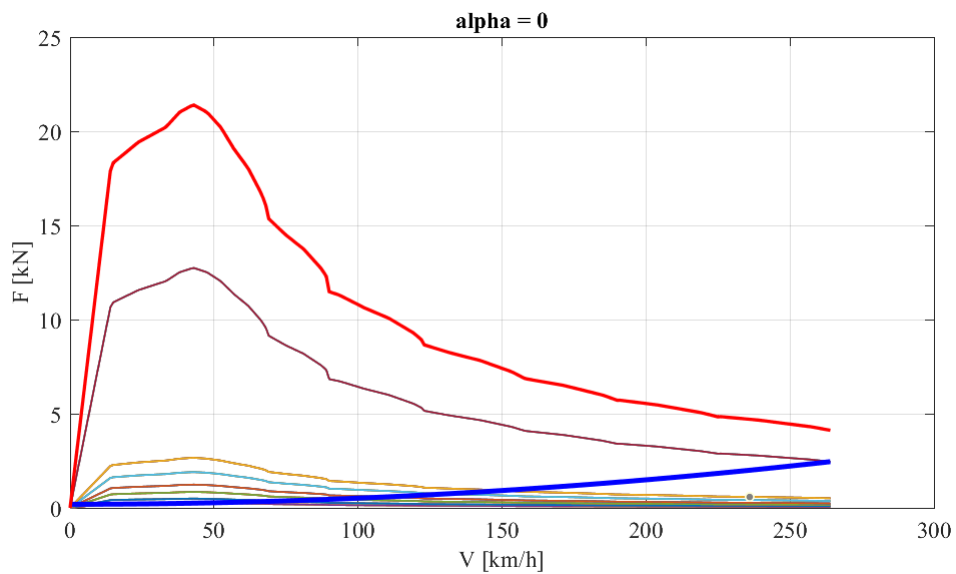




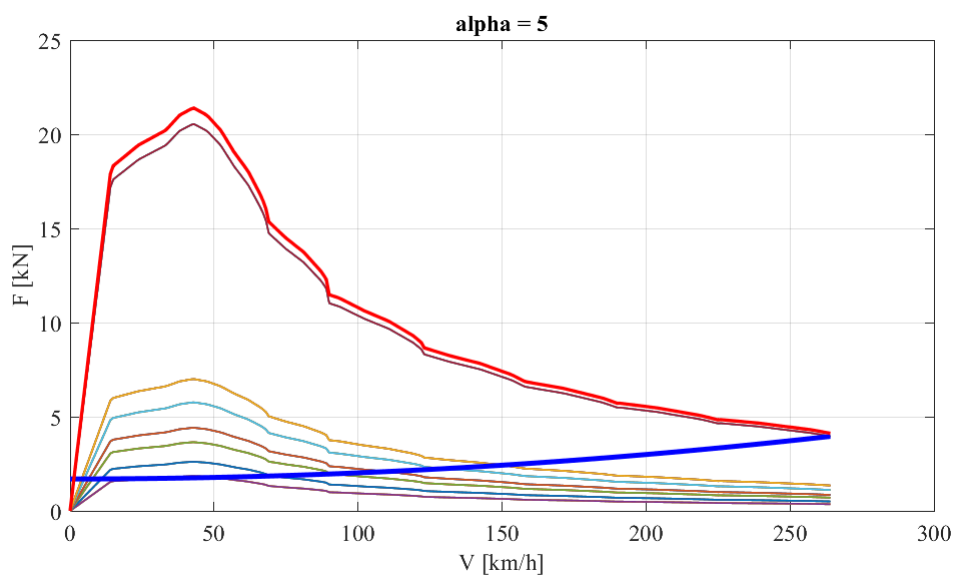
Vehicle Dynamics and Control: Longitudinal Dynamics Lab

Car segment: **Sedan car**

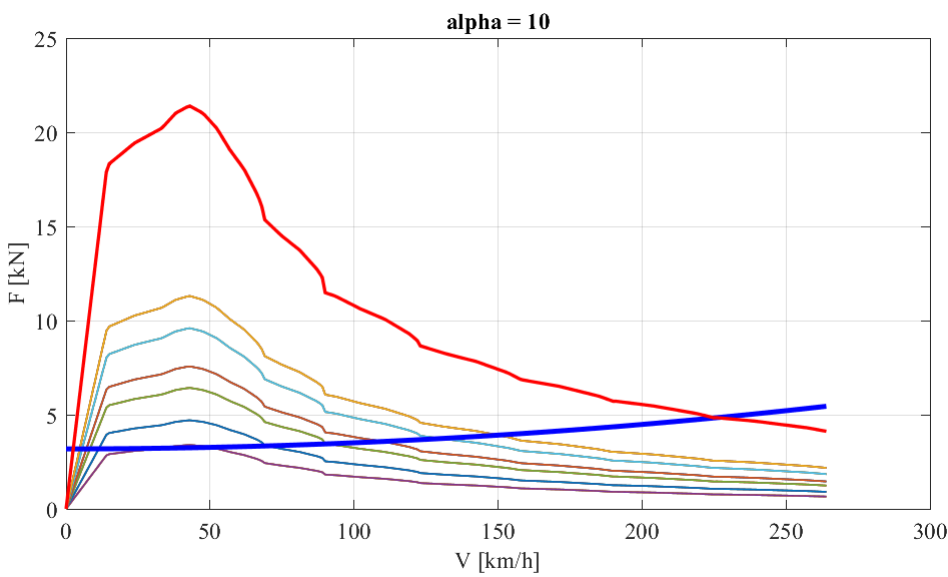
Motor working points at $\alpha = 0^\circ$



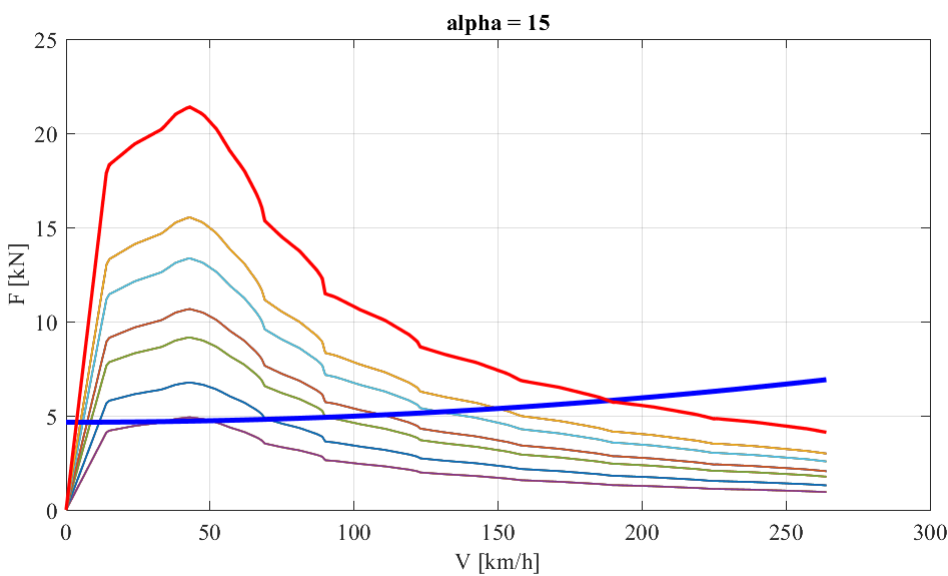
Motor working points at $\alpha = 5^\circ$



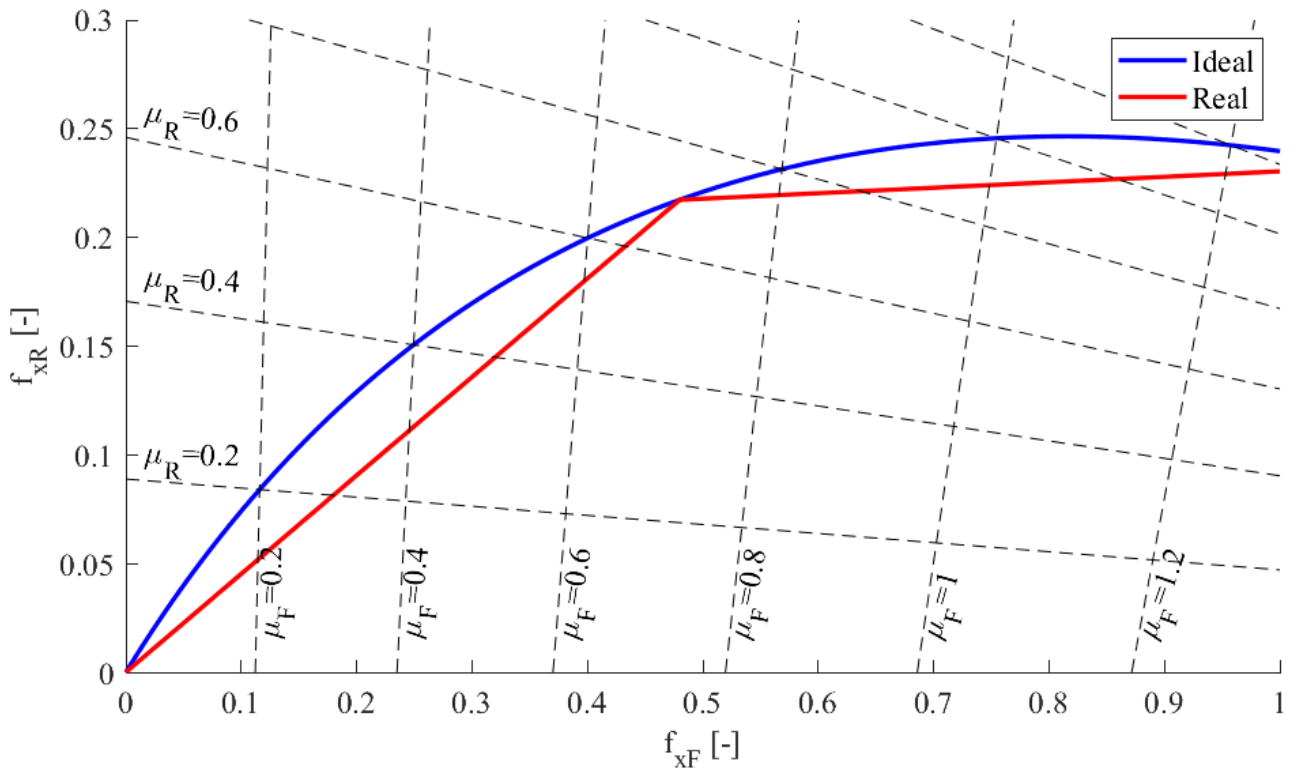
Motor working points at $\alpha = 10^\circ$



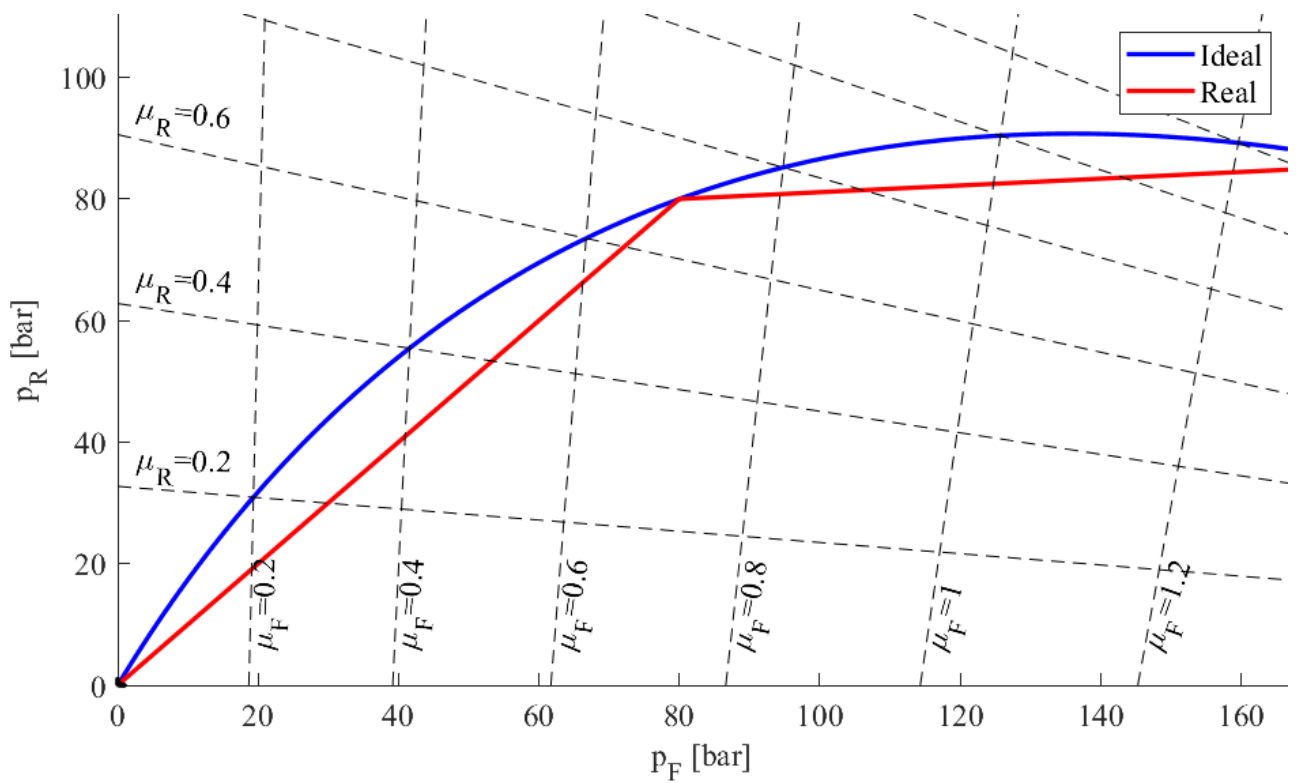
Motor working points at $\alpha = 15^\circ$



Front-rear ideal force distribution



Front-rear ideal pressure distribution



Working points

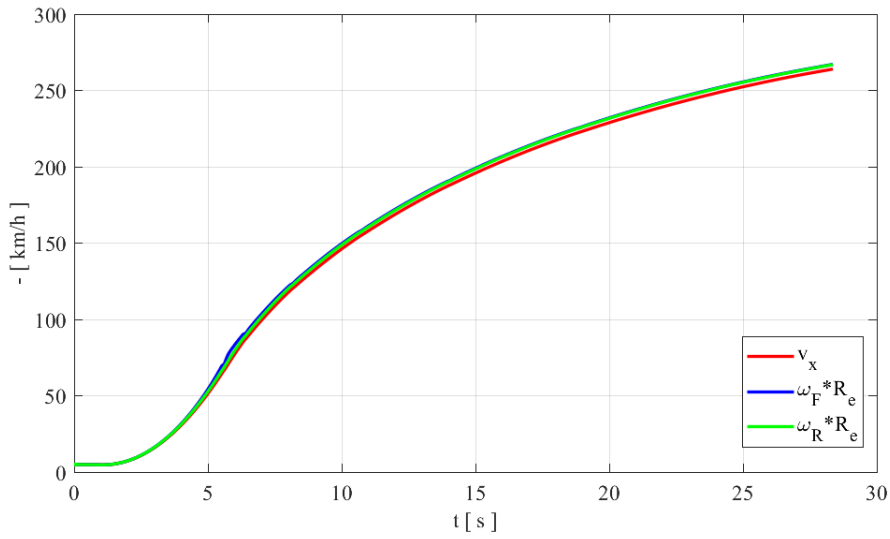
alpha [deg]	vx [Km/h]	Fx [kN]	gamma [%]
0	50	0.2730	1,325
0	70	0.3510	2,308
0	90	0.4548	3,962
0	110	0.5847	5,779
0	130	0.7405	8,893
0	150	0.9223	12,442
0	264	24.543	59,551
5	50	17.766	8,623
5	70	18.545	12,195
5	90	19.584	17,059
5	110	20.882	20,64
5	130	22.440	26,949
5	150	24.258	32,727
5	264	39.579	96,034
10	50	32.672	15,859
10	70	33.452	21,996
10	90	34.490	30,044
10	110	35.789	35,374
10	130	37.347	44,85
10	150	39.165	52,838
10	226.09	54.485	100
15	50	47.336	22,976
15	70	48.116	31,639
15	90	49.154	42,817
15	110	50.453	49,868
15	130	52.011	62,461
15	150	53.829	72,621
15	188.37	69.149	100

Pressure brake distributor: 80 [bar]

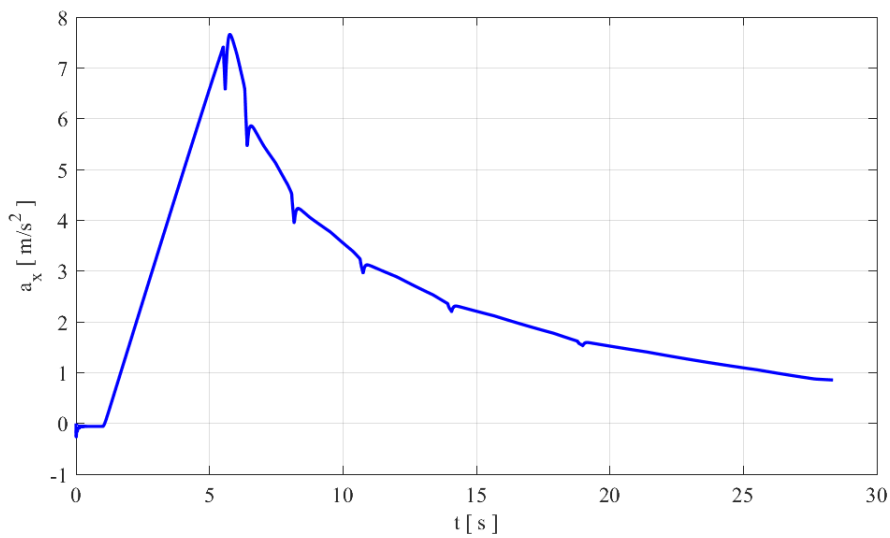
Pressure ratio brake distributor pf/pr: 0.05556 (= 1/18) [bar/bar]

Acceleration maneuver:

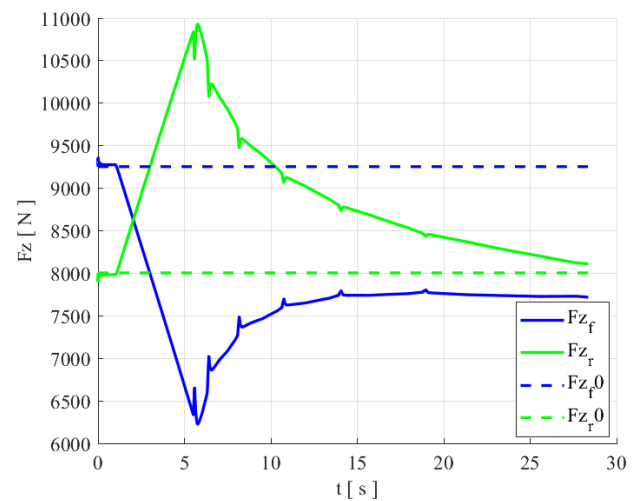
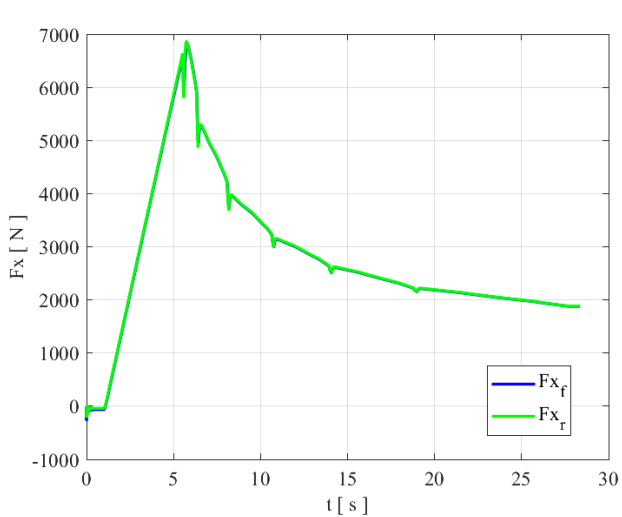
Longitudinal speed



Longitudinal acceleration

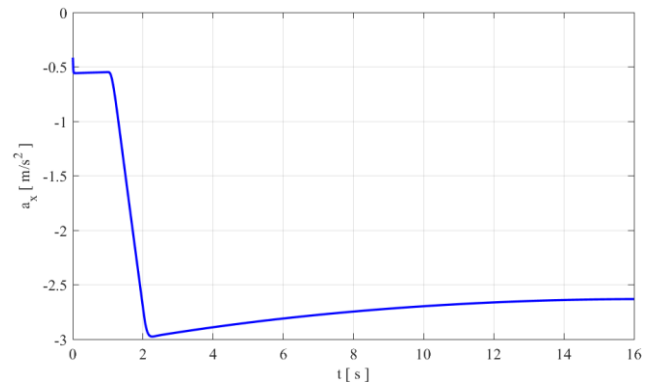
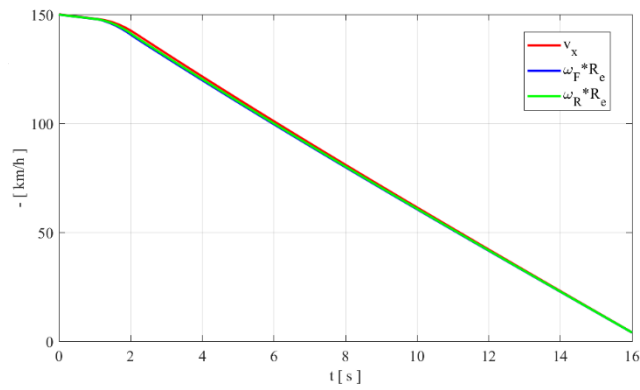


Longitudinal and vertical forces

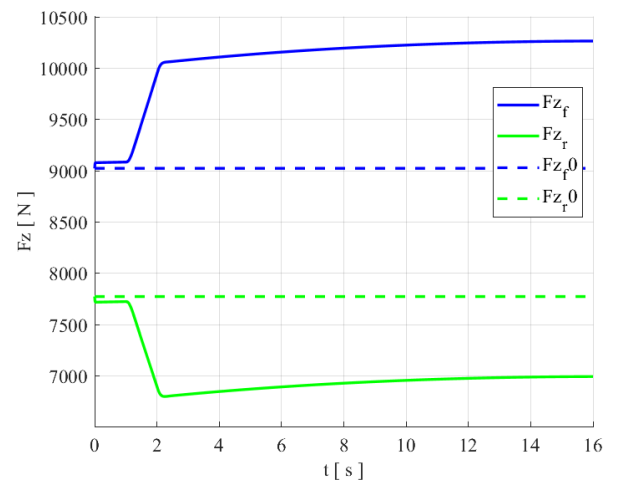
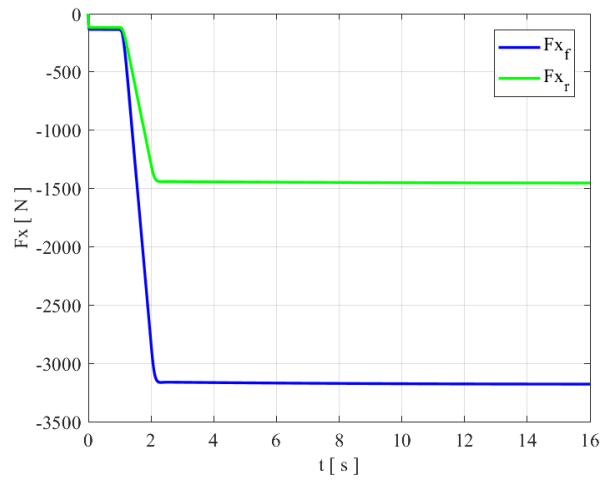


Braking maneuver:

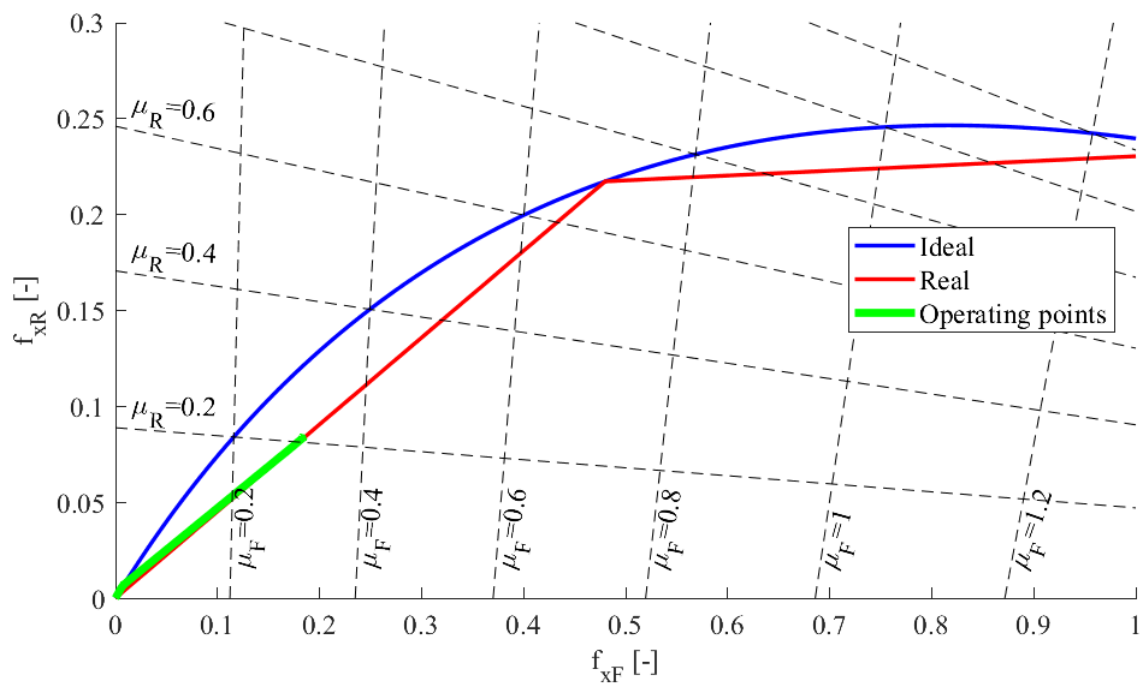
Longitudinal speed and acceleration at **30 [bar]**



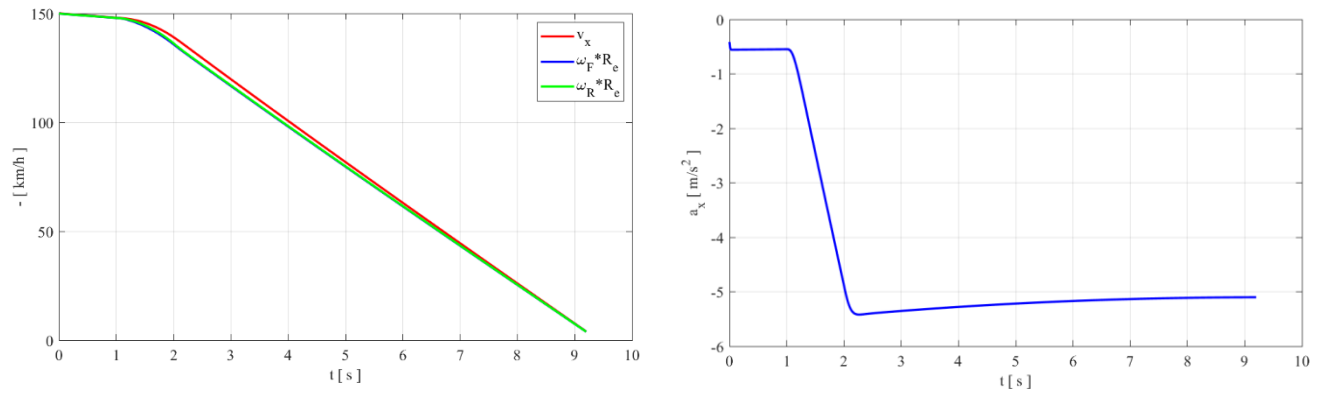
Longitudinal and vertical forces



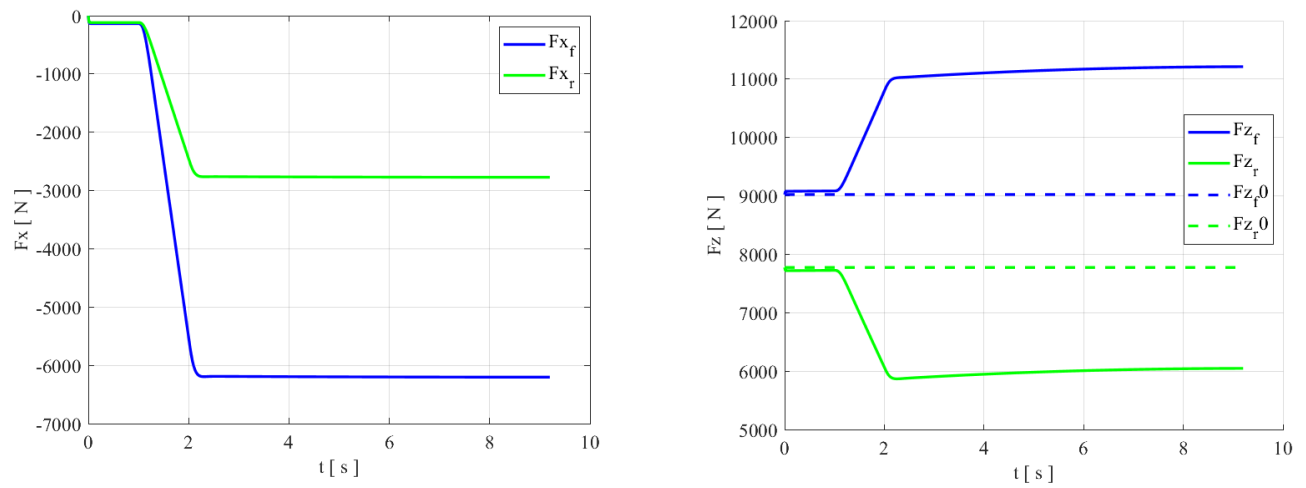
Longitudinal front and rear ideal forces



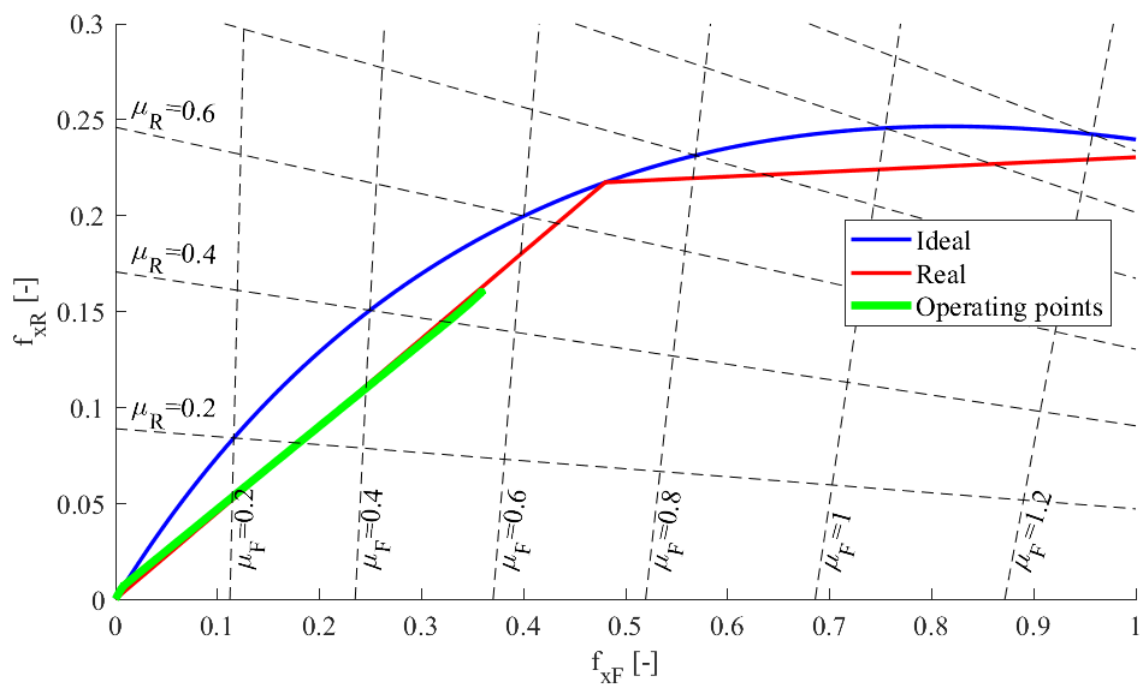
Longitudinal speed and acceleration at 60 [bar]



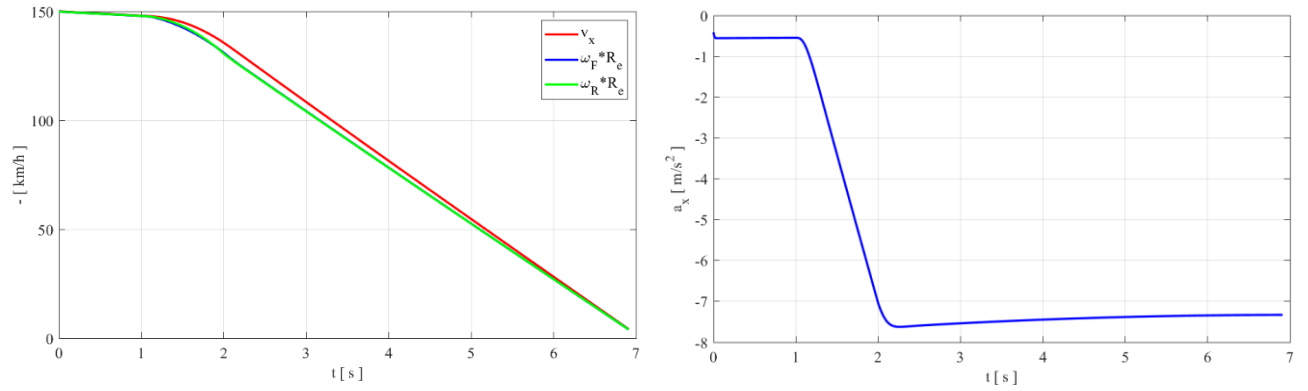
Longitudinal and vertical forces



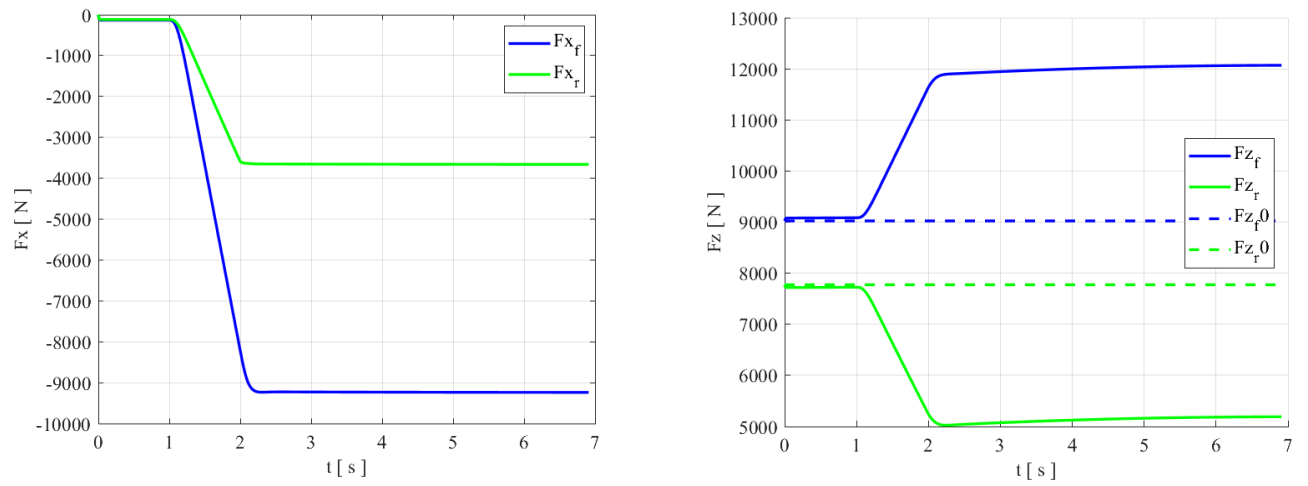
Longitudinal front and rear ideal forces



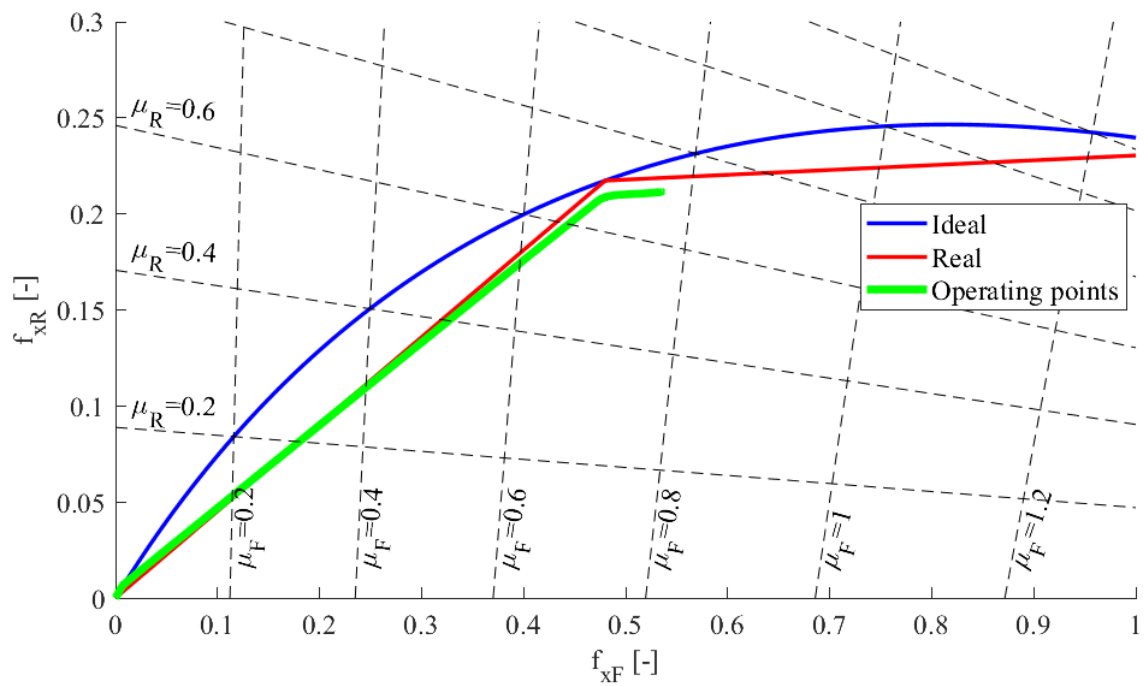
Longitudinal speed and acceleration at 90 [bar]



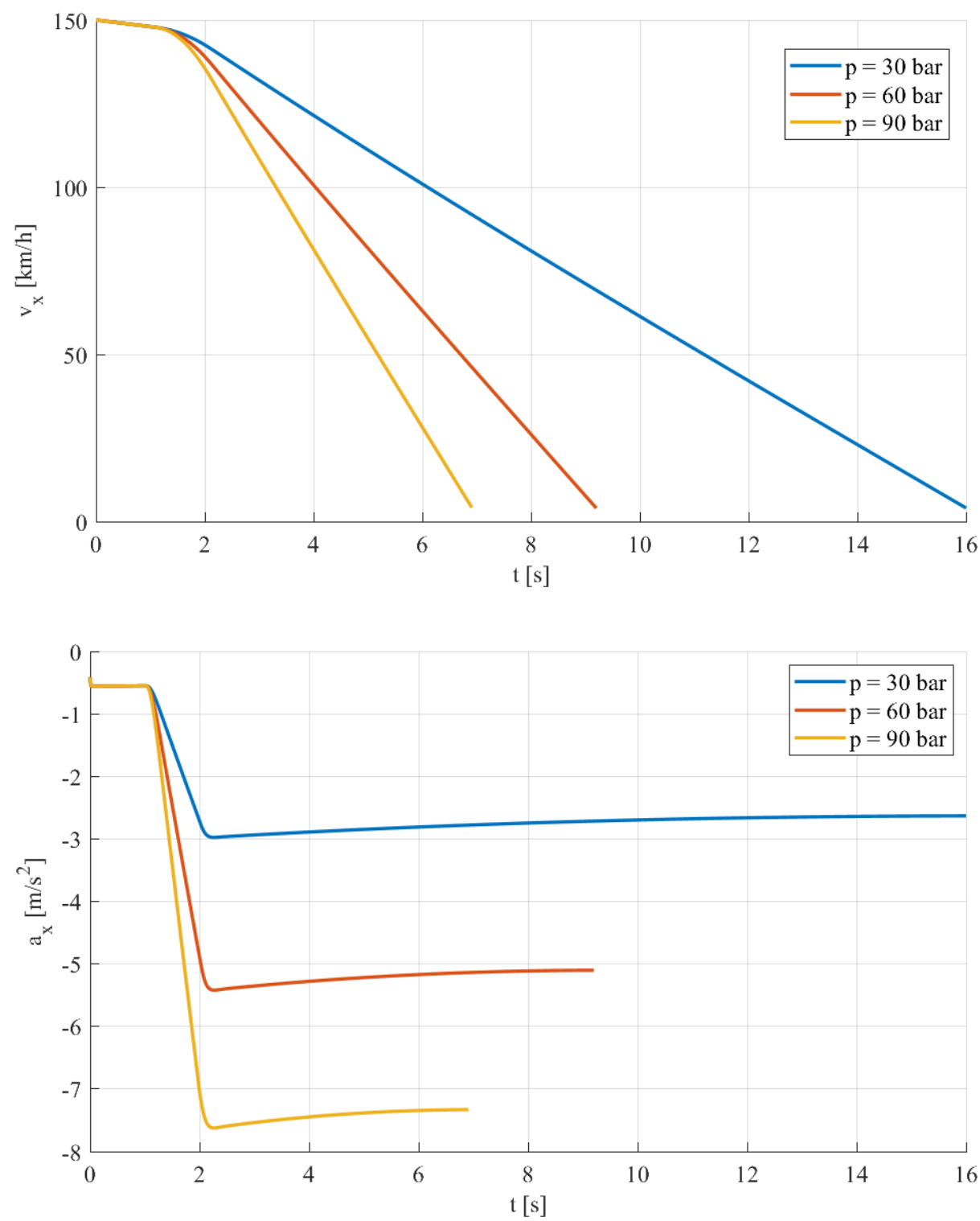
Longitudinal and vertical forces



Longitudinal front and rear ideal forces



Comparison between the 3 braking maneuvers:



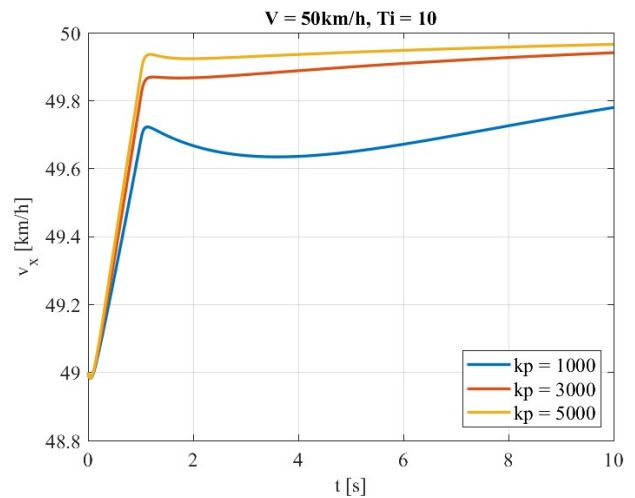
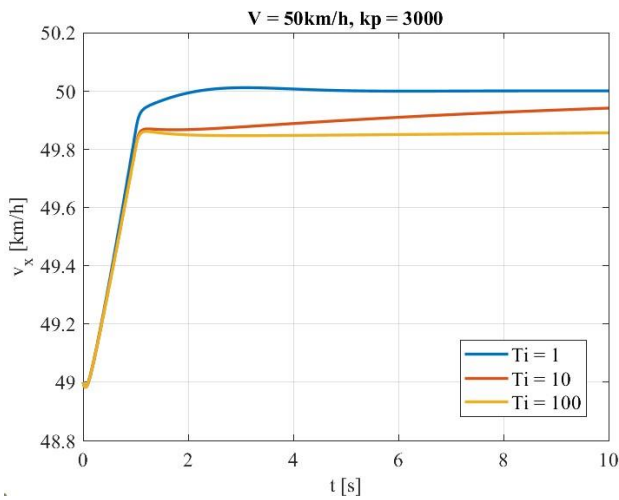
Cruise control

The tuning procedure was conducted through a step maneuver, increasing the velocity by 1 km/h until the top speed used to evaluate the bests k_p and T_i was reached.

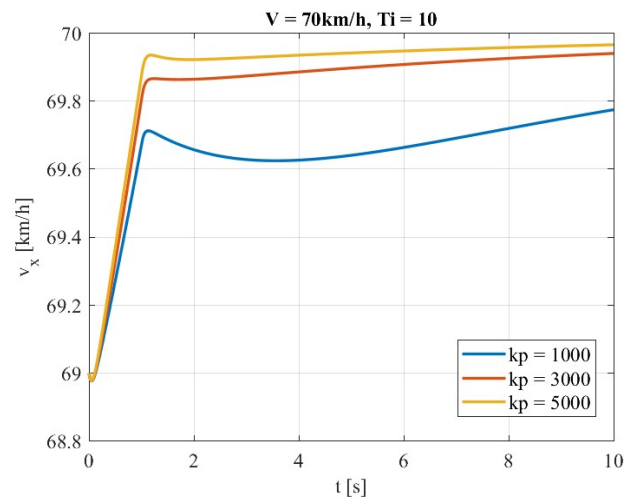
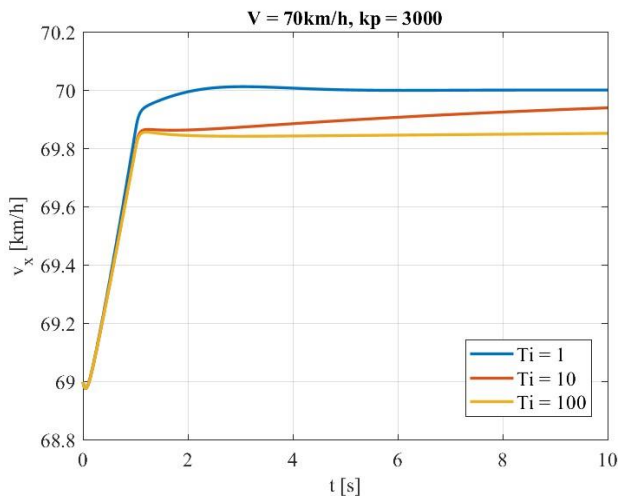
To find the best gains for every velocity, it was decided to plot the car response in function of T_i with fixed k_p and vice versa. According to the results, it was chosen to use a unique value of k_p and T_i for every velocity.

The chosen gain values are $k_p = 3000$ and $T_i = 10$.

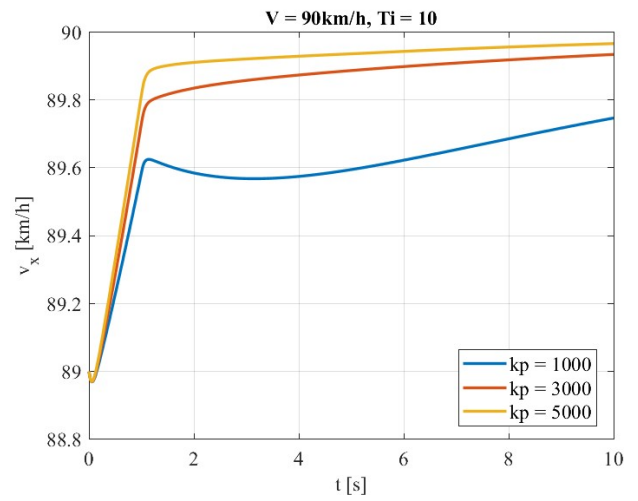
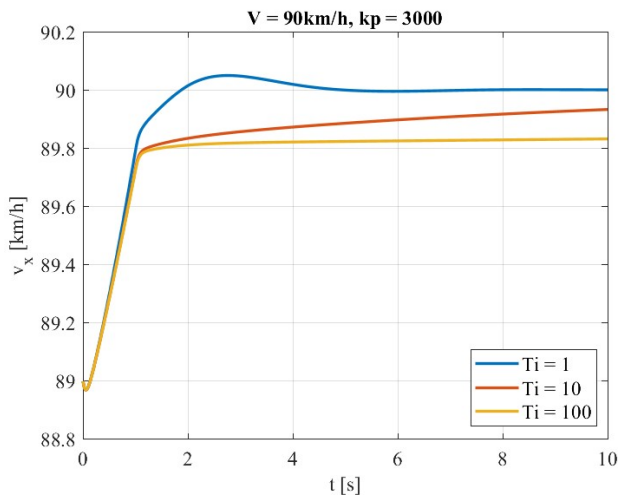
Gain tuning at $v = 50$ [km/h]



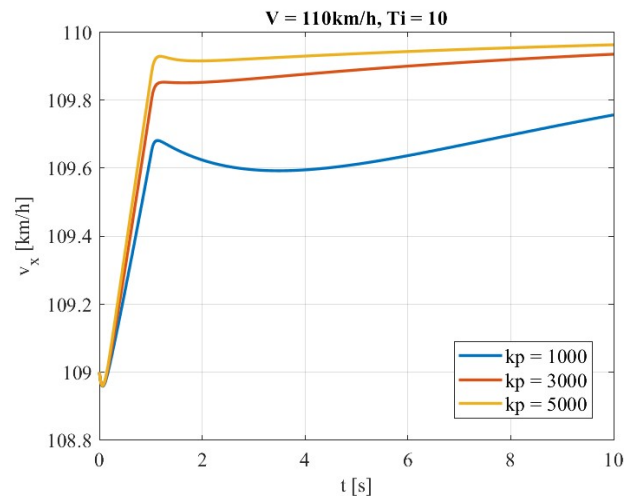
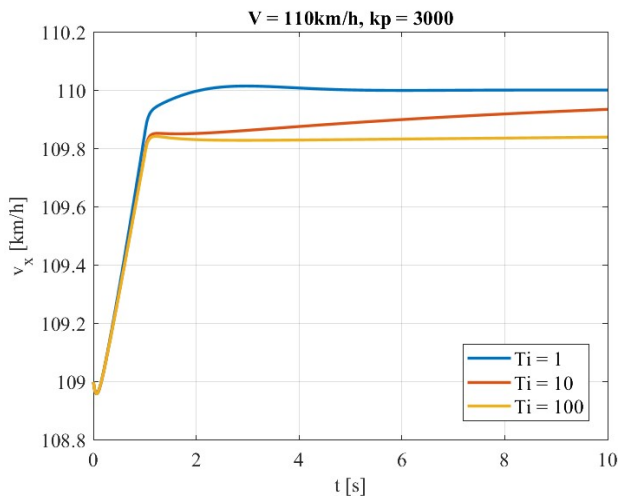
Gain tuning at $v = 70$ [km/h]



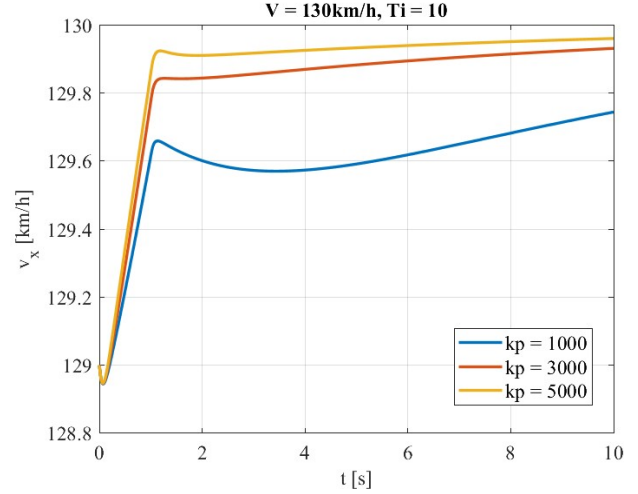
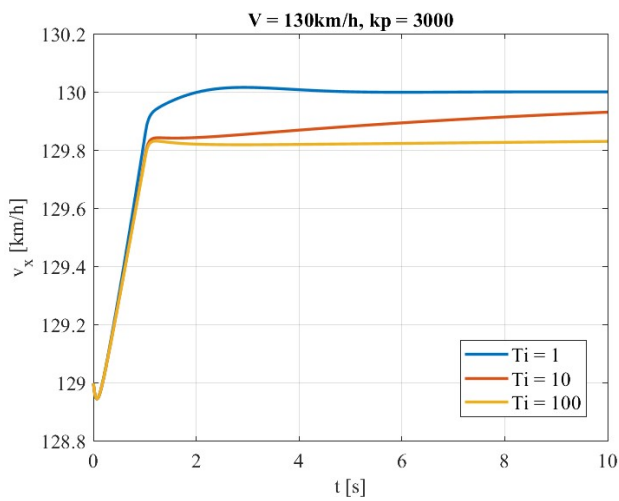
Gain tuning at $v = 90$ [km/h]



Gain tuning at $v = 110$ [km/h]



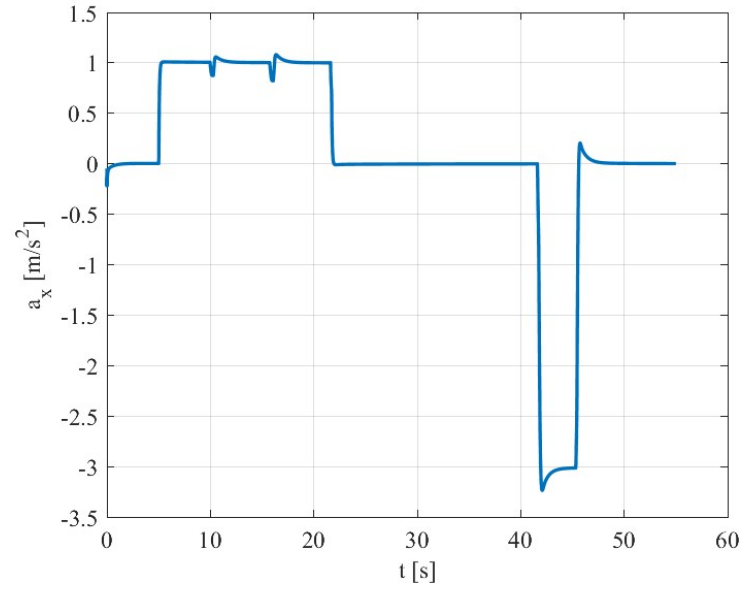
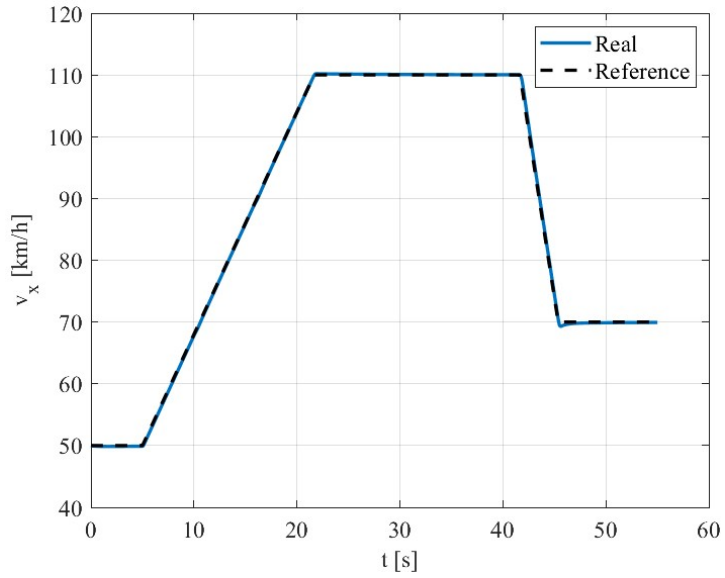
Gain tuning at $v = 130$ [km/h]



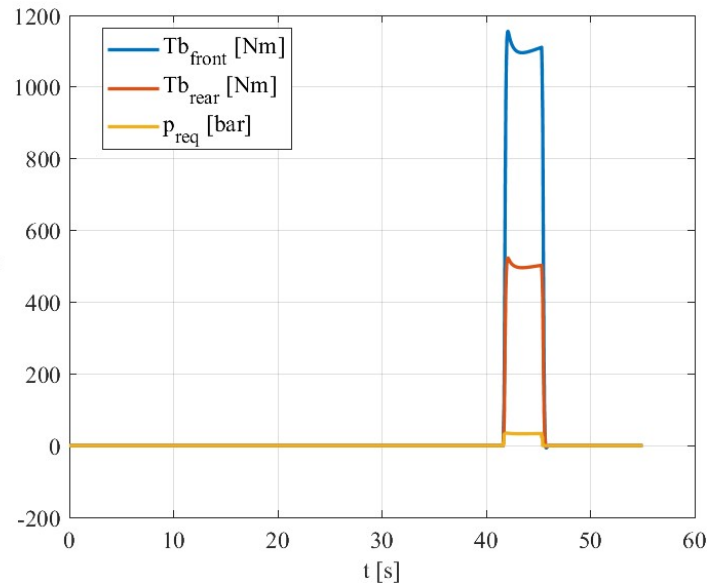
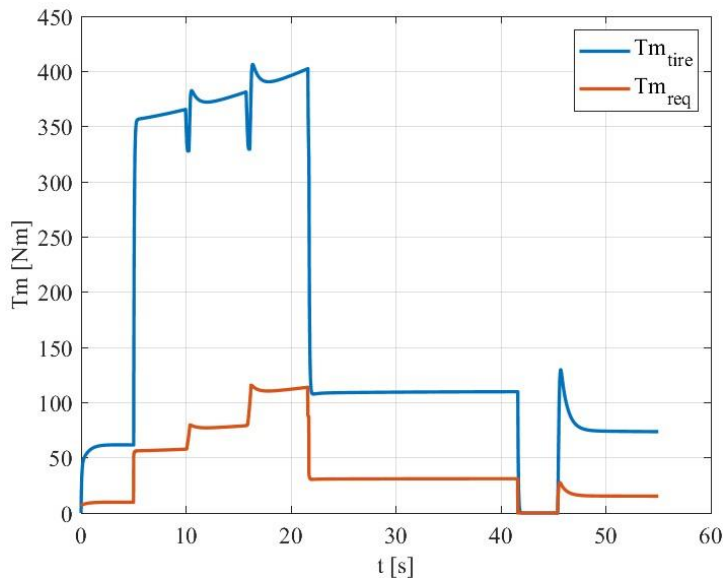
Cruise control performance testing:

1st case: $a^+ = 1 \text{ m/s}^2$, $a^- = -3 \text{ m/s}^2$, no wind speed, no road grade

Longitudinal speed and acceleration:



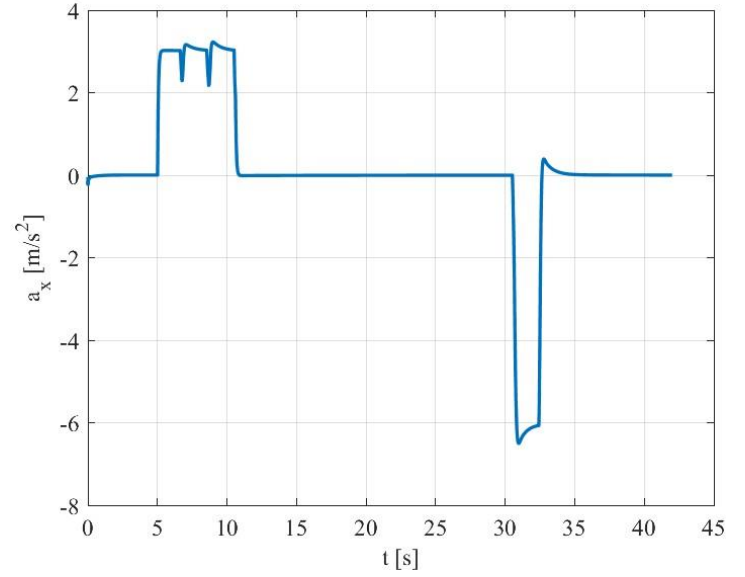
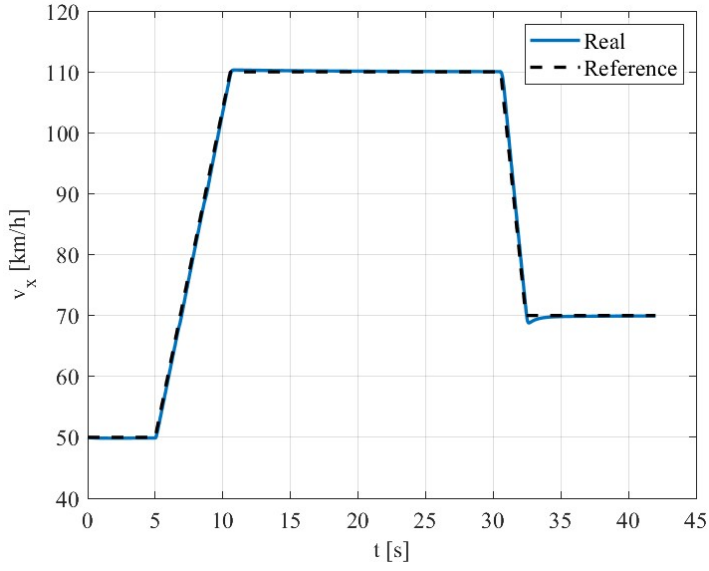
Driving and braking torque:



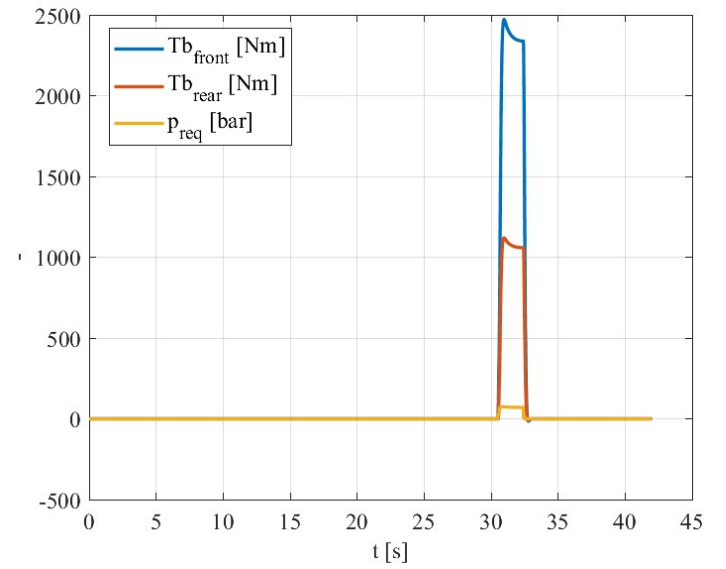
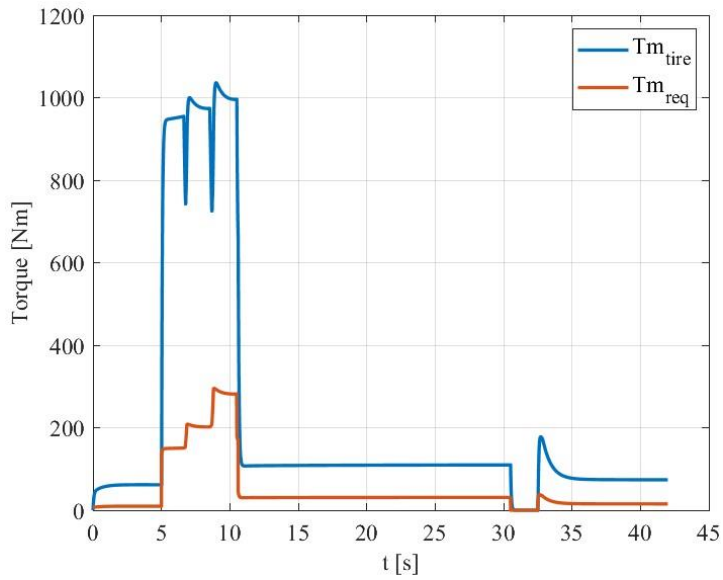
($T_{m, req}$ is the torque required by the motor)

2nd case: $a^+ = 3 \text{ m/s}^2$, $a^- = -6 \text{ m/s}^2$, no wind speed, no road grade

Longitudinal speed and acceleration:

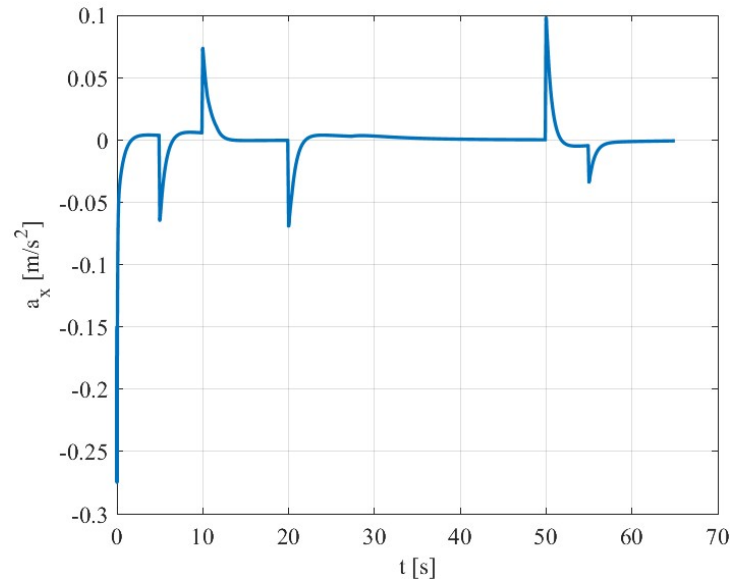
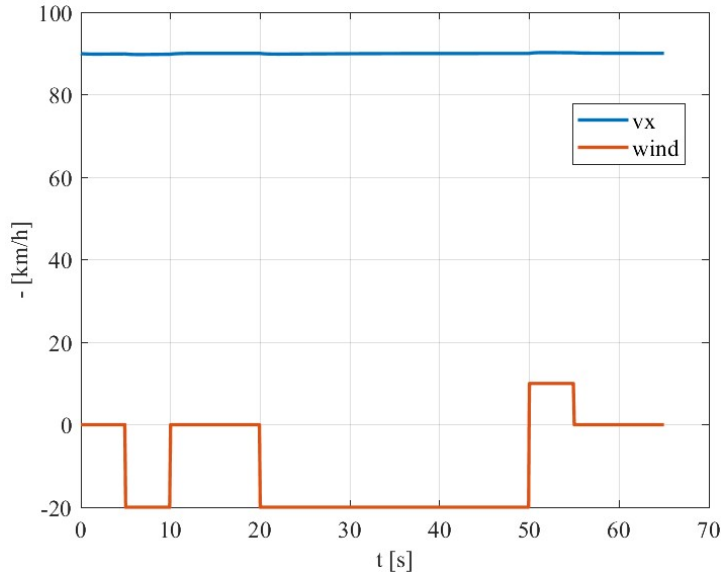


Driving and braking torque:

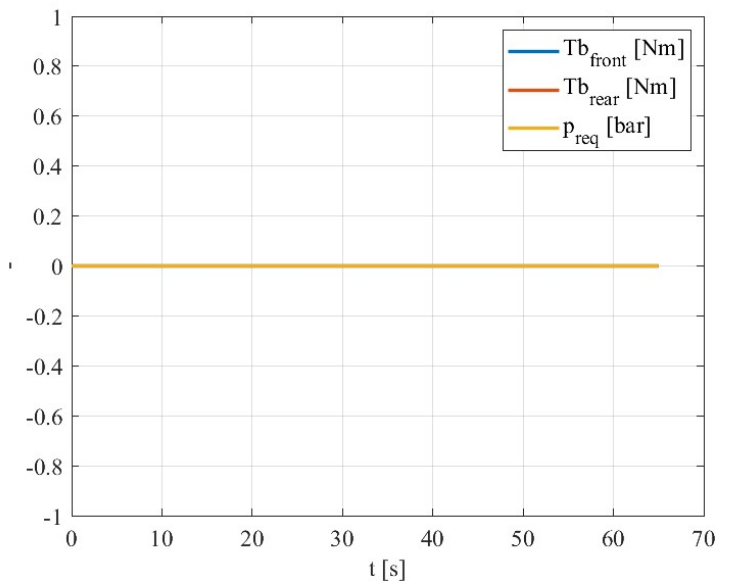
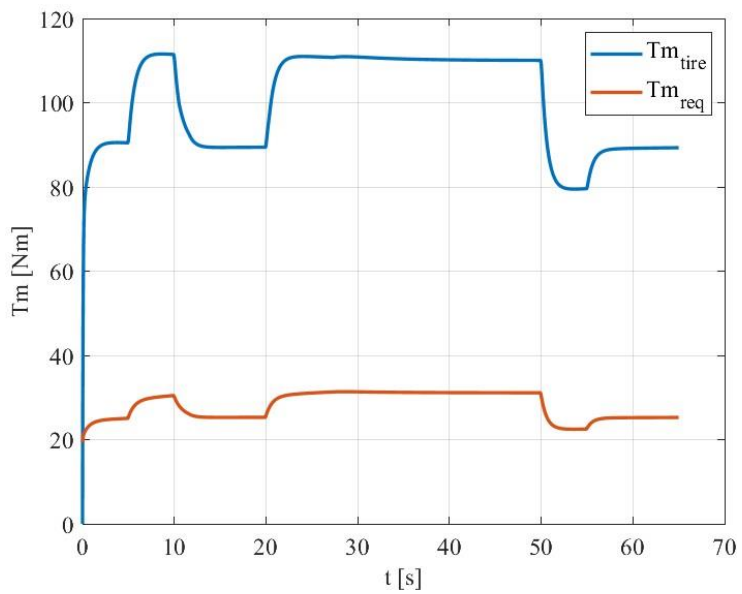


3rd case: constant reference speed $v_x = 90$ [km/h], wind perturbation considered, no road grade

Longitudinal speed and acceleration:

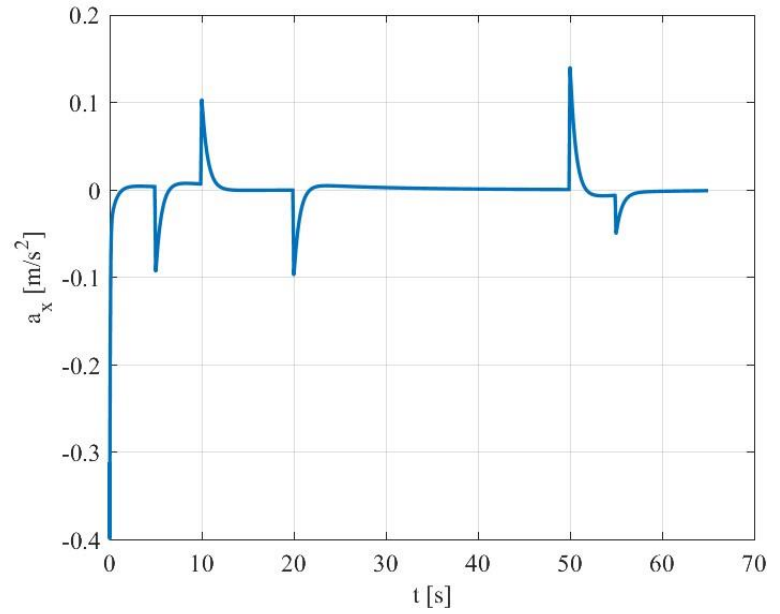
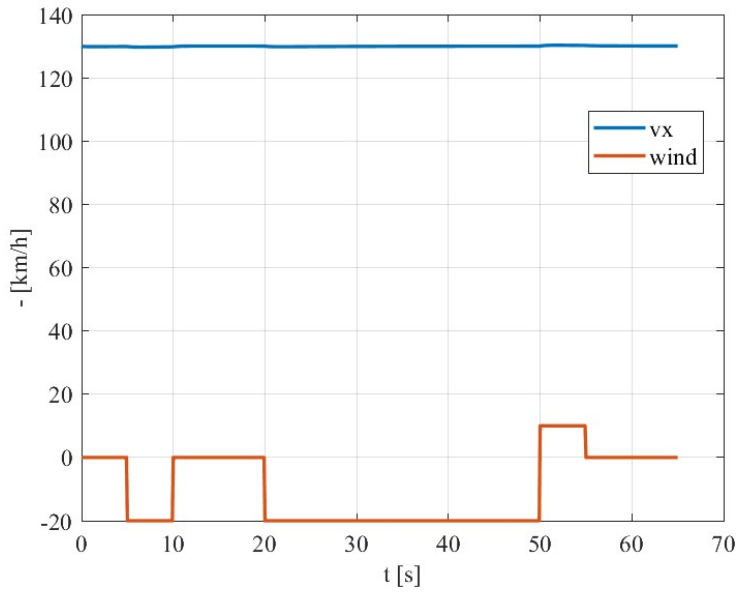


Driving and braking torque:

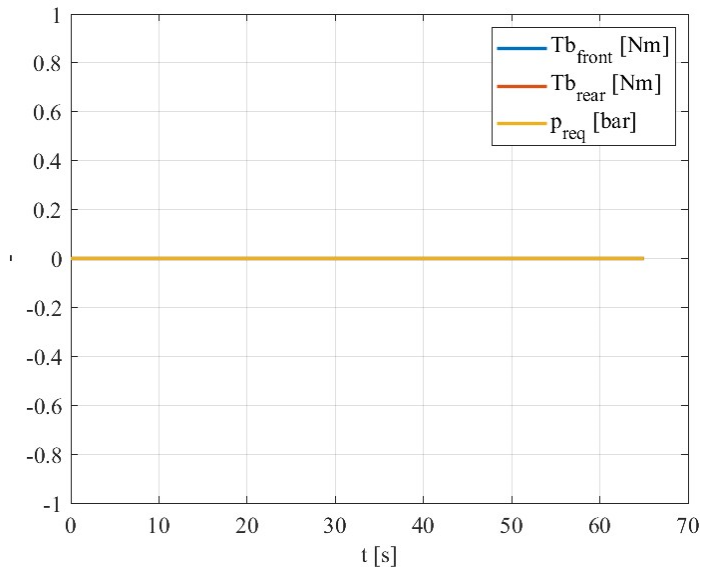
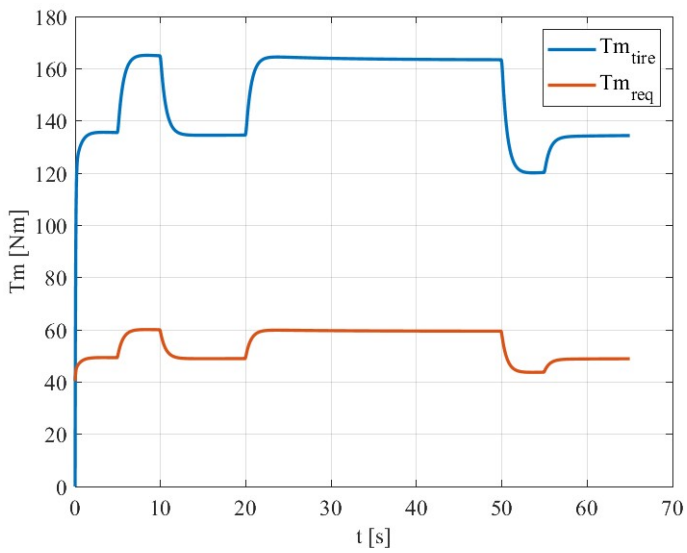


3rd case: constant reference speed $v_x = 130$ [km/h], wind perturbation considered, no road grade

Longitudinal speed and acceleration:



Driving and braking torque

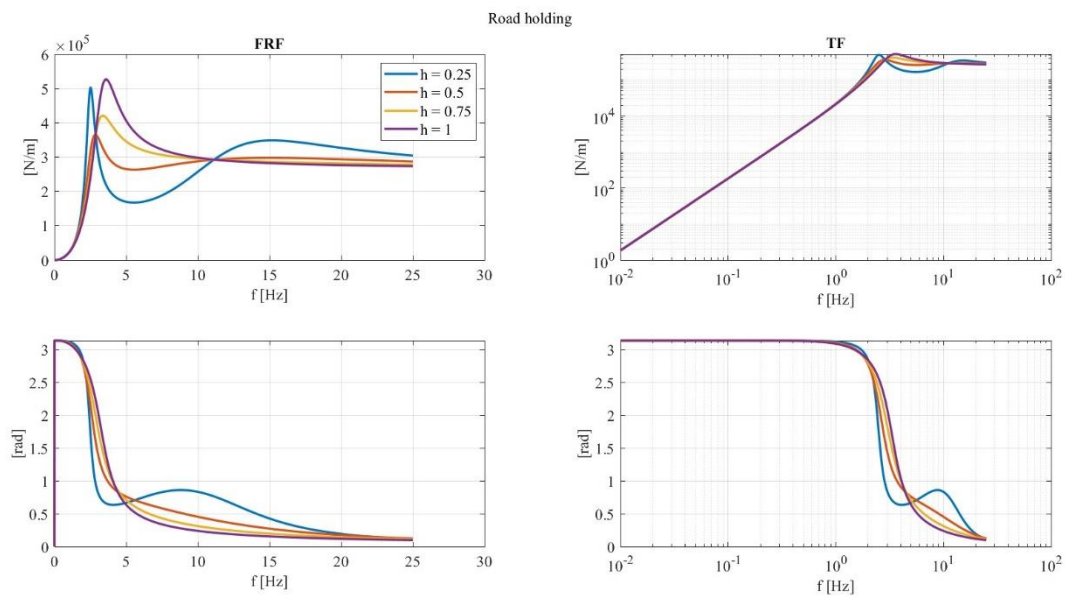


Vehicle Dynamics and Control: Vertical Dynamics Lab

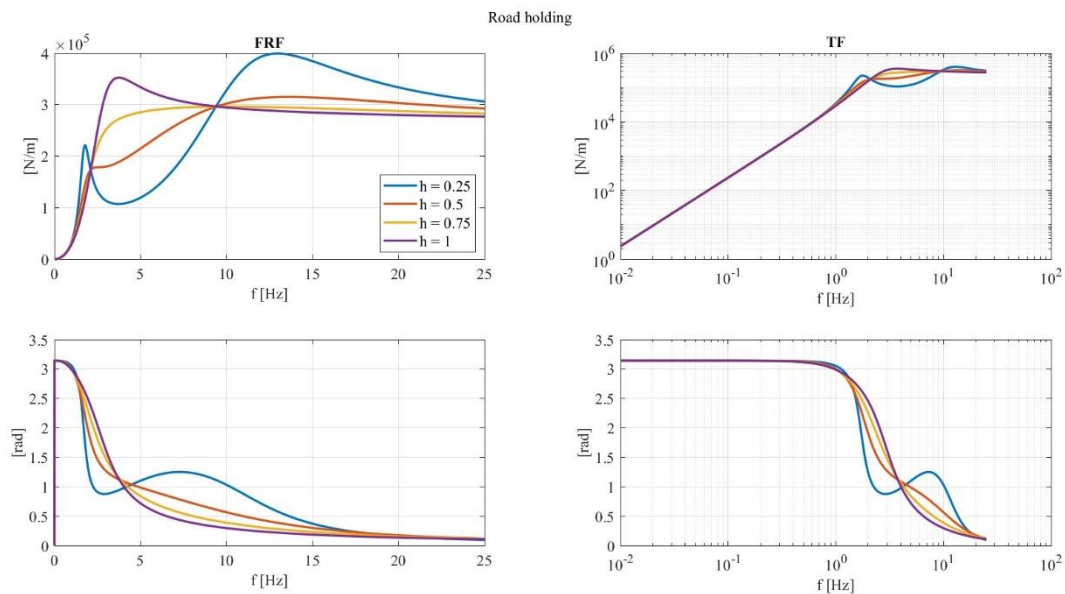
Car segment: SUV

Comparing four different damping coefficients to find the best one for front and rear suspensions, by evaluating discomfort, road holding, and working space FRF

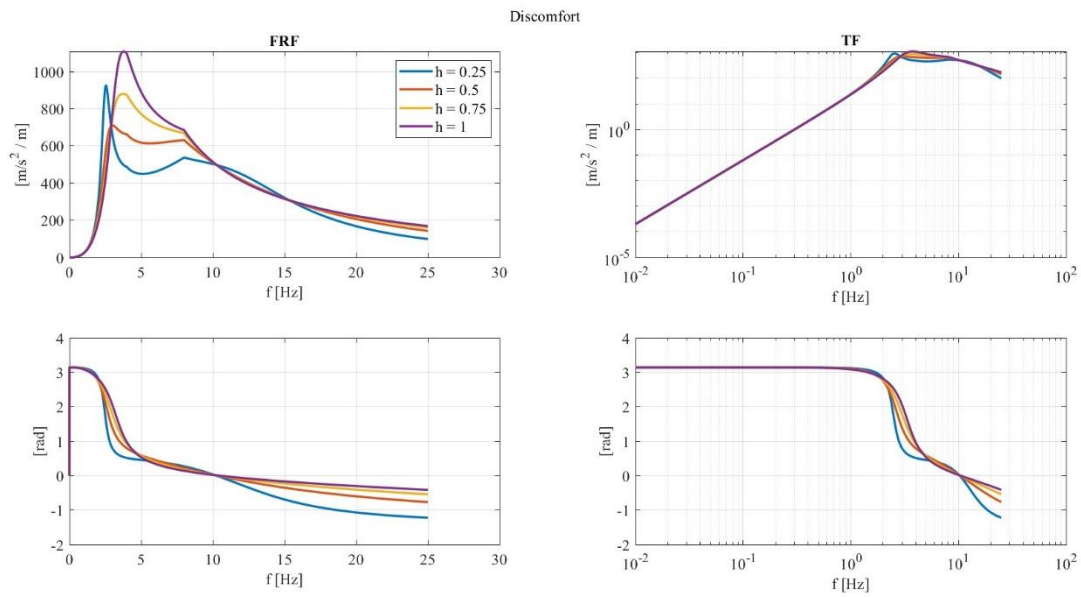
Road holding FRF for the front suspension:



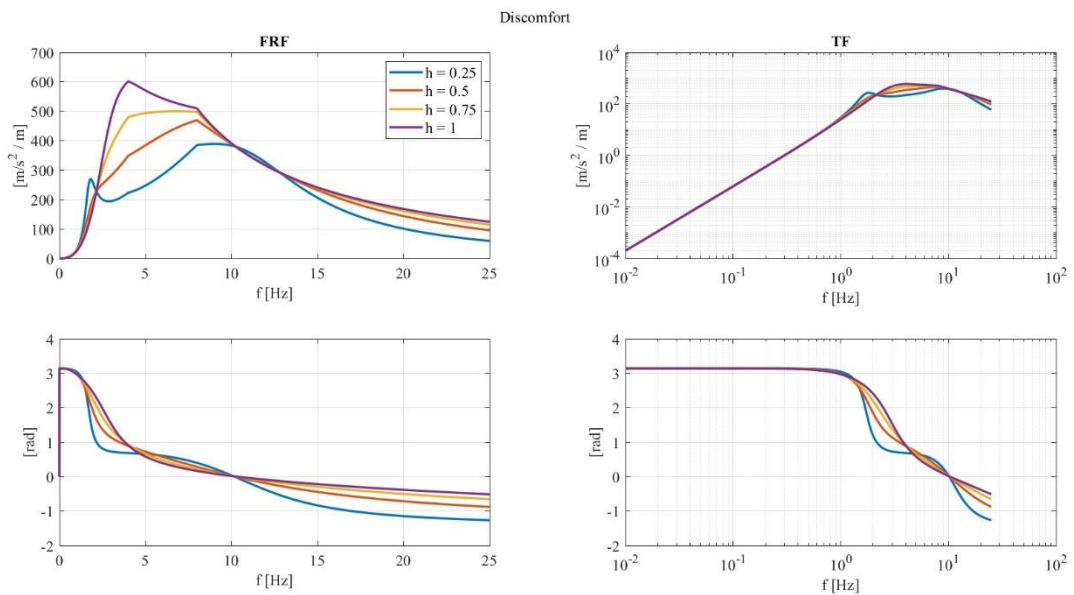
Road holding FRF for the rear suspension:



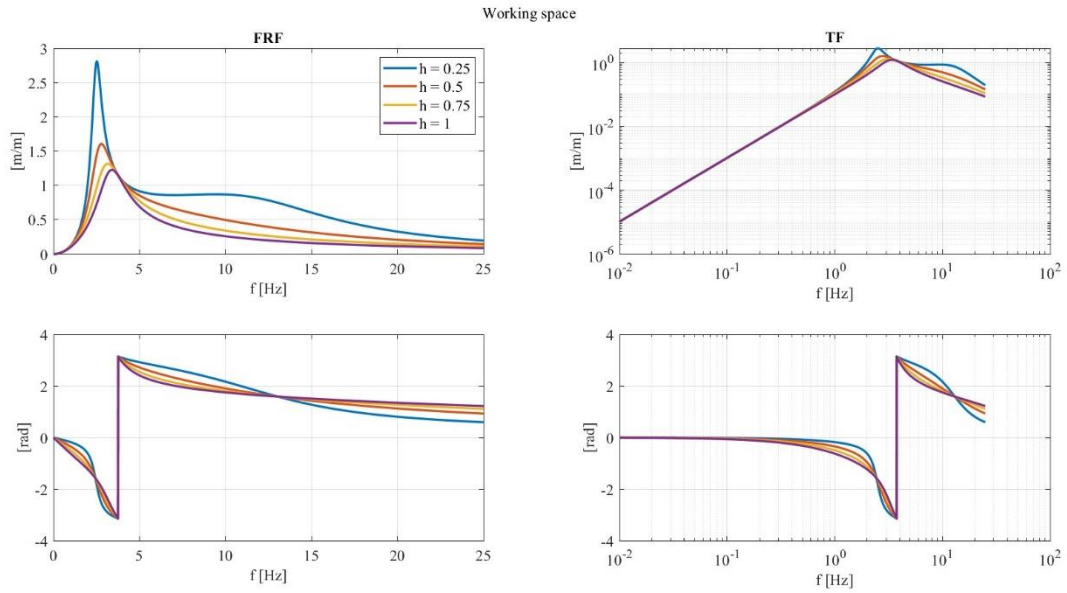
Discomfort FRF for the front suspension:



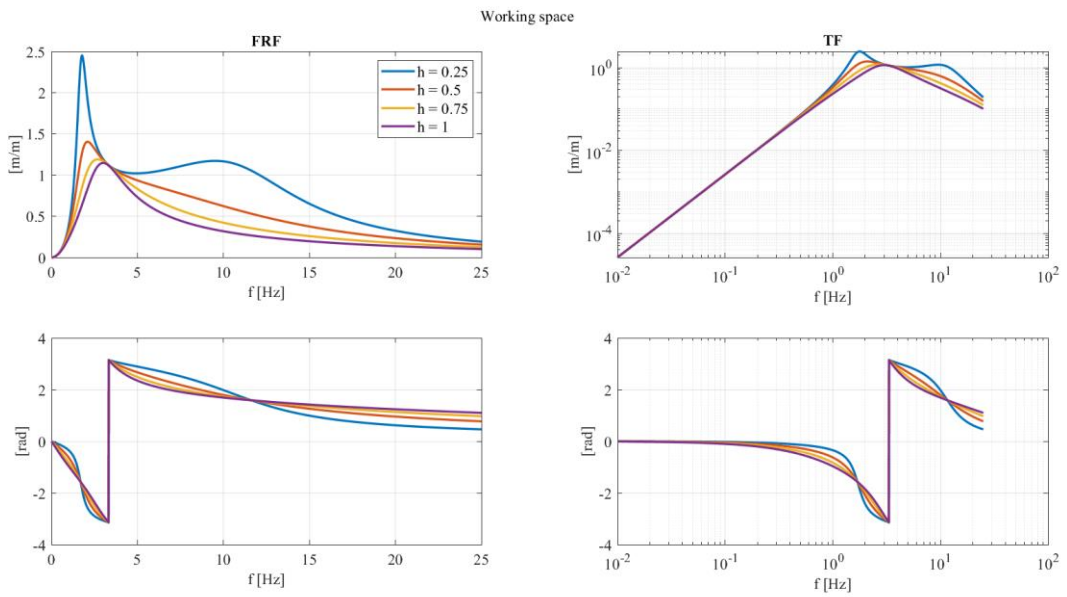
Discomfort FRF for the rear suspension:



Working space FRF for the front suspension:



Working space FRF for the rear suspension:



Best damping coefficient for the front suspension (h_f): 0.3

Best damping coefficient for the rear suspension (h_r): 0.35

$$c_{sf} = h_f * c_{cf} \quad c_{pasf} = \frac{c_{sf}}{2} \quad c_{skyf} = c_{sf}$$

$$c_{sr} = h_r * c_{cr} \quad c_{pasr} = \frac{c_{sr}}{2} \quad c_{skyr} = c_{sr}$$

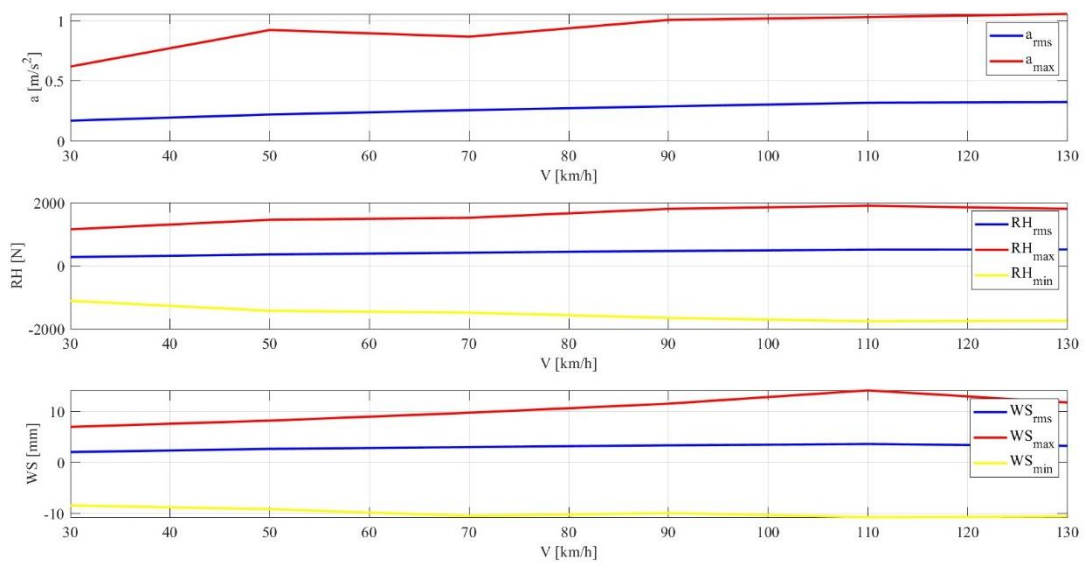
Road irregularity profile: **class B**

Comfort parameters:

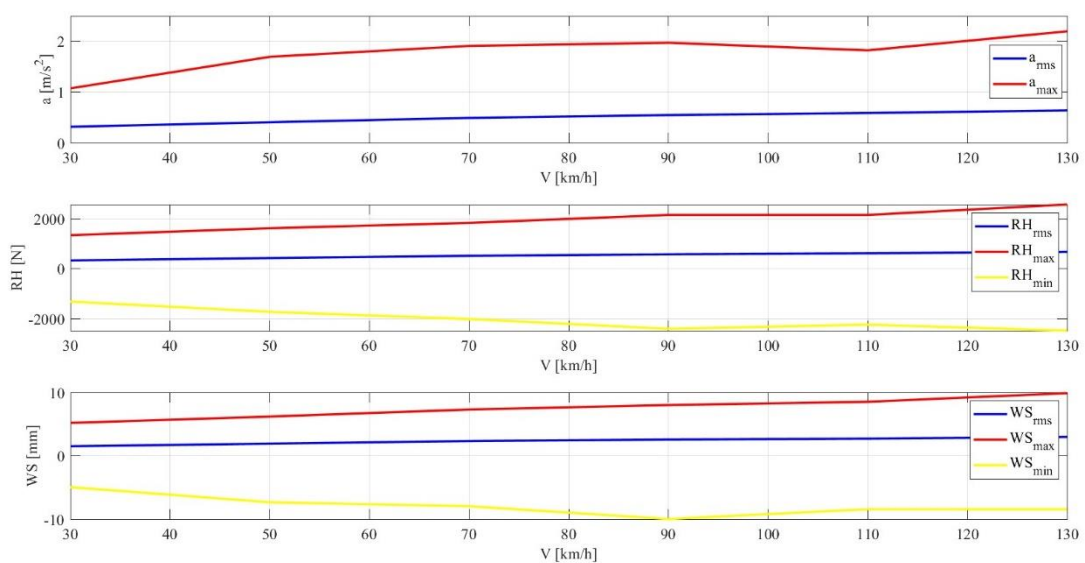
Comfort Index
Road Holding
Working Space

Passive system:

Front suspension:

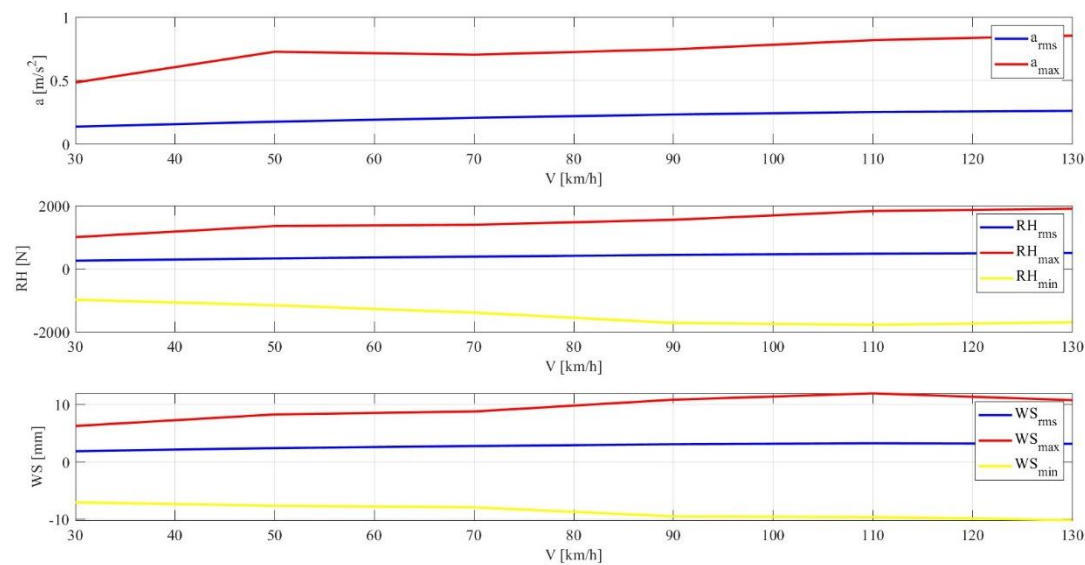


Rear suspension:

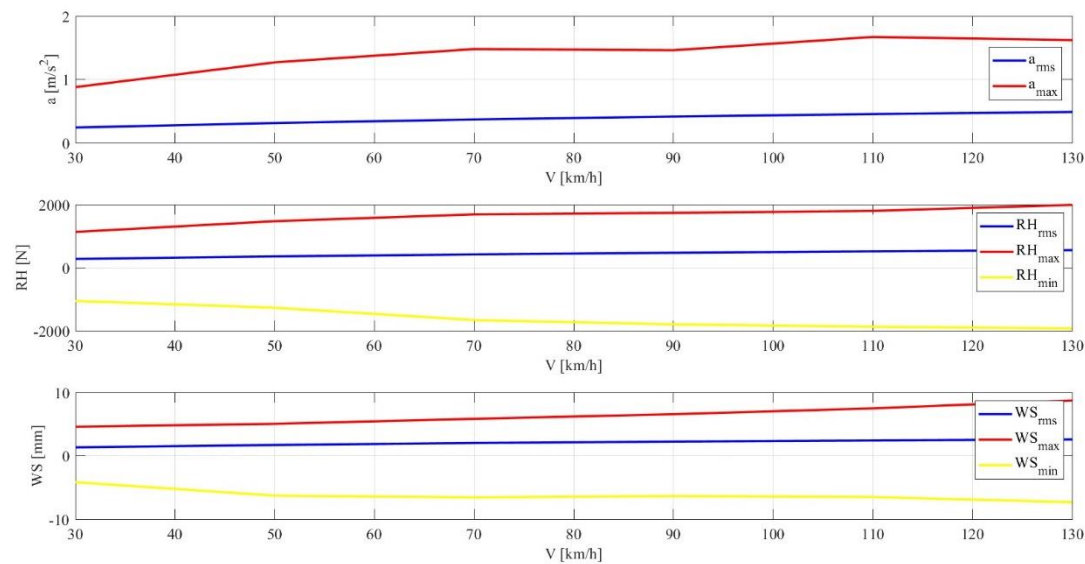


Semi-active system:

Front suspension:

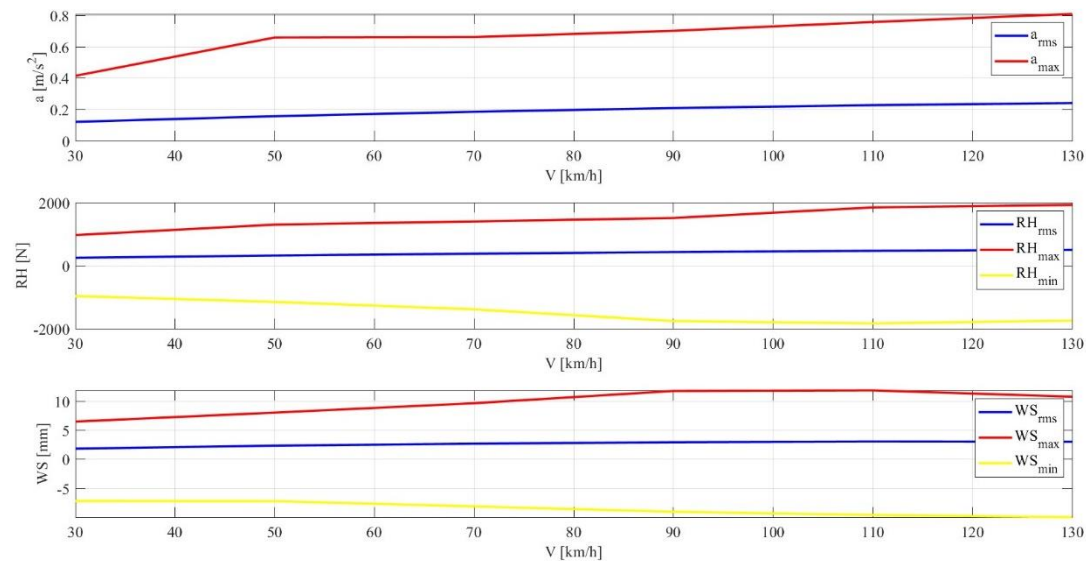


Rear suspension:

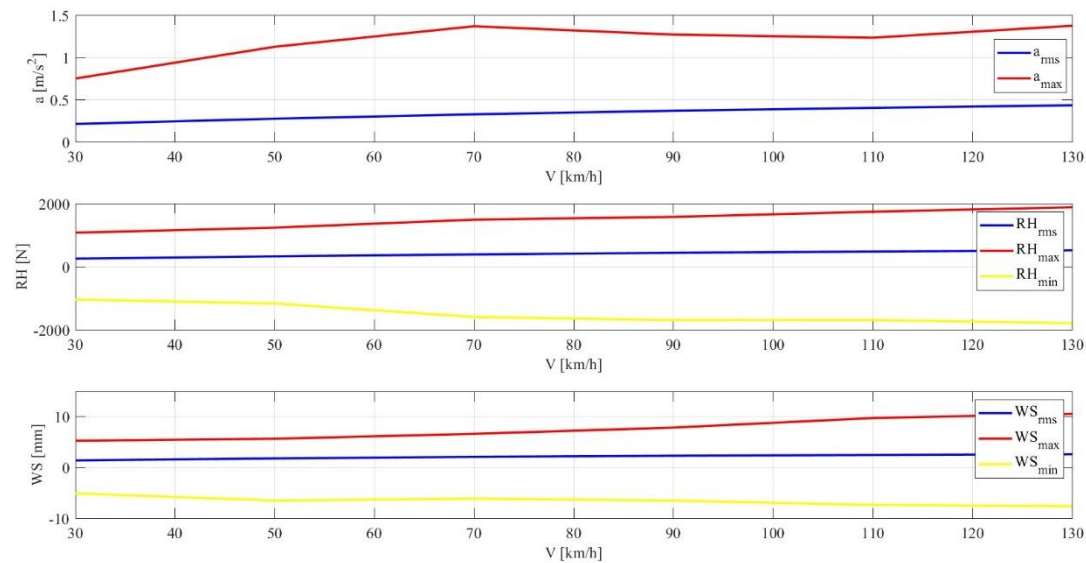


Active system:

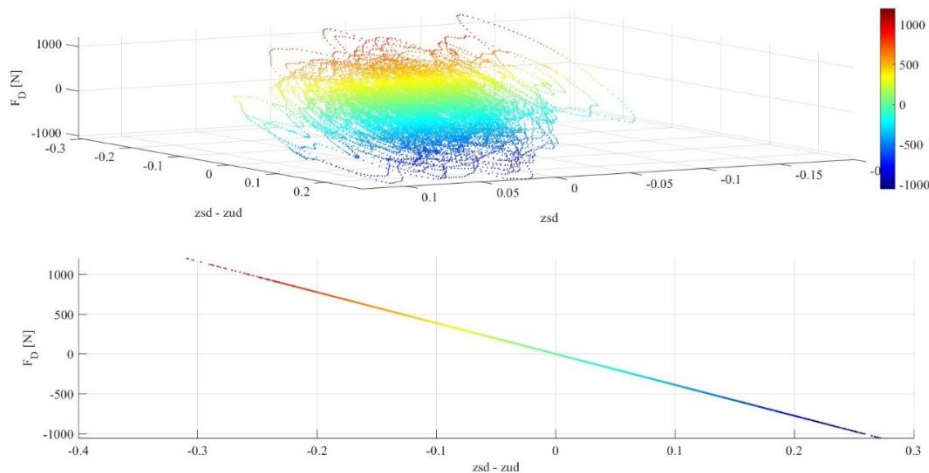
Front suspension:



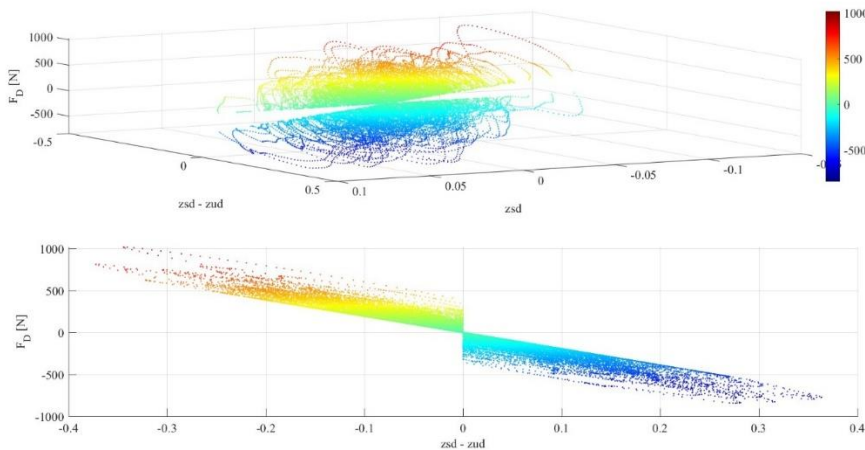
Rear suspension:



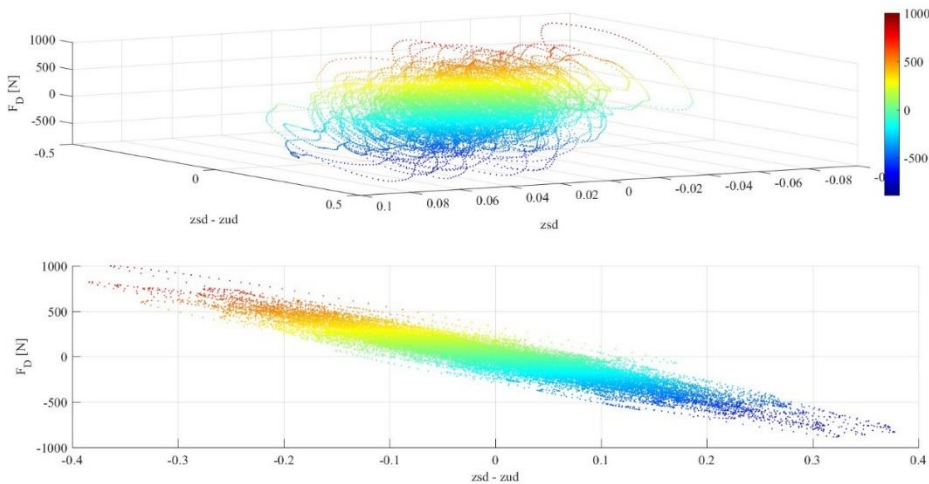
Front damper characteristic for:
passive system:



semi-active system:

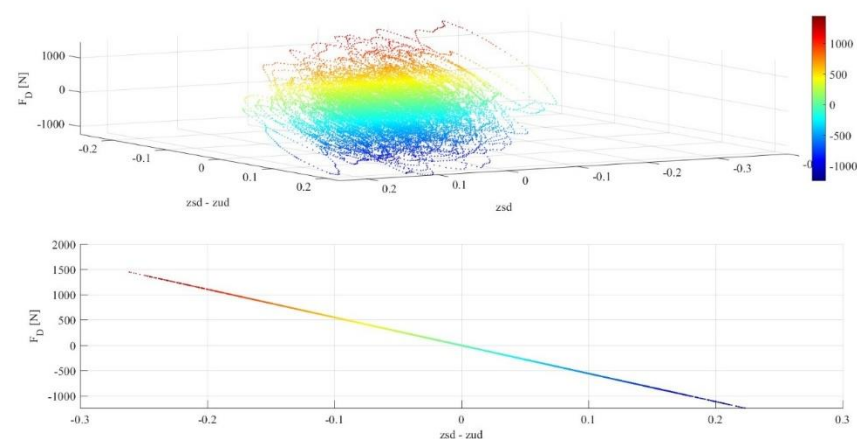


active system:

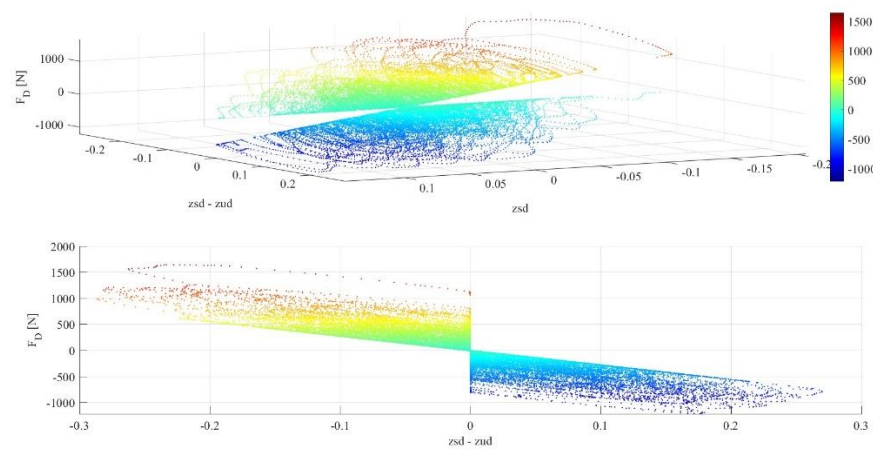


Rear damper characteristic:

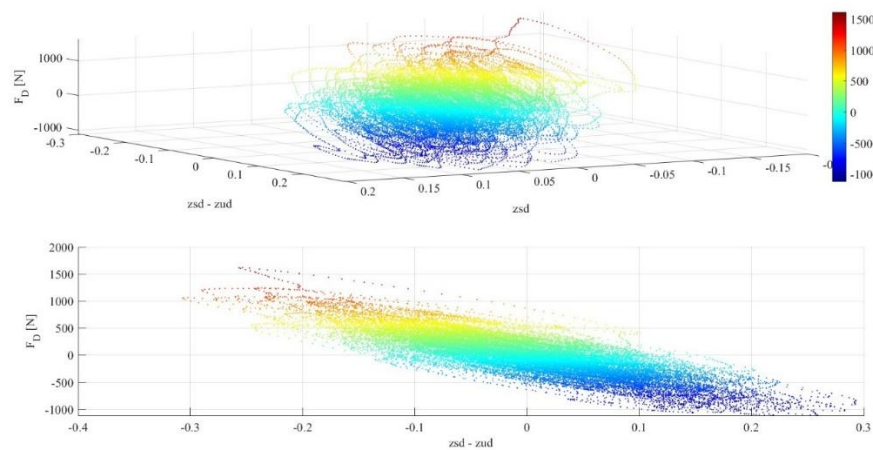
passive system:



semi-active system:

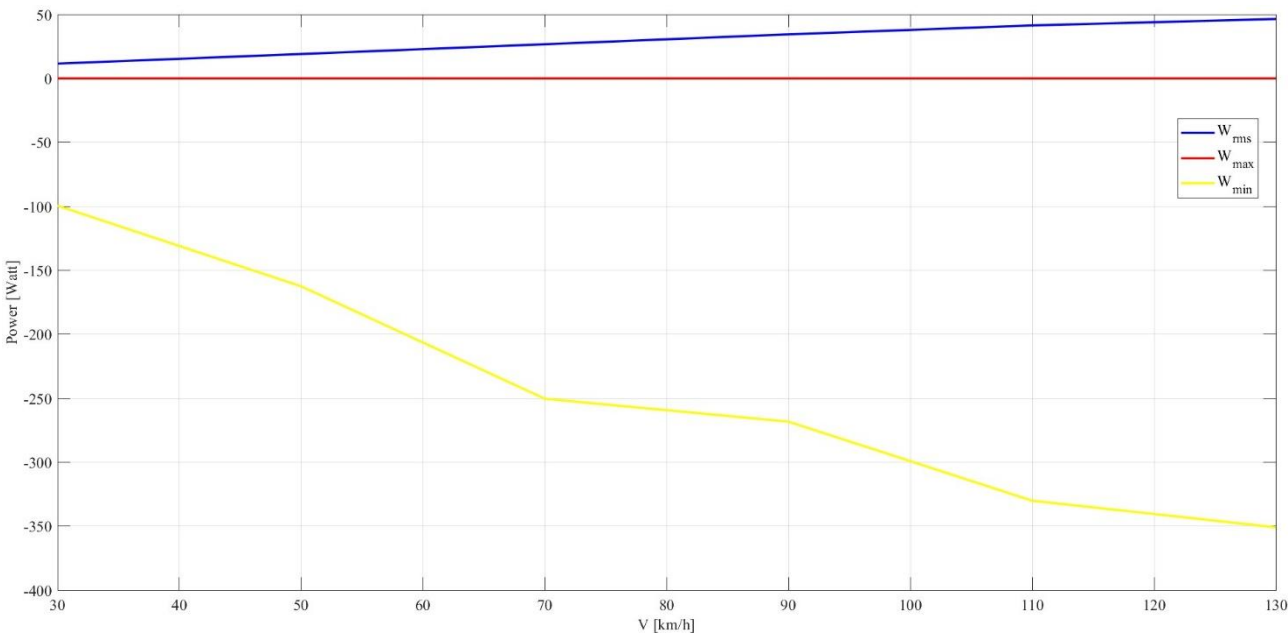


active system:

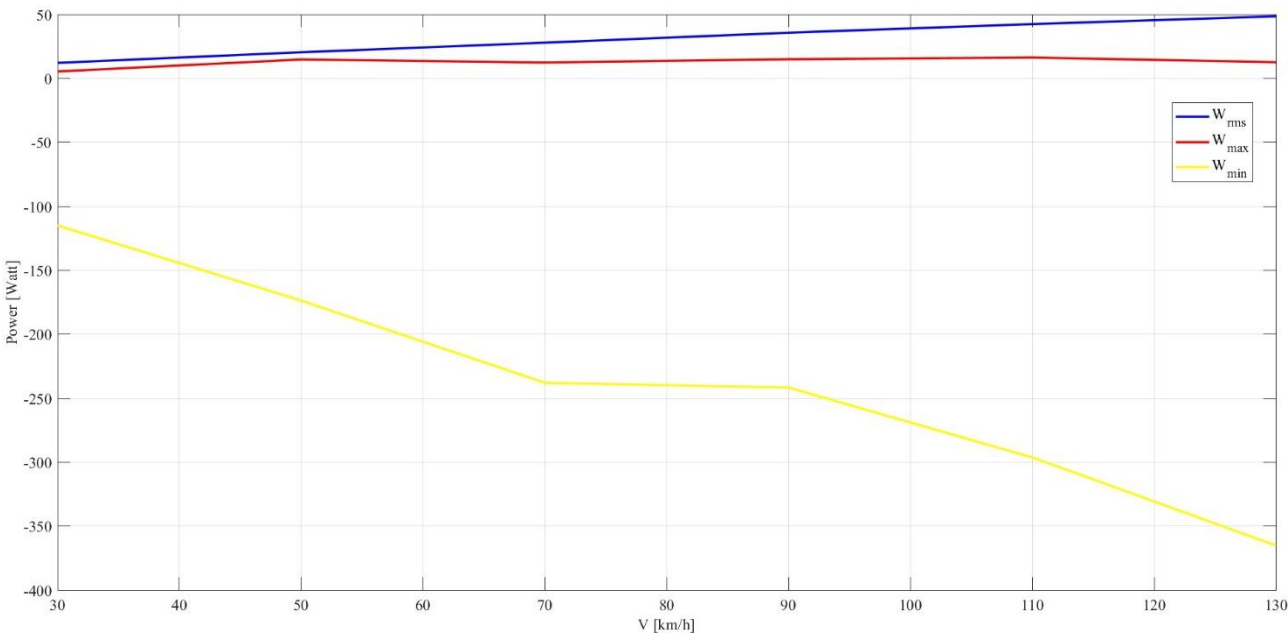


The power dissipated or introduced by the passive suspension and actuator in the **front**:

Semi-active:

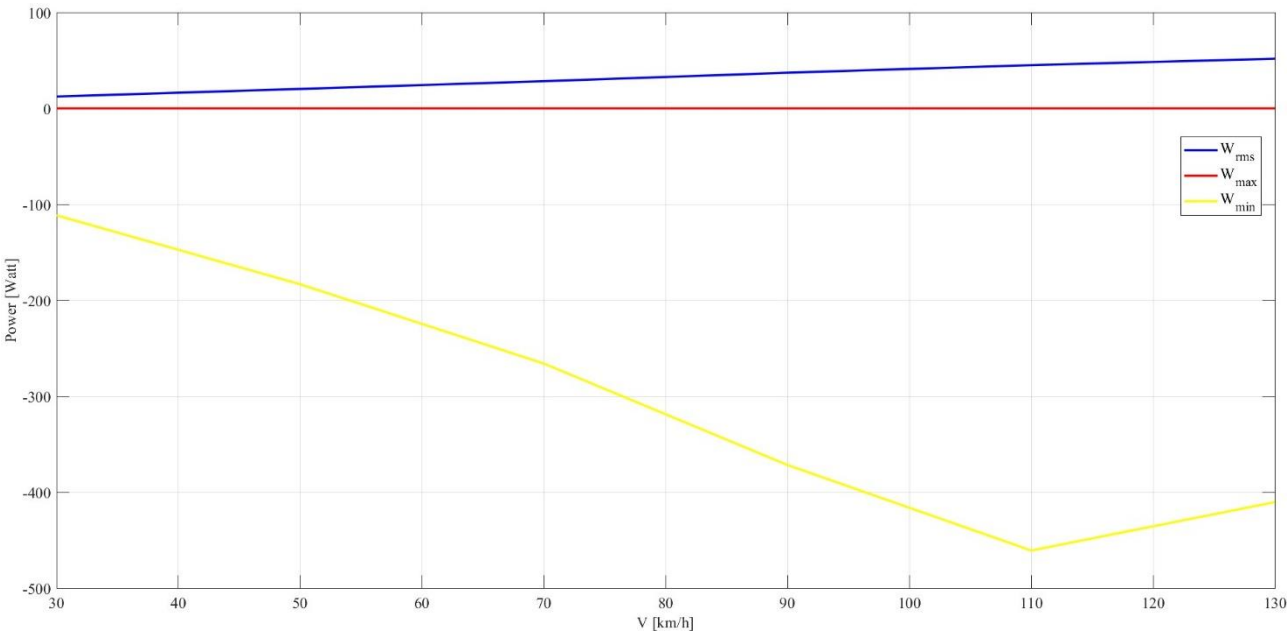


Active:

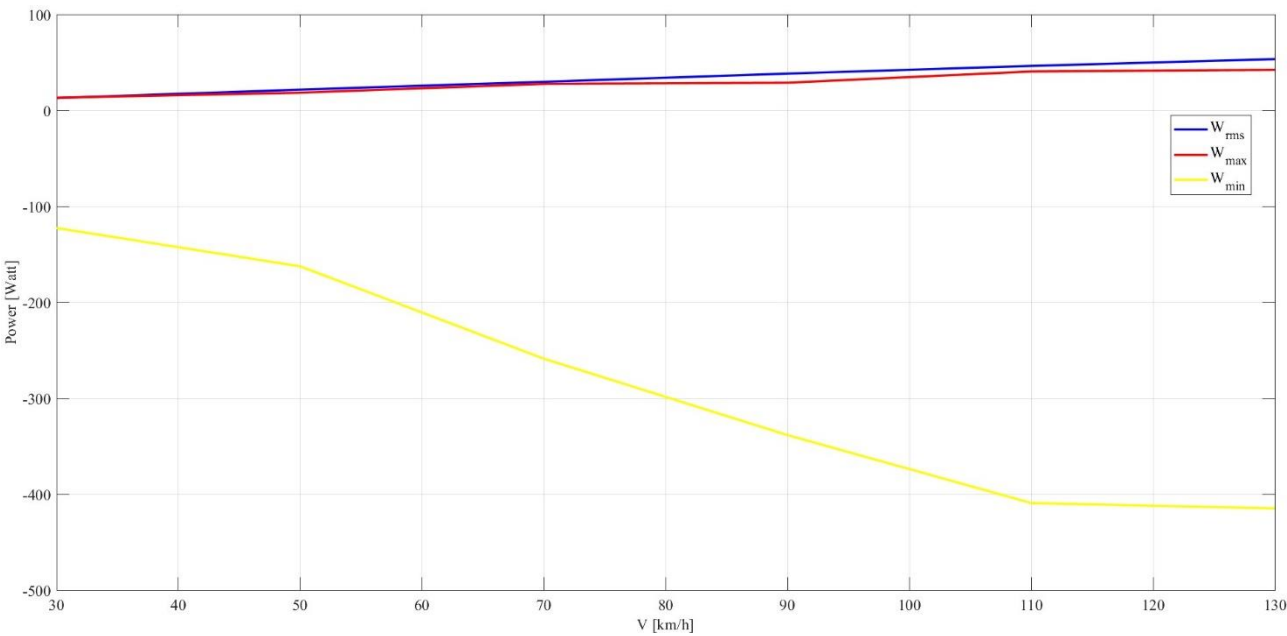


The power dissipated or introduced by the passive suspension and actuator in the **rear**:

Semi-active:



Active:



Road irregularity profile: **class C**

Comfort parameters:

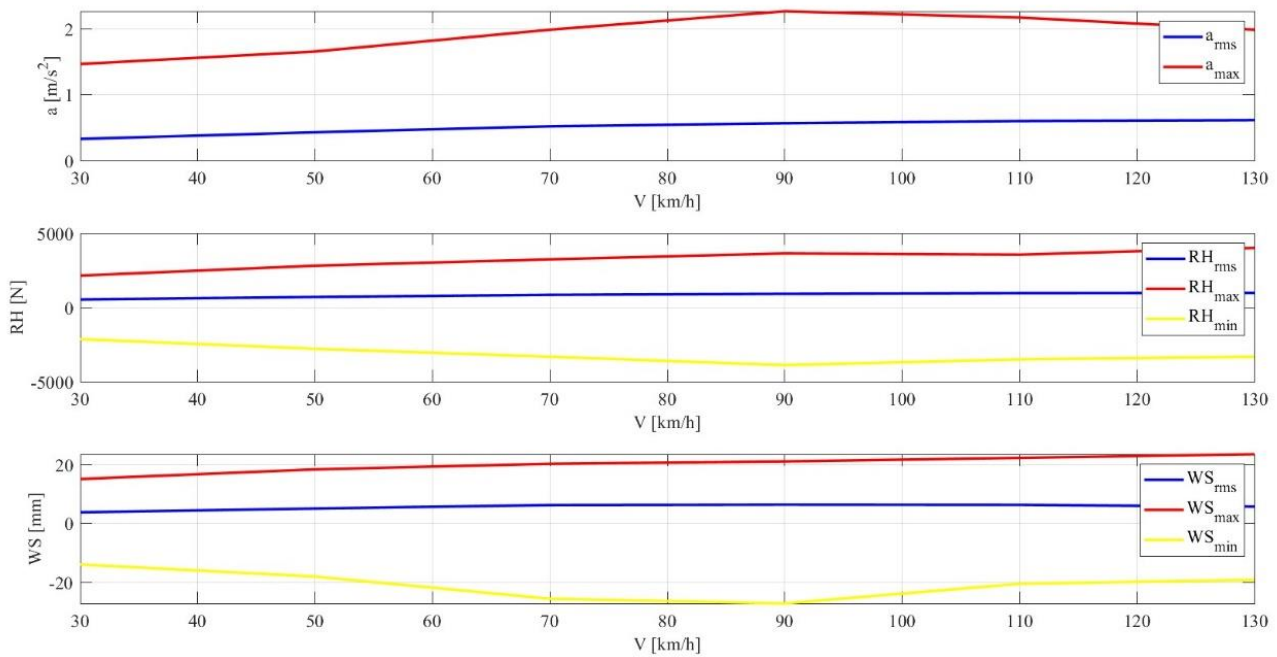
Comfort Index

Road Holding

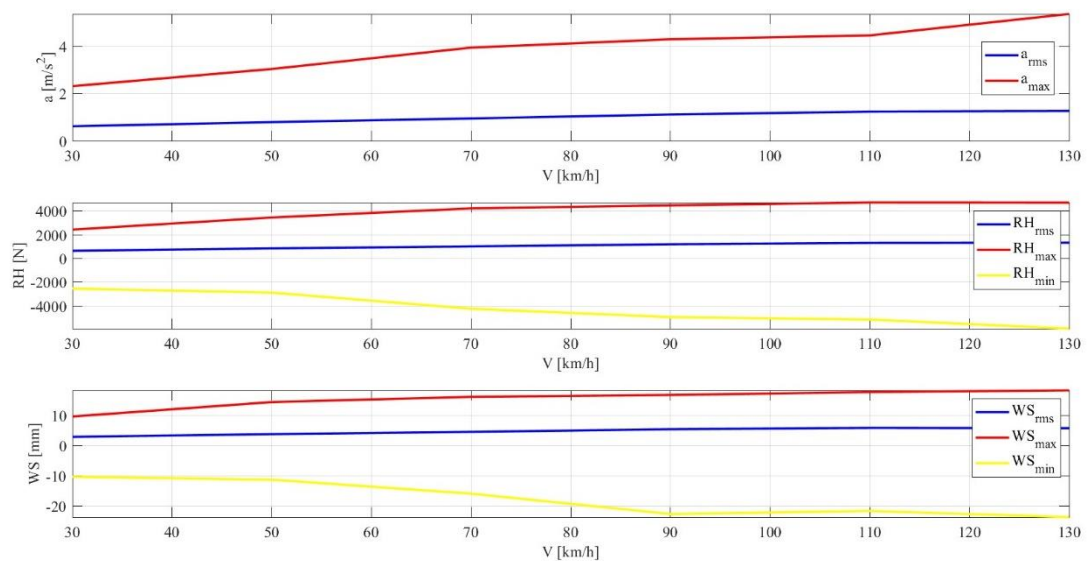
Working Space

Passive system:

Front suspension:

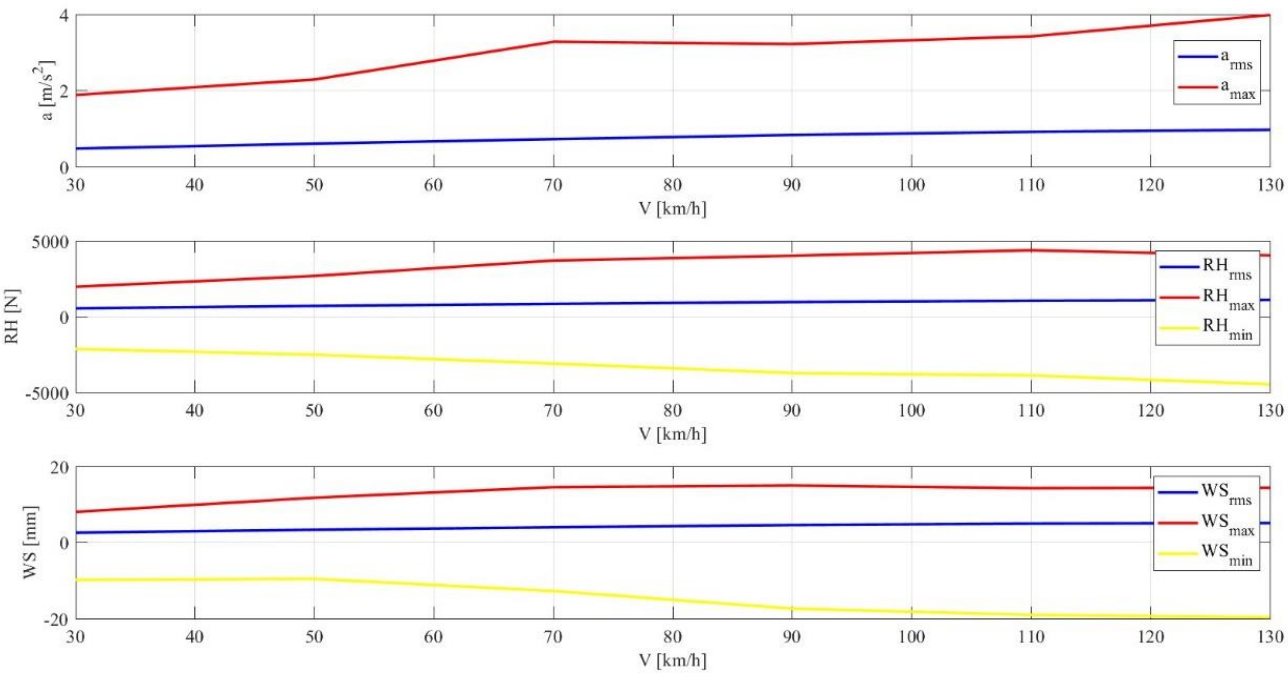


Rear suspension:

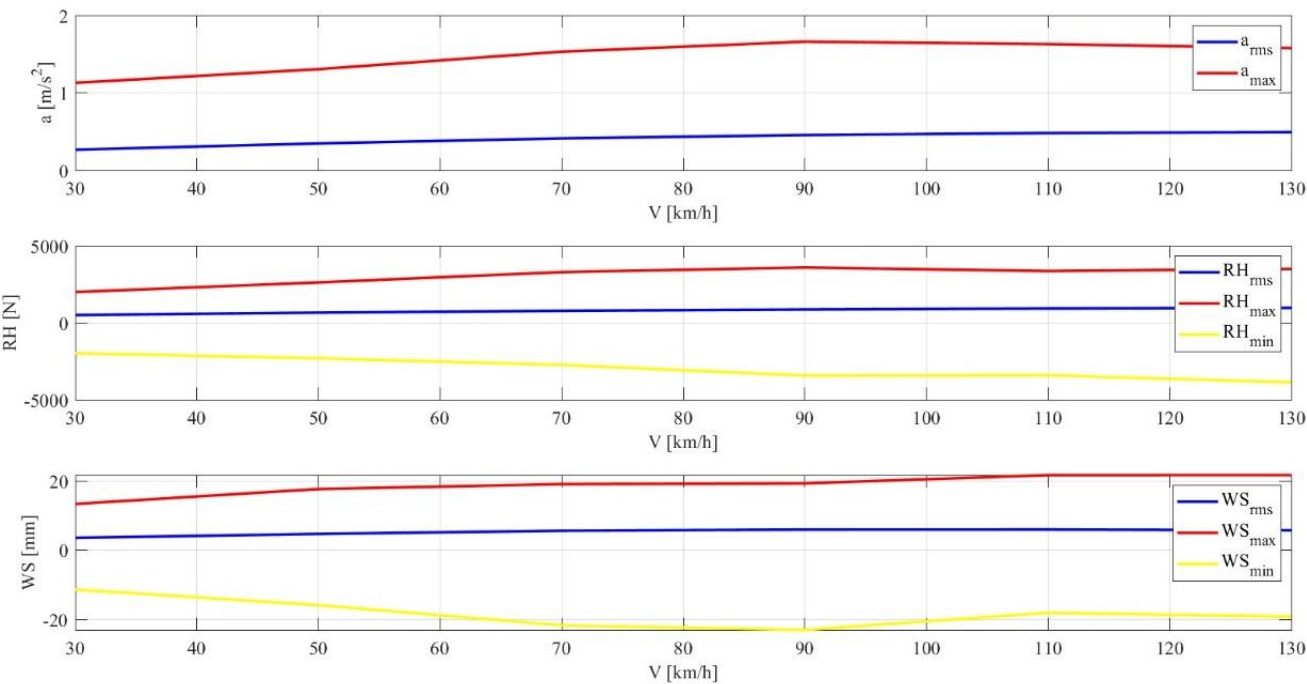


Semi-active system:

Front suspension:

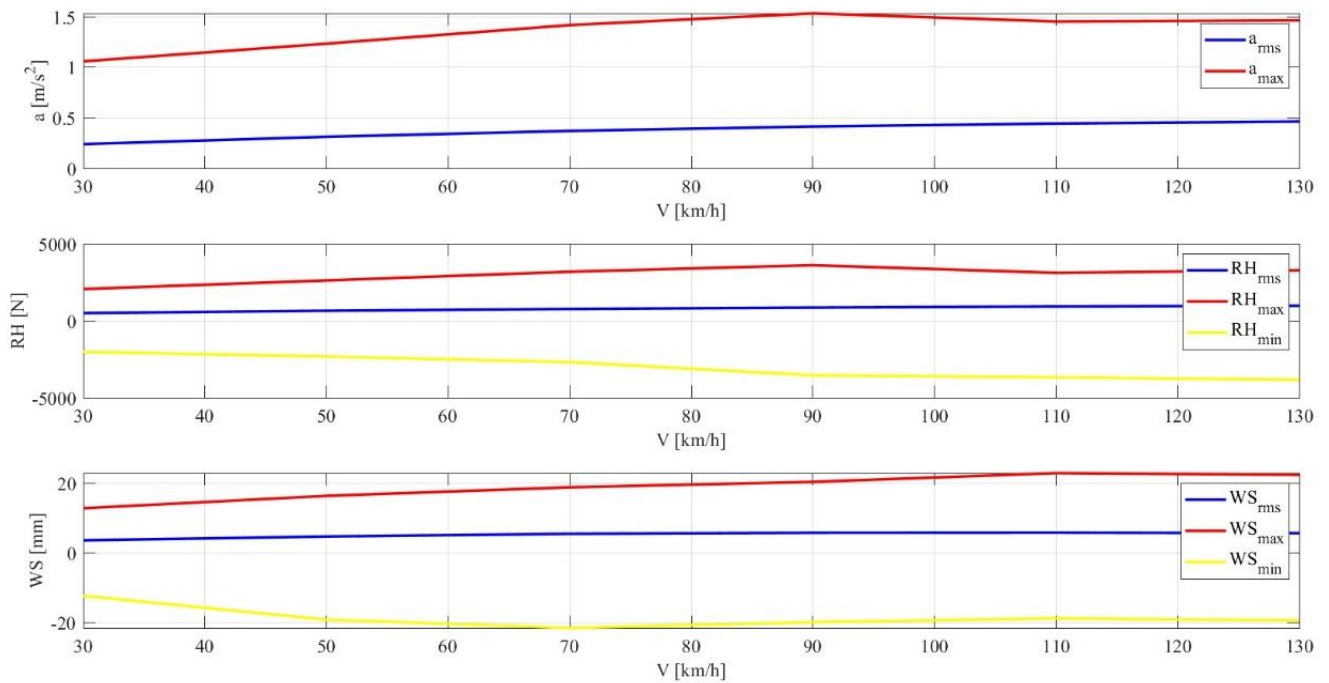


Rear suspension:

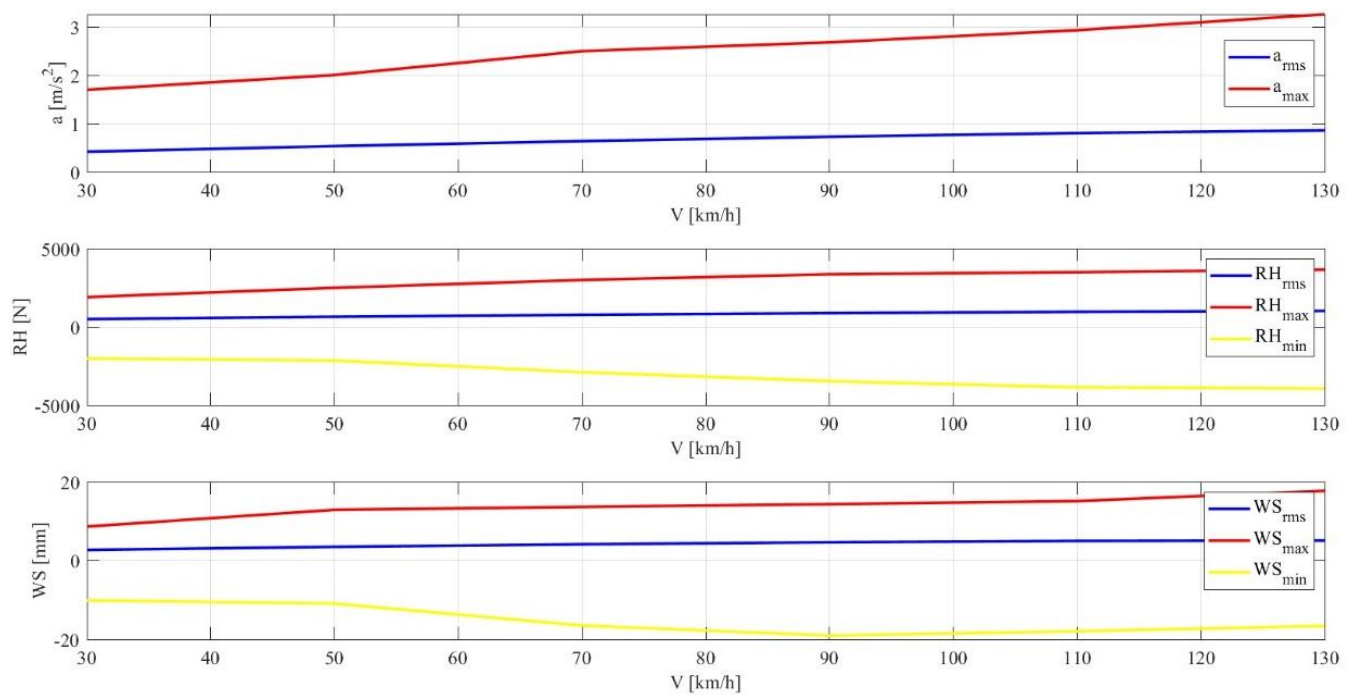


Active system:

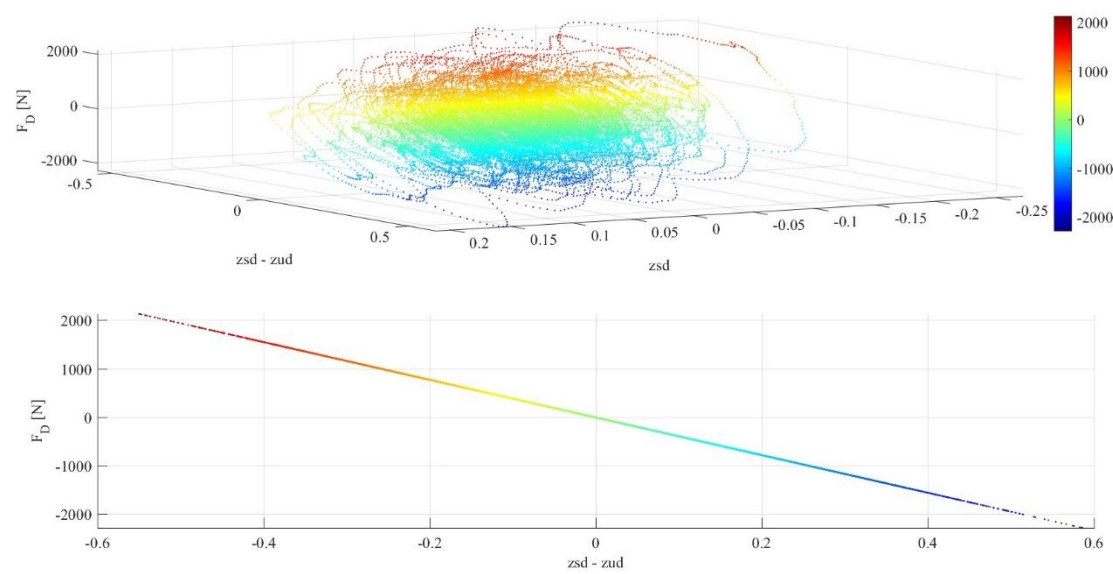
Front suspension:



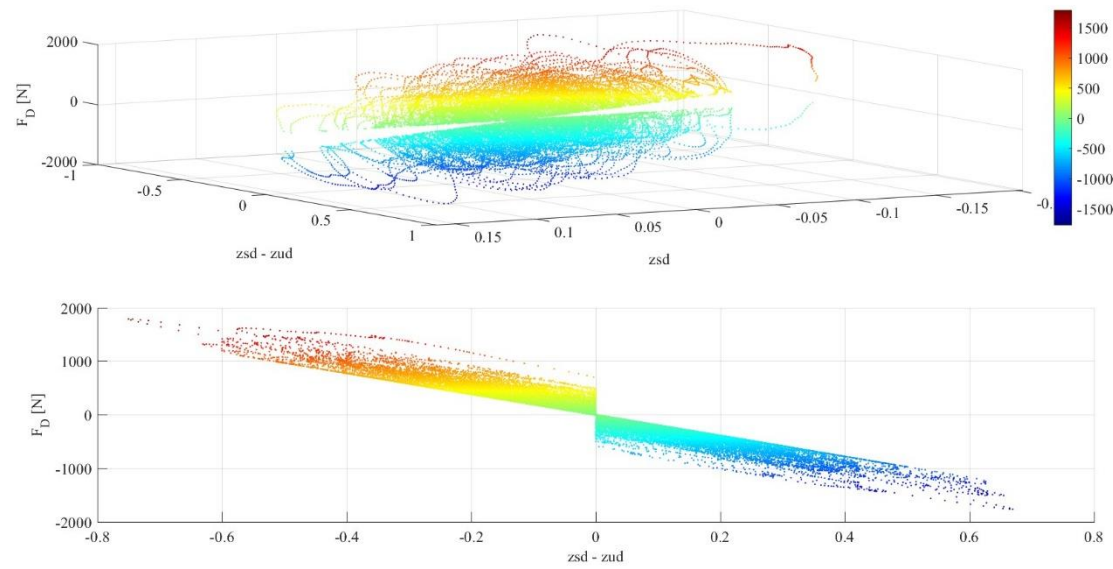
Rear suspension:



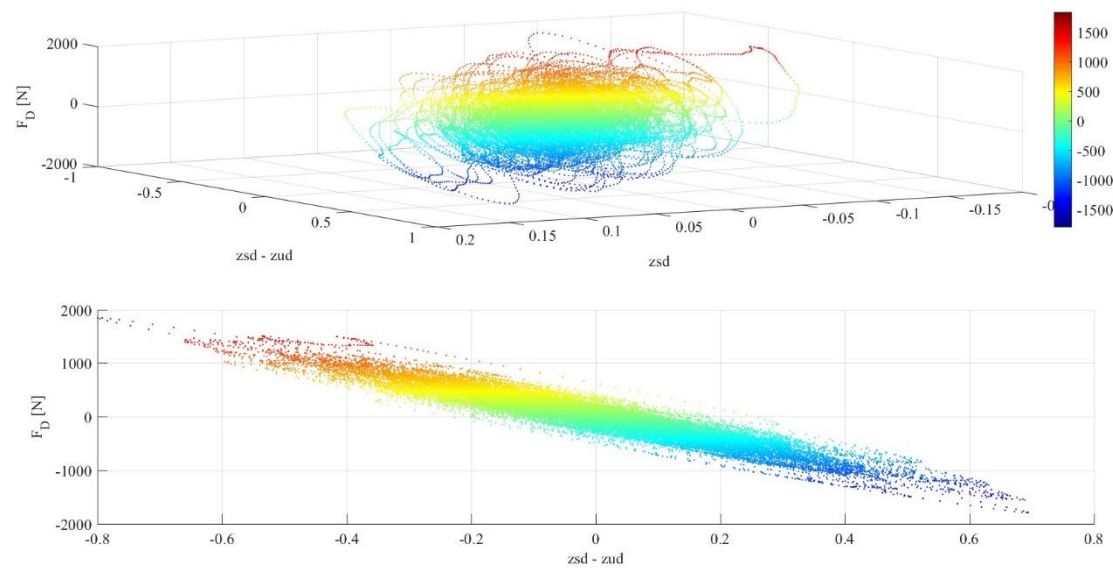
Front damper characteristic for:
passive system:



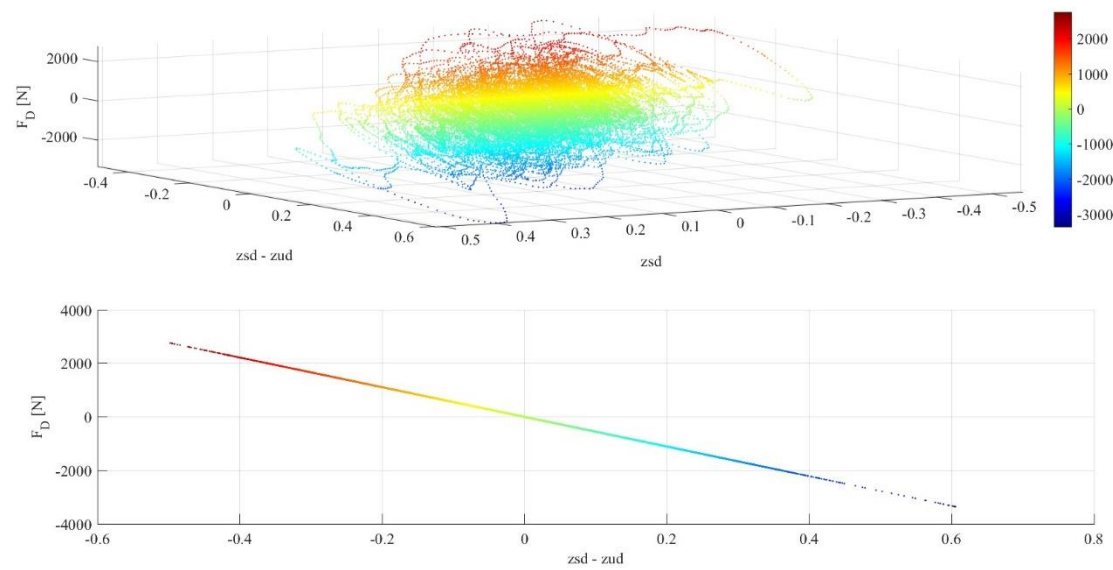
semi-active system:



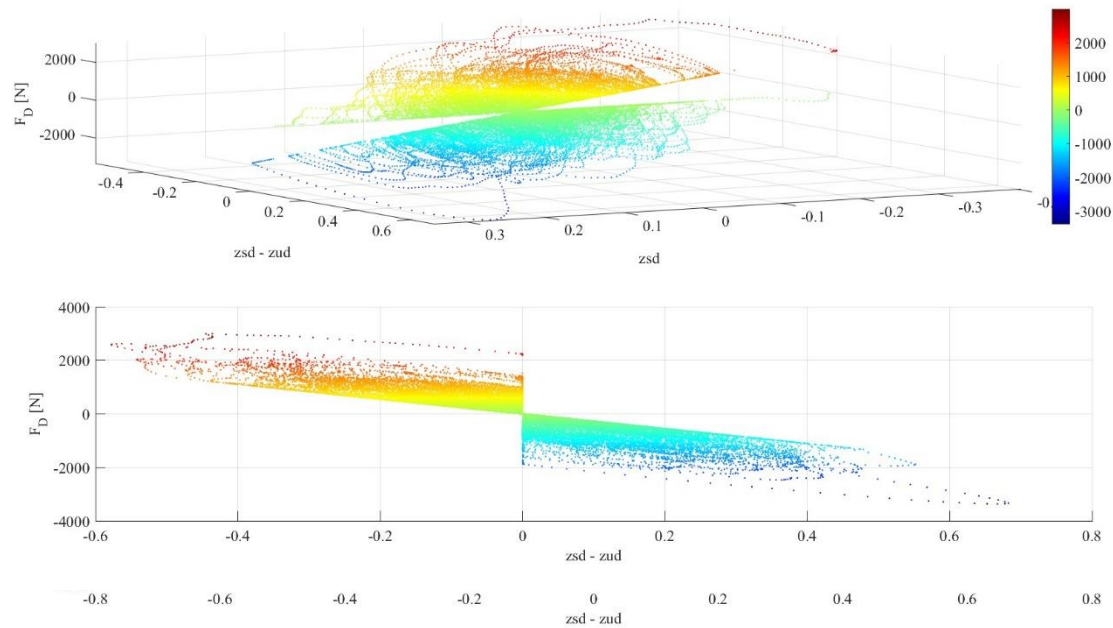
Active system:



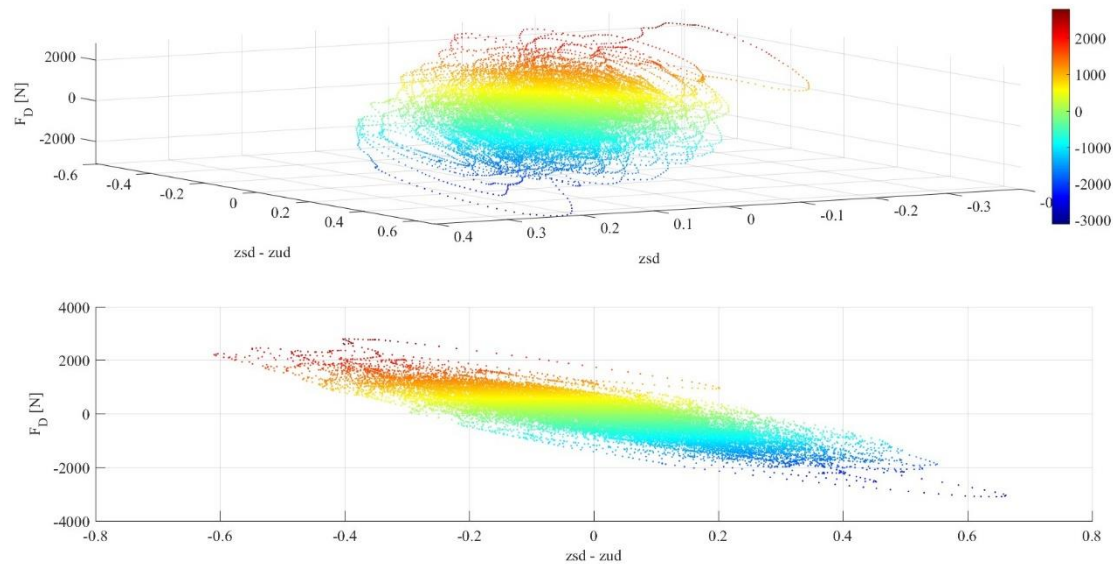
Rear damper characteristic for:
passive system



semi-active system:

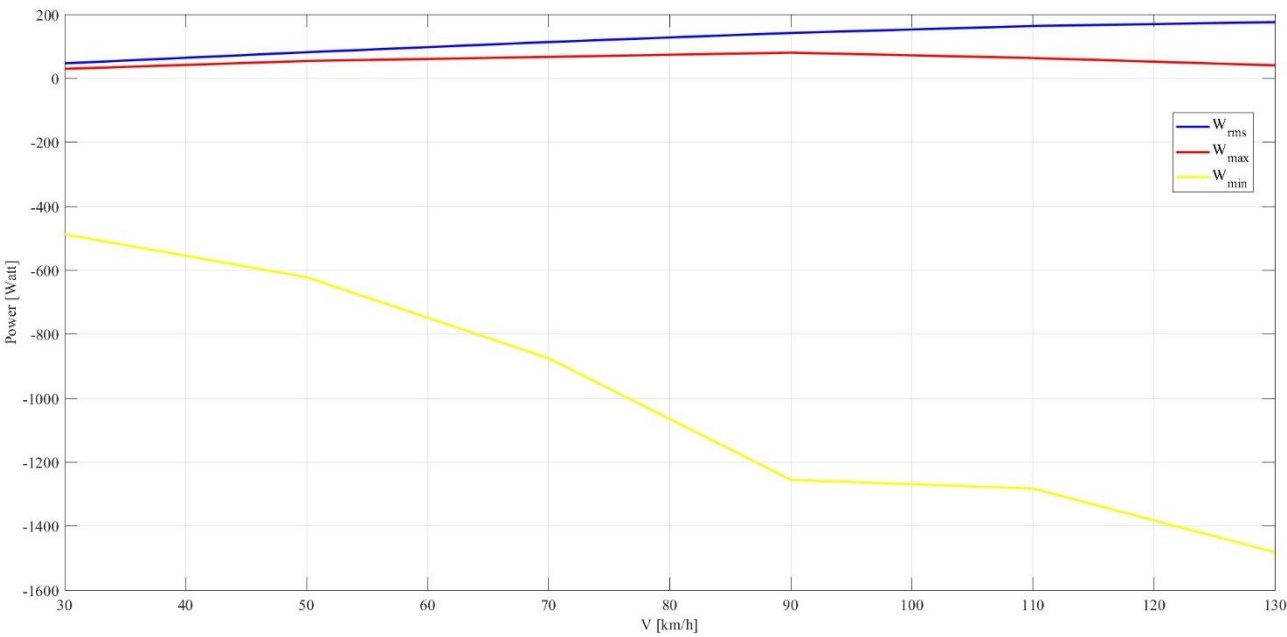


Active system:

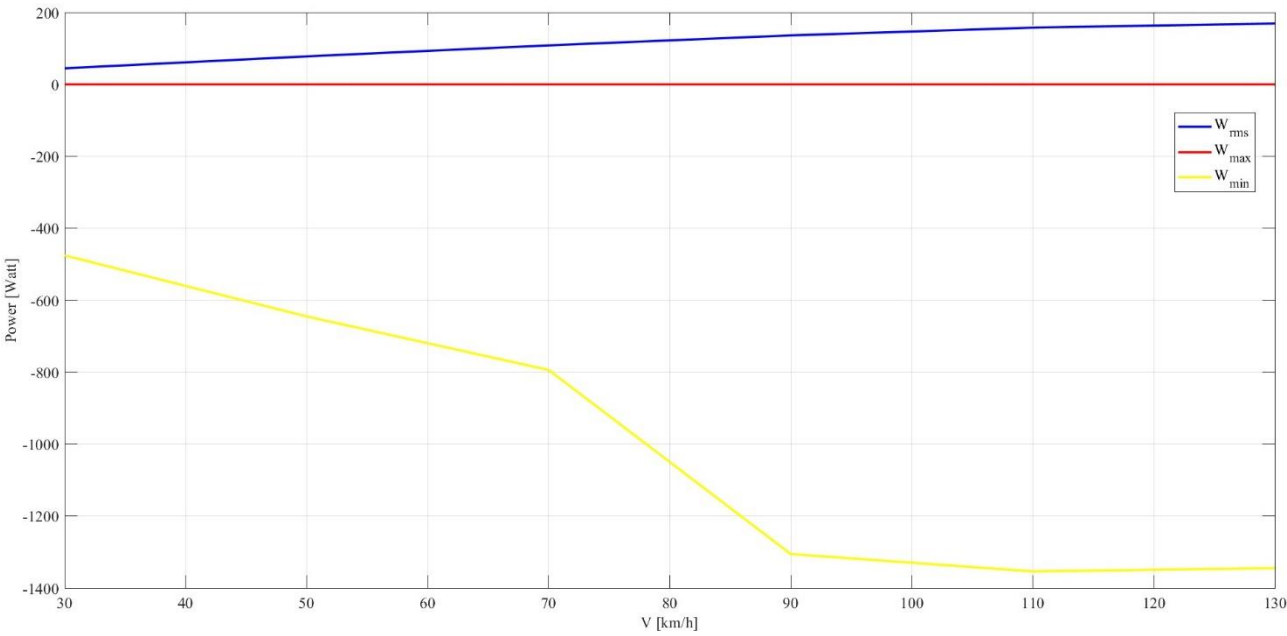


The power dissipated or introduced by the passive suspension and actuator in the **front**:

Semi-active

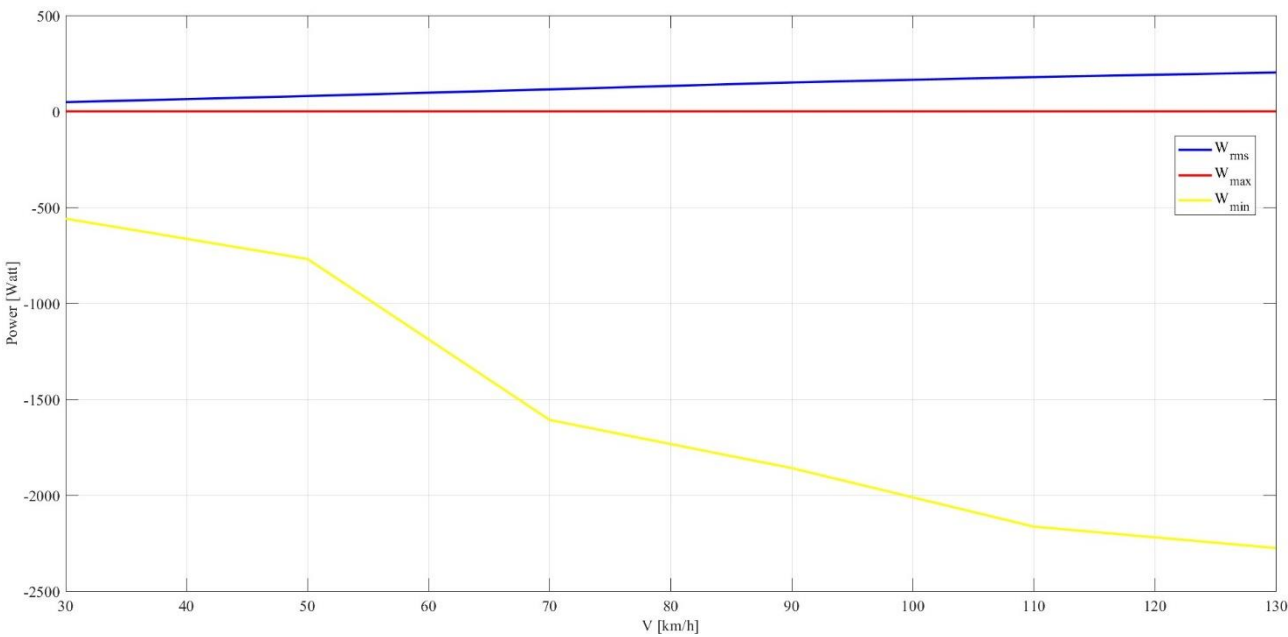


Active

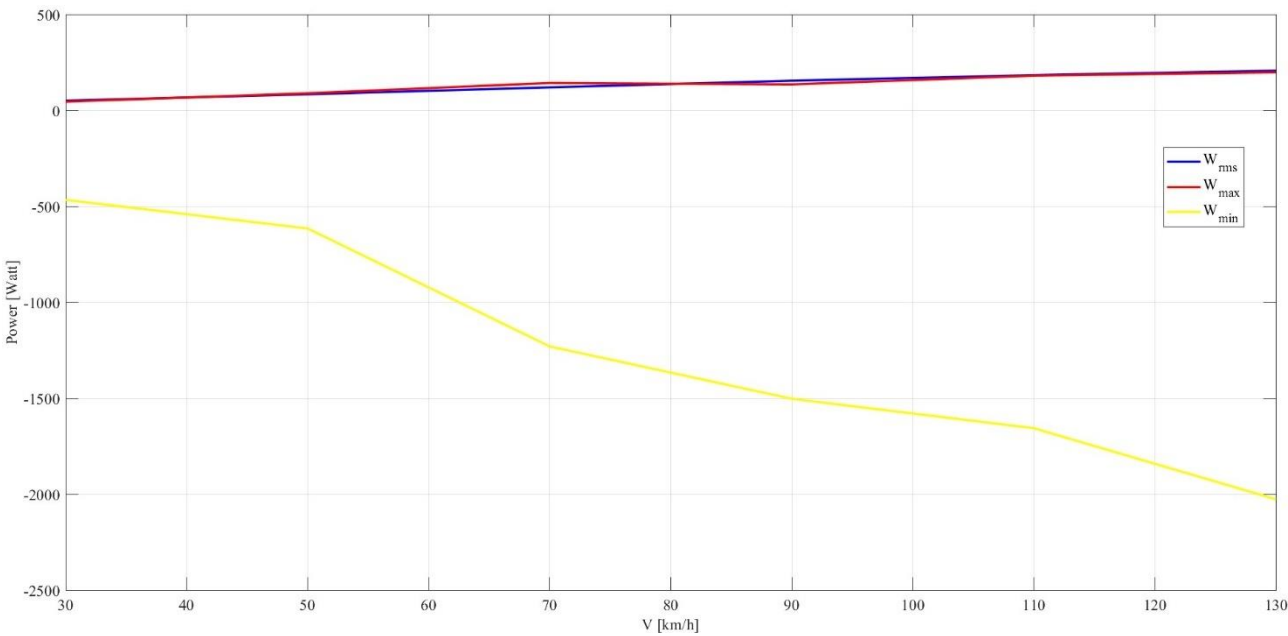


The power dissipated or introduced by the passive suspension and actuator in the **rear**:

Semi-active

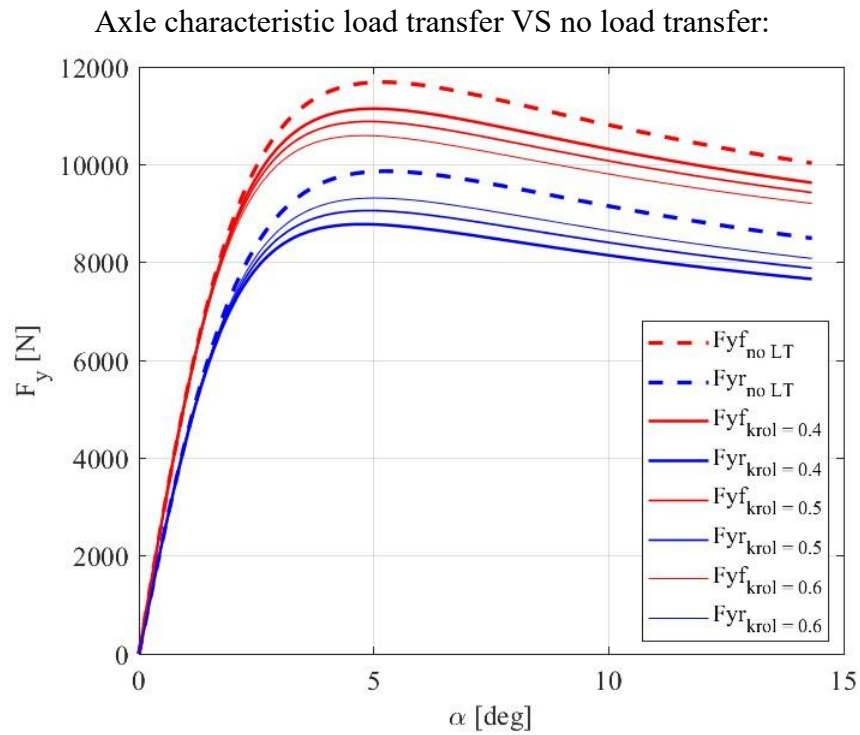


Active:

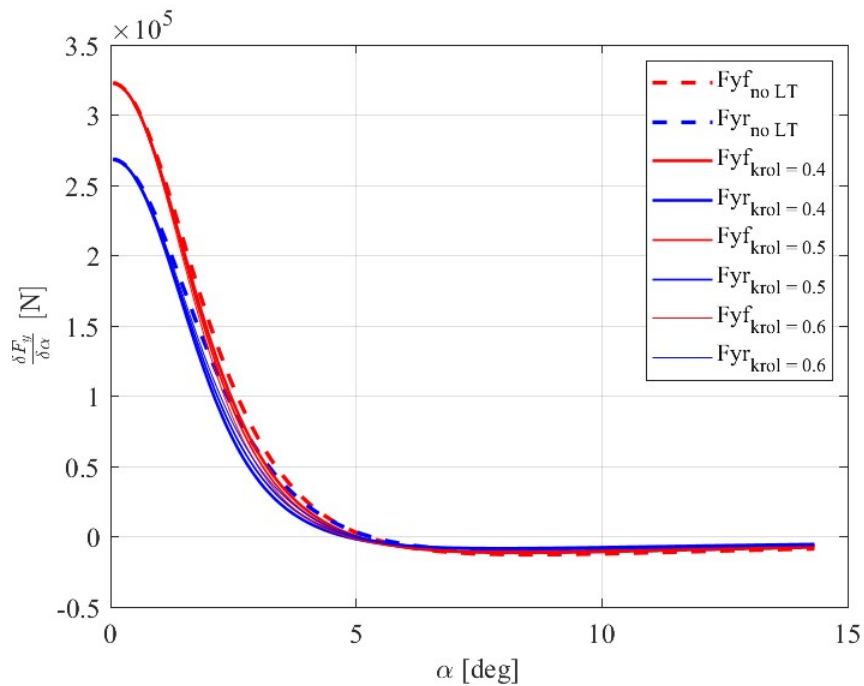


Vehicle Dynamics and Control: Lateral Dynamics Lab

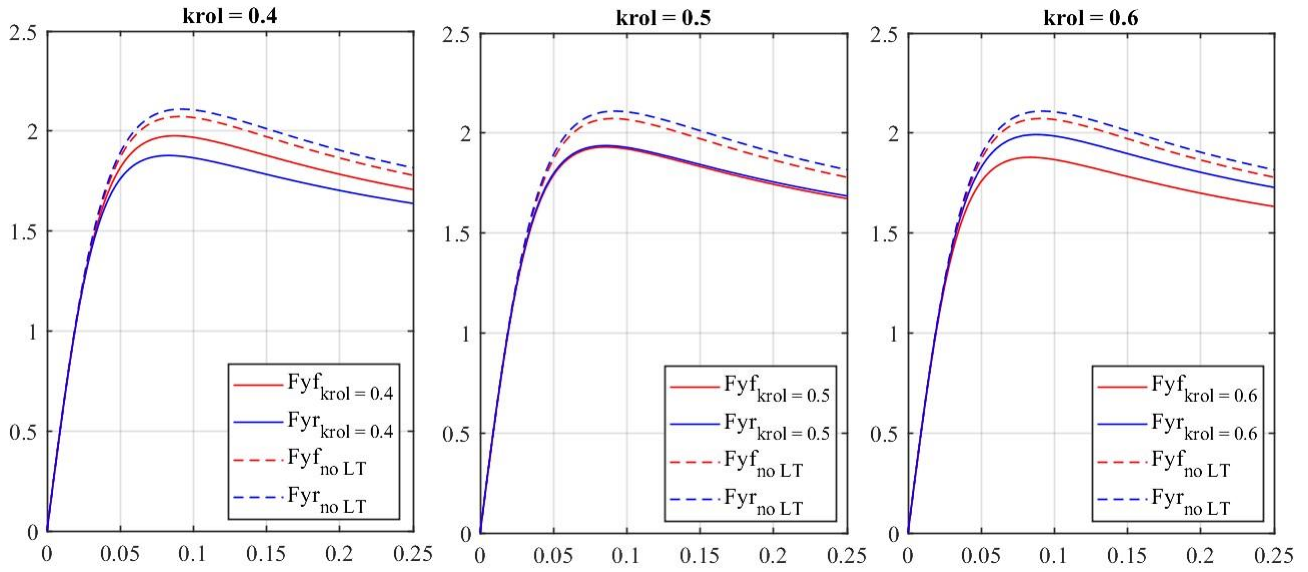
Car segment: **Race car**



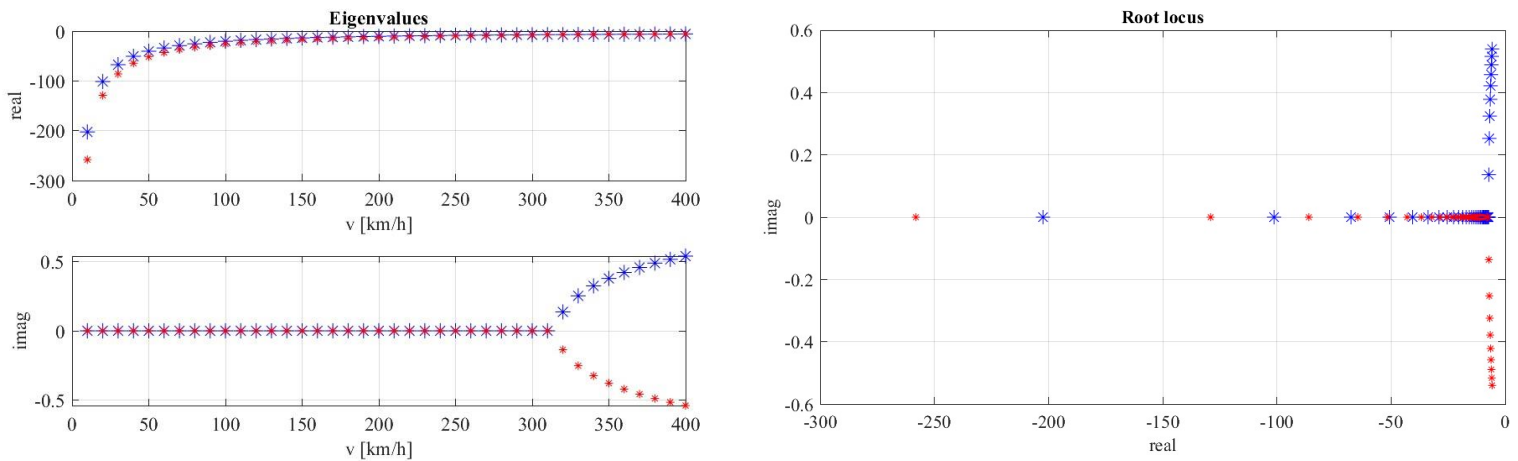
Derivative of lateral axle characteristic load transfer VS no load transfer:



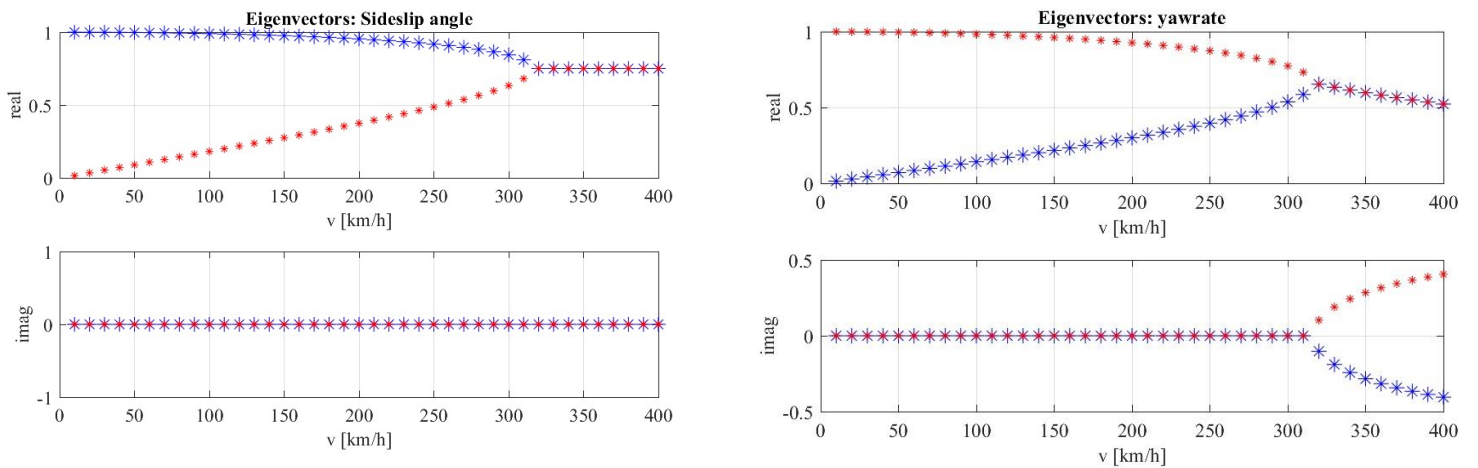
Normalized to vertical load axle characteristic load transfer VS no load transfer:



Eigenvalues and root locus $K_{rol} = 0.5$



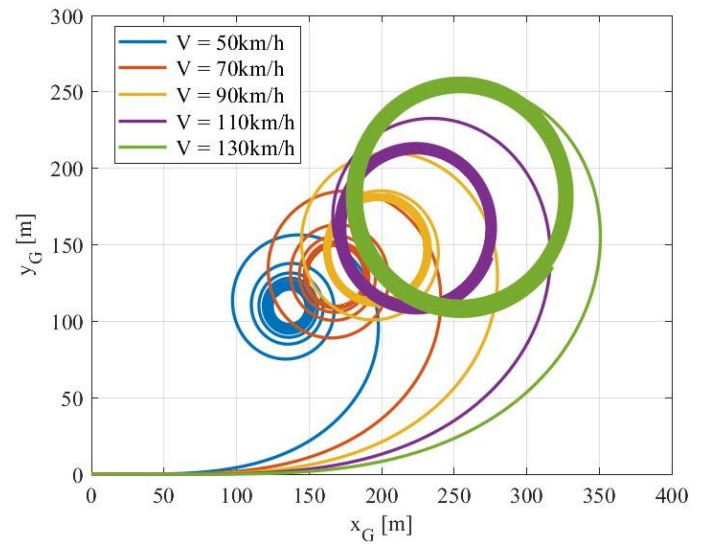
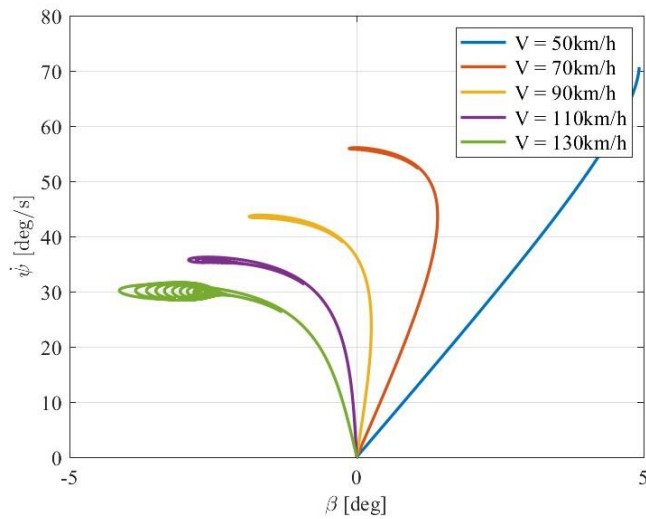
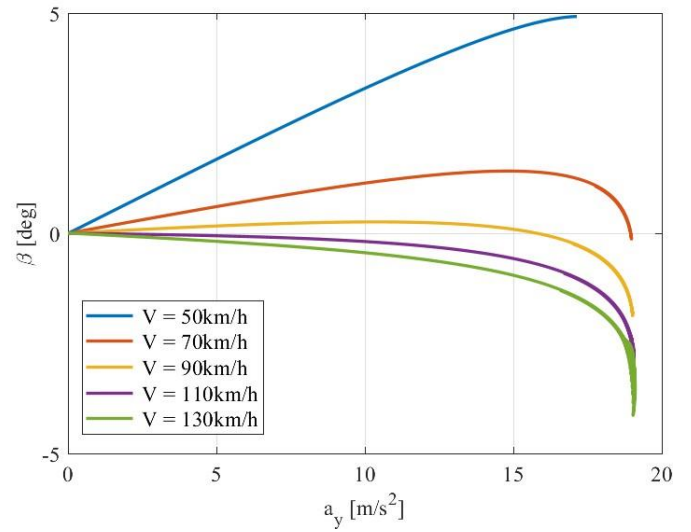
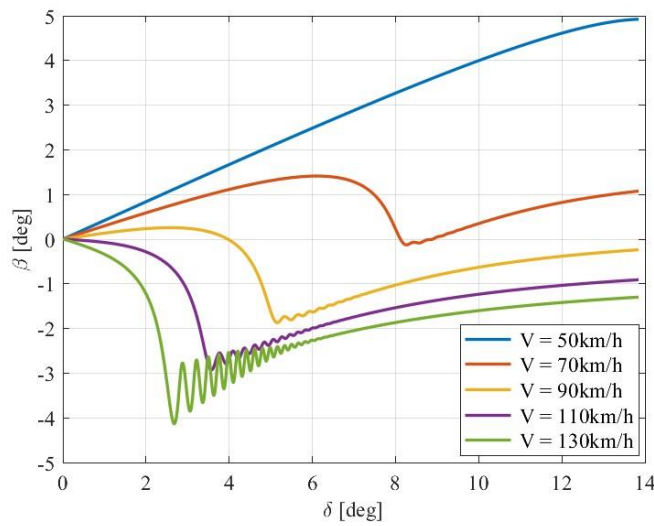
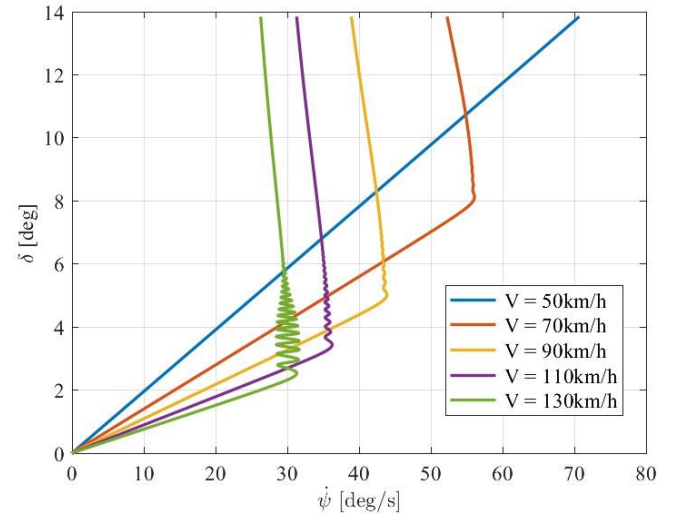
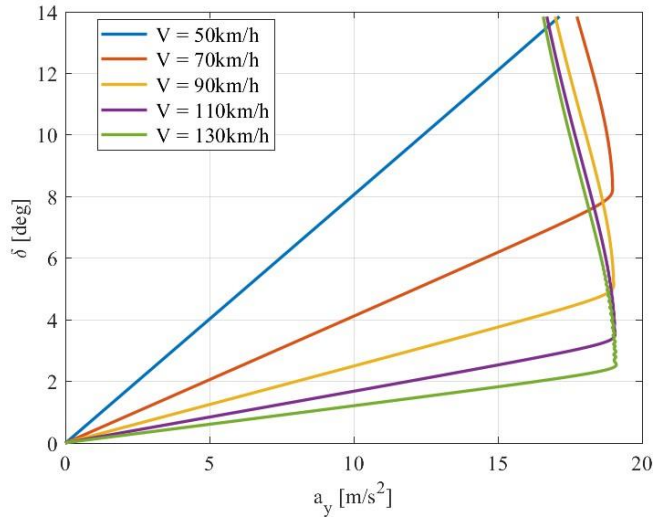
Eigenvectors:



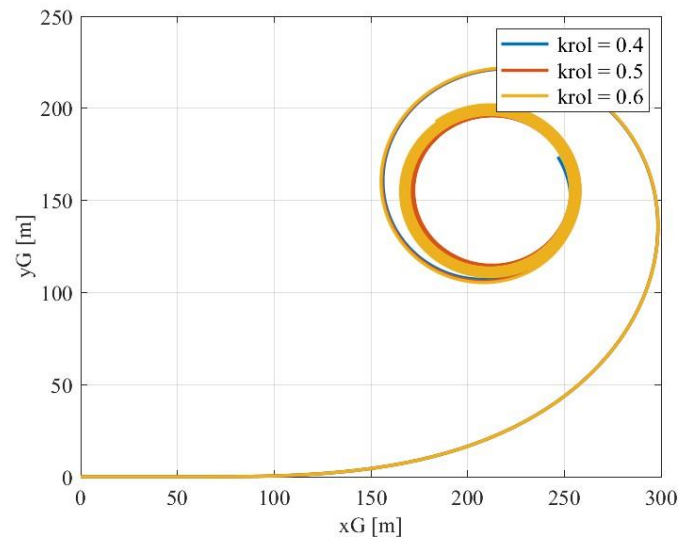
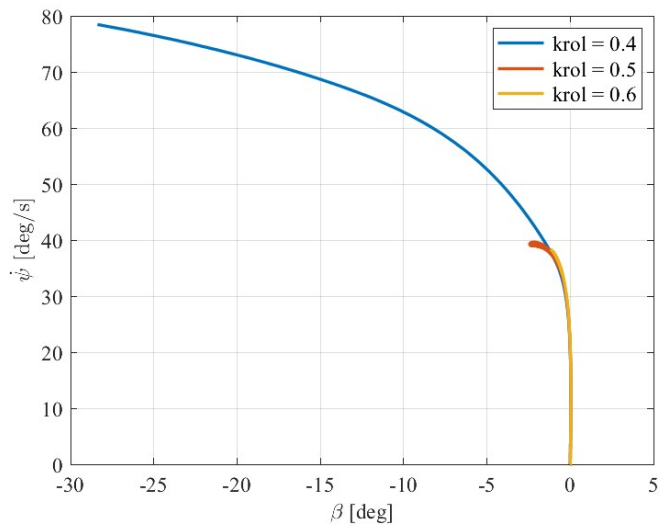
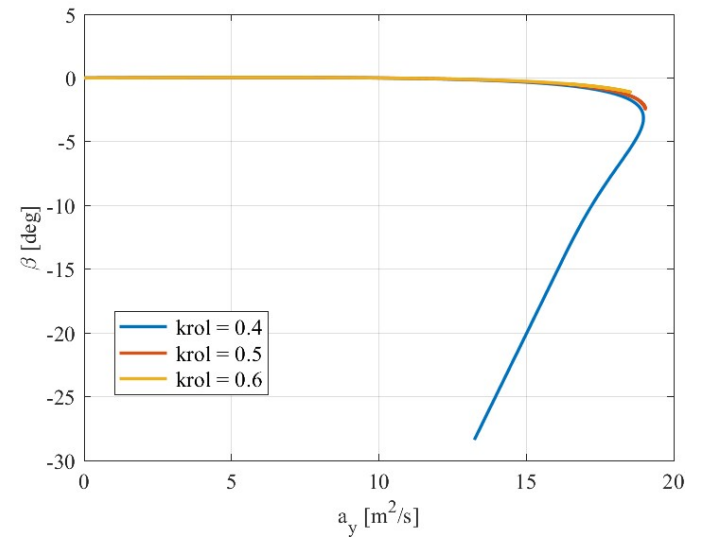
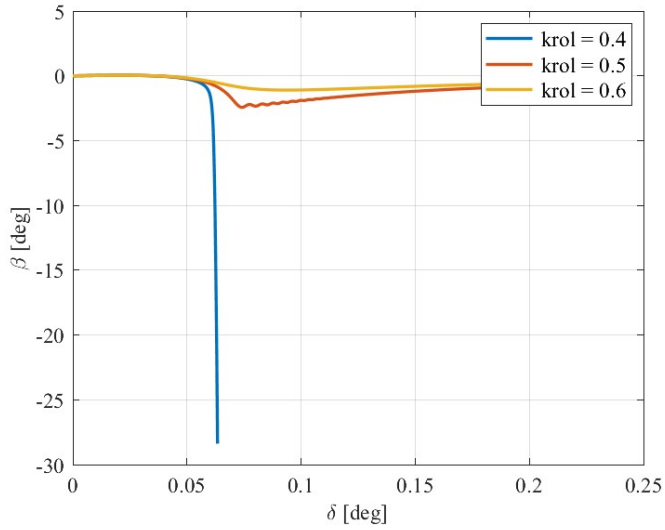
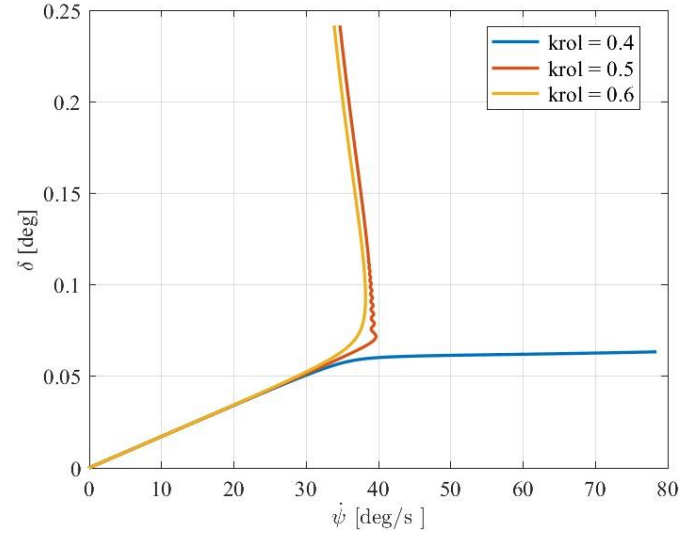
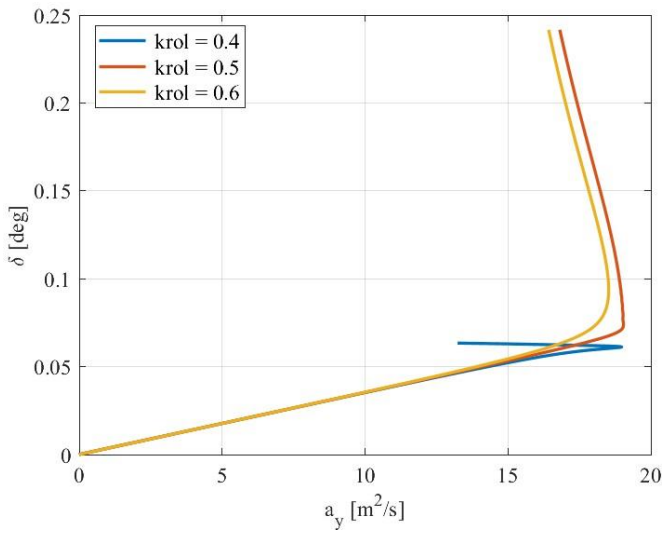
Steady-state maneuvers

Constant speed steering pad:

$K_{rol} = 0.5$ and $V = [50; 70; 90; 110; 130]$ km/h

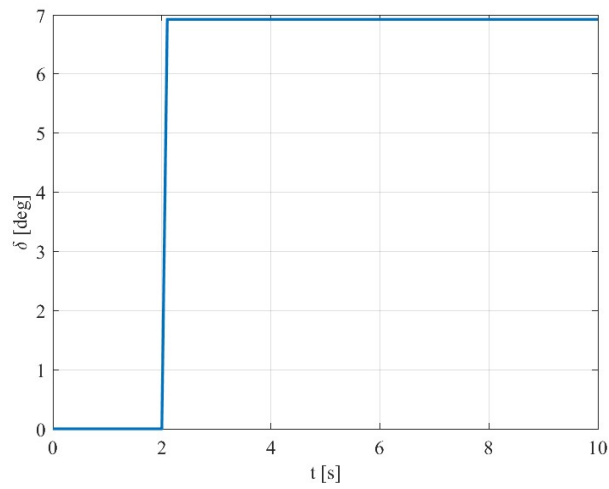


Krol = [0.4;0.5;0.6] and V = 100 km/h



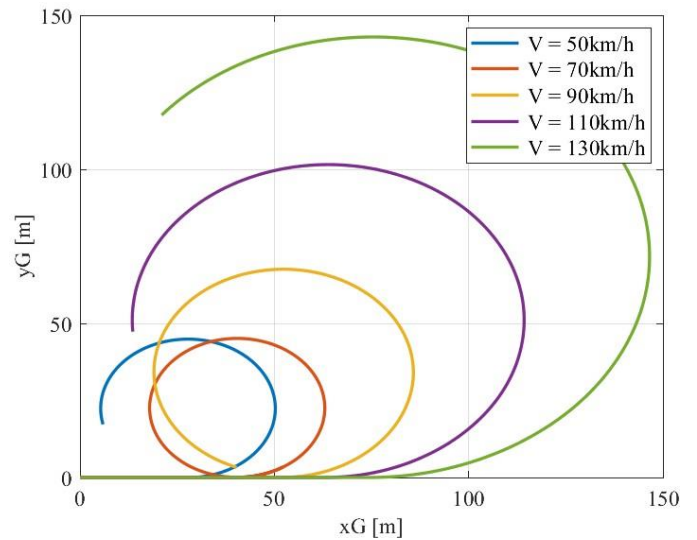
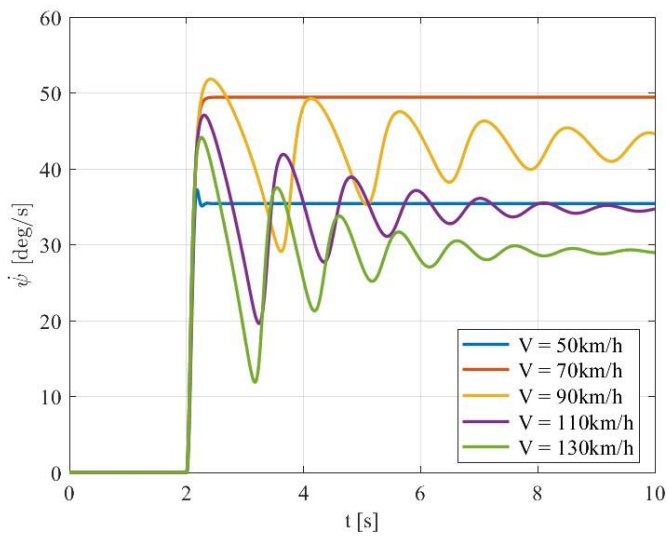
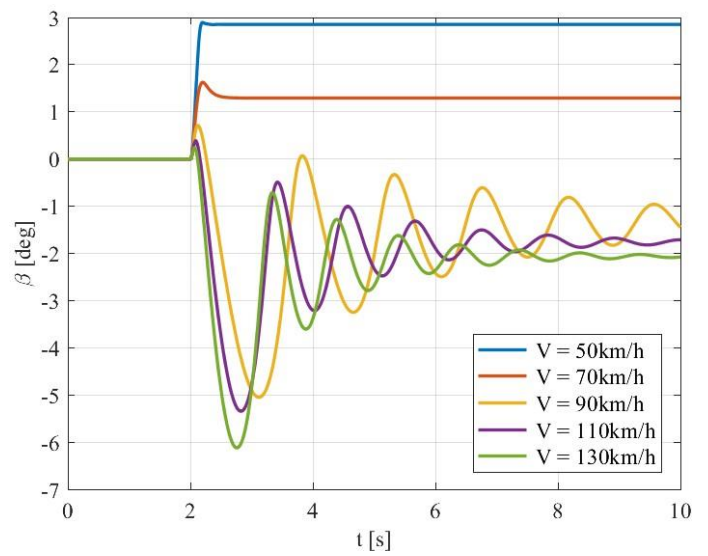
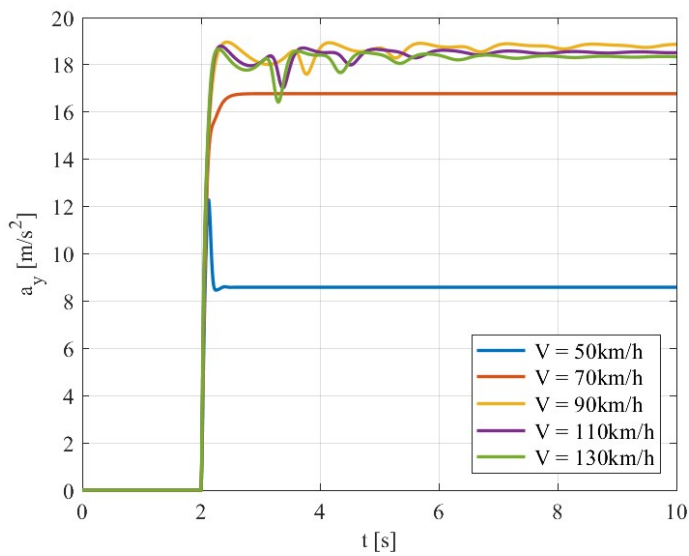
Transient maneuvers:

Step Steer ($\delta = 0^\circ - 90^\circ$):

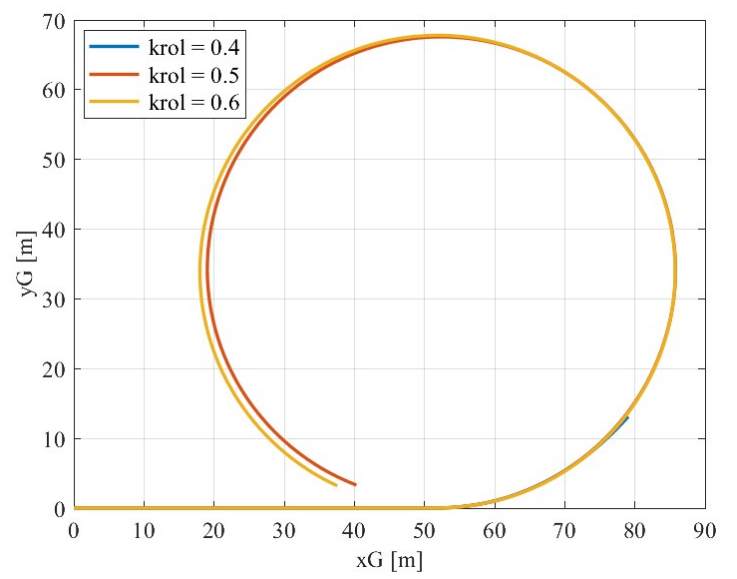
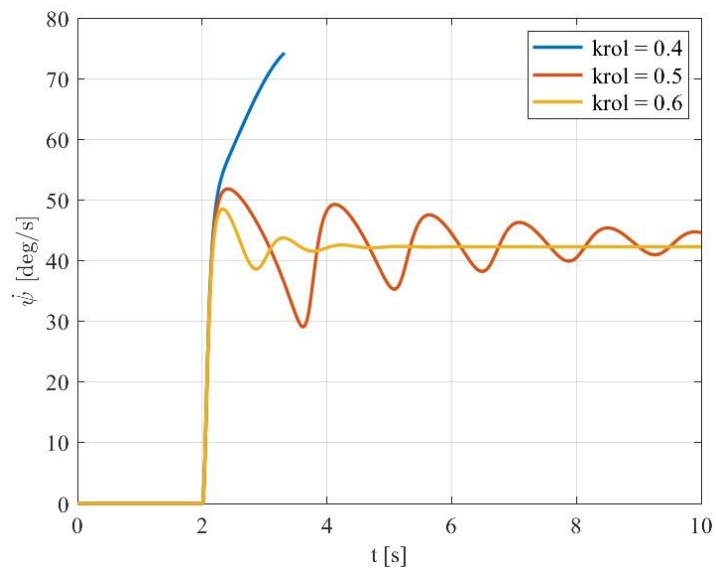
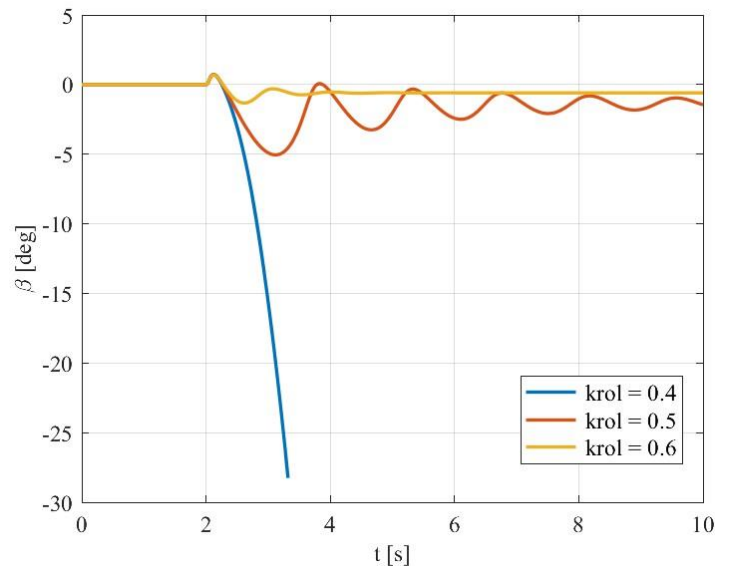
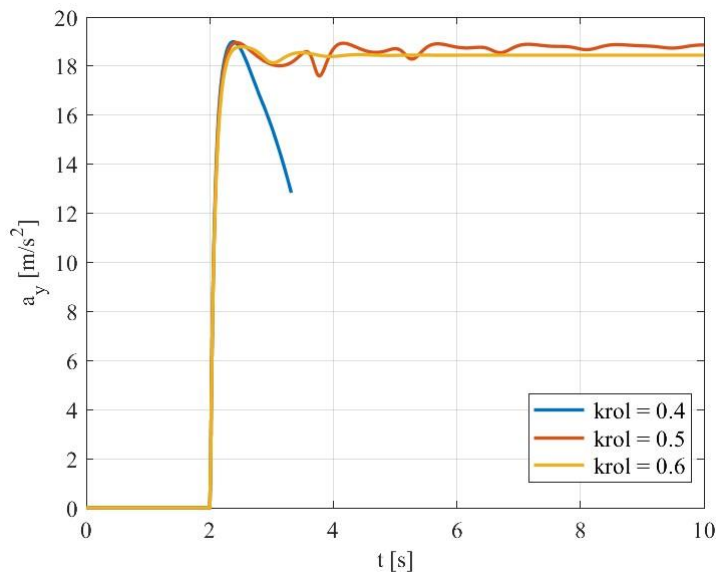


Steering angle on the tires

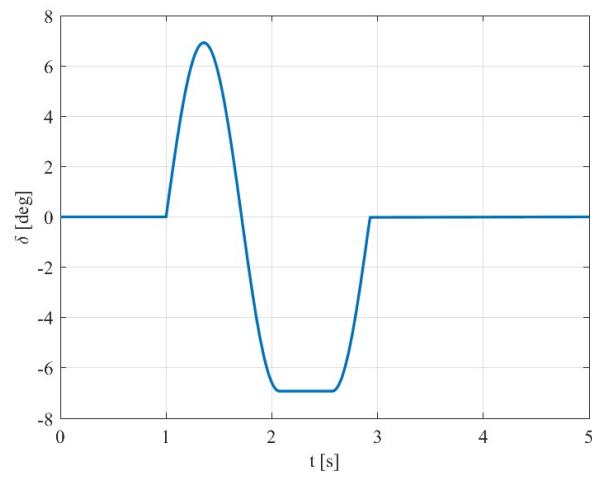
$K_{rol} = 0.5$ and $V = [50; 70; 90; 110; 130]$ km/h



$K_{rol} = [0.4; 0.5; 0.6]$ and $V = 90 \text{ km/h}$

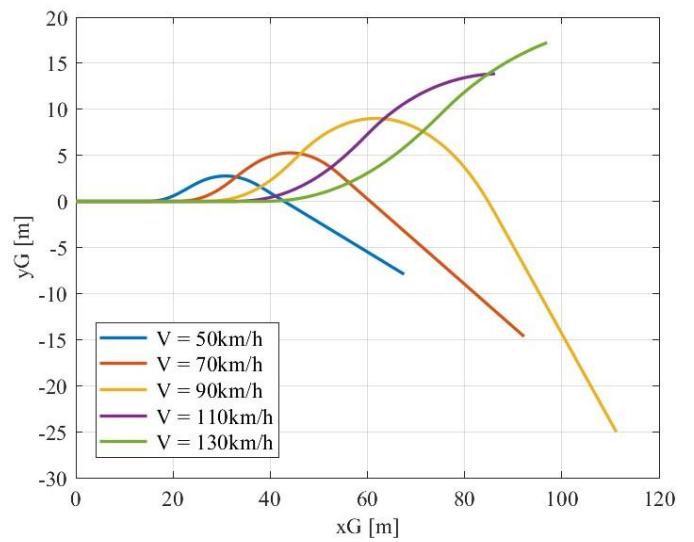
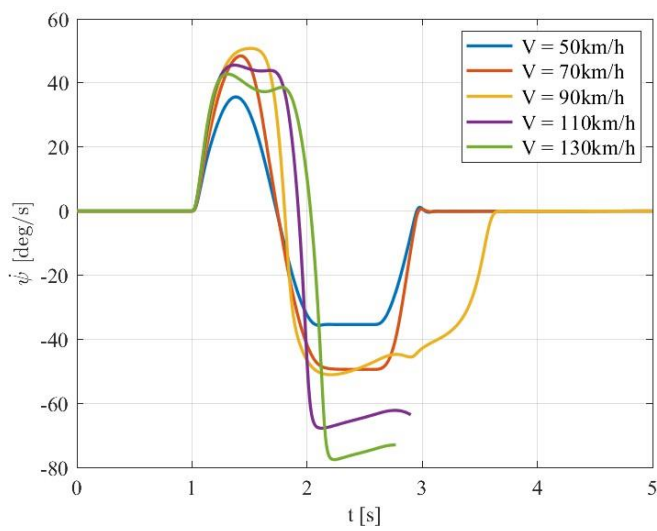
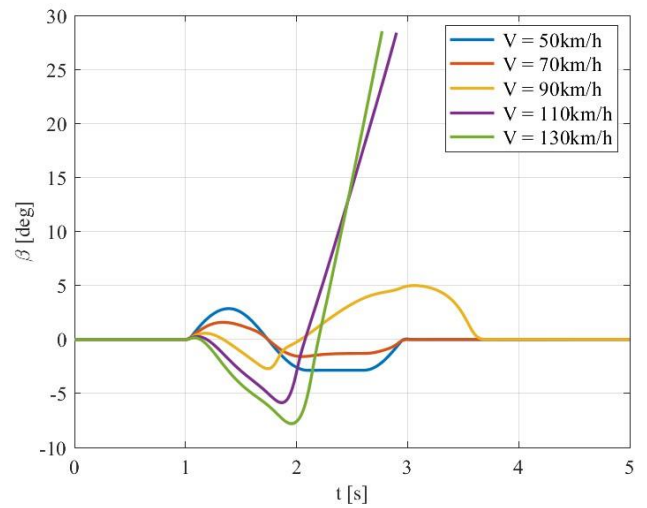
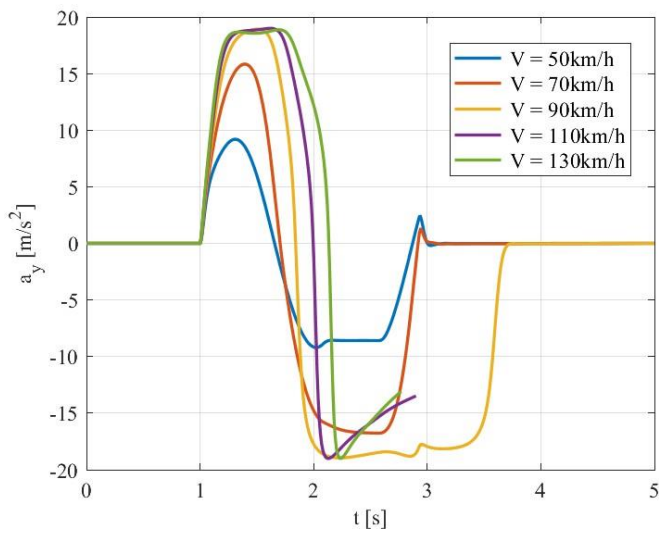


Sine dwell:

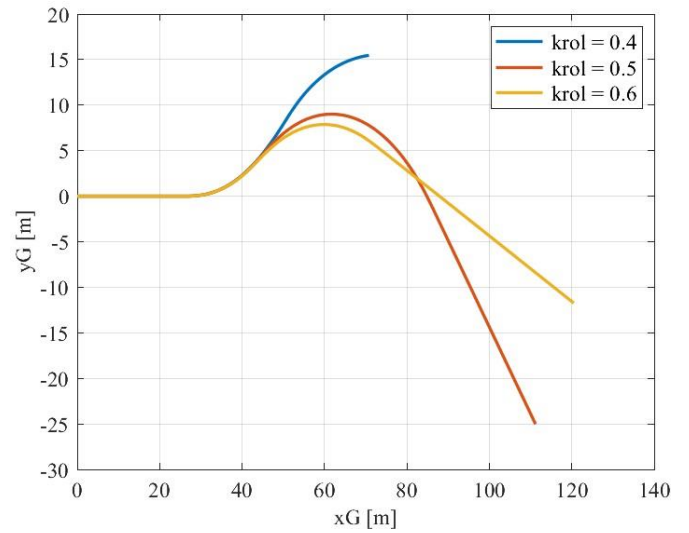
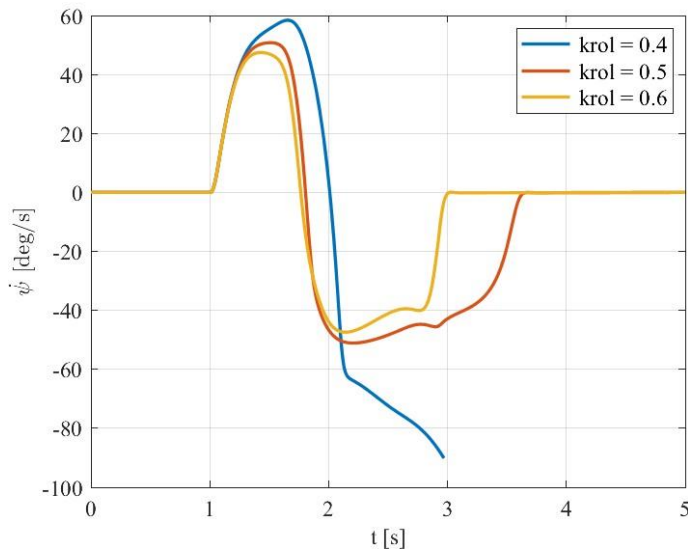
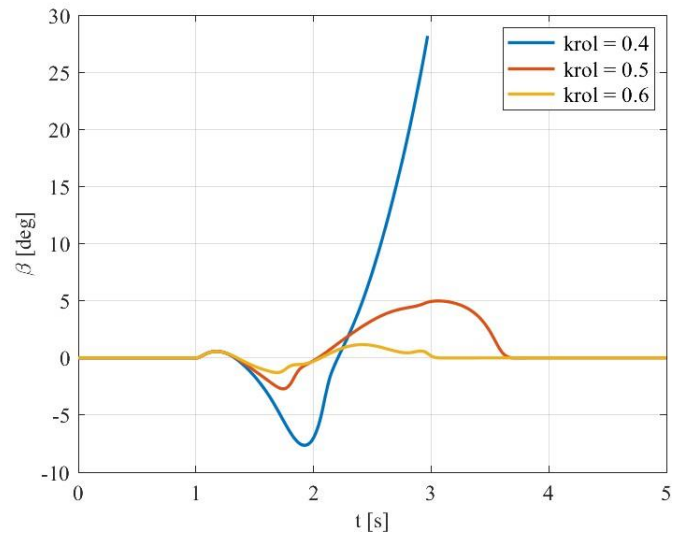
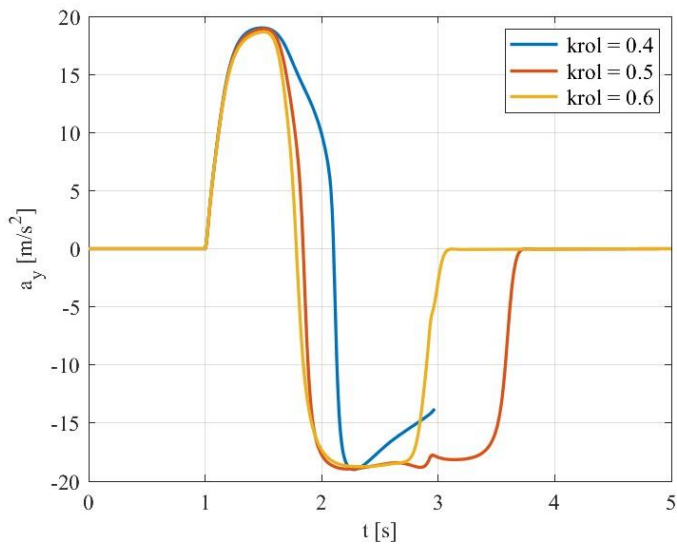


Steering angle on the tires

Krol = 0.5 and $V = [50;70;90;110;130]$ km/h



Krol = [0.4;0.5;0.6] and V = 90 km/h



Maneuvers stop criteria:

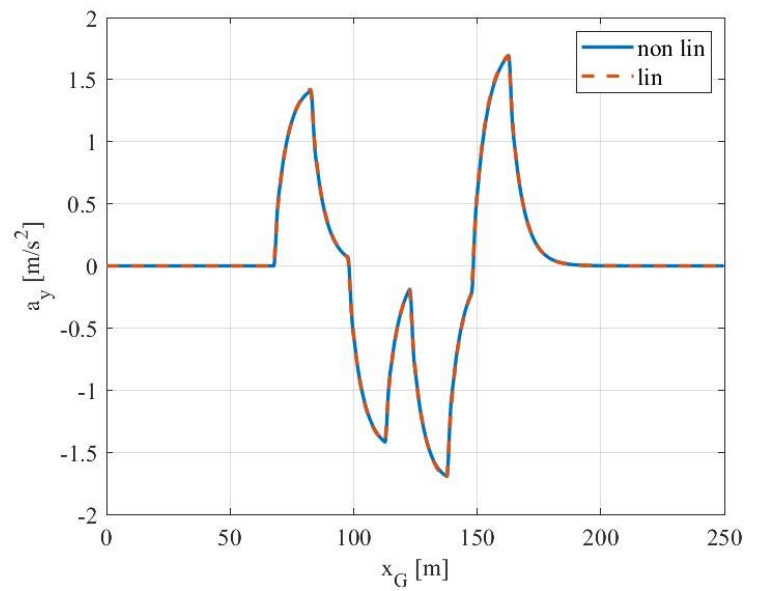
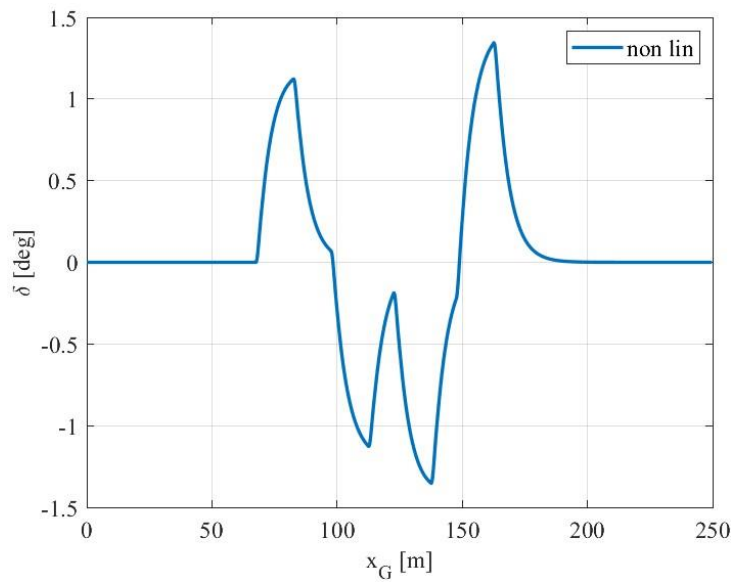
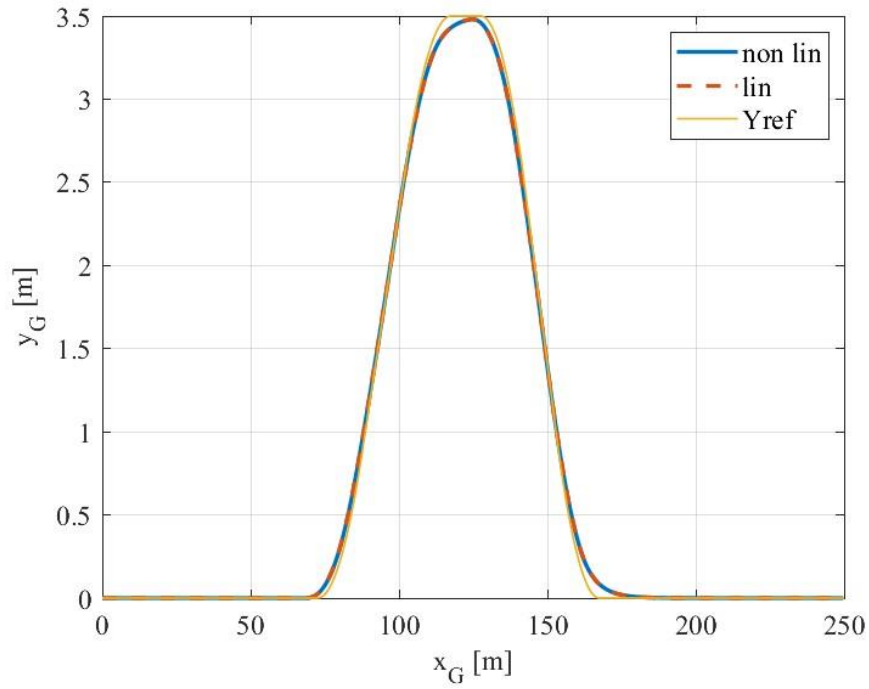
- $a_y > 20 \text{ m/s}^2$
- $|\beta| > 0.5 \text{ rad}$

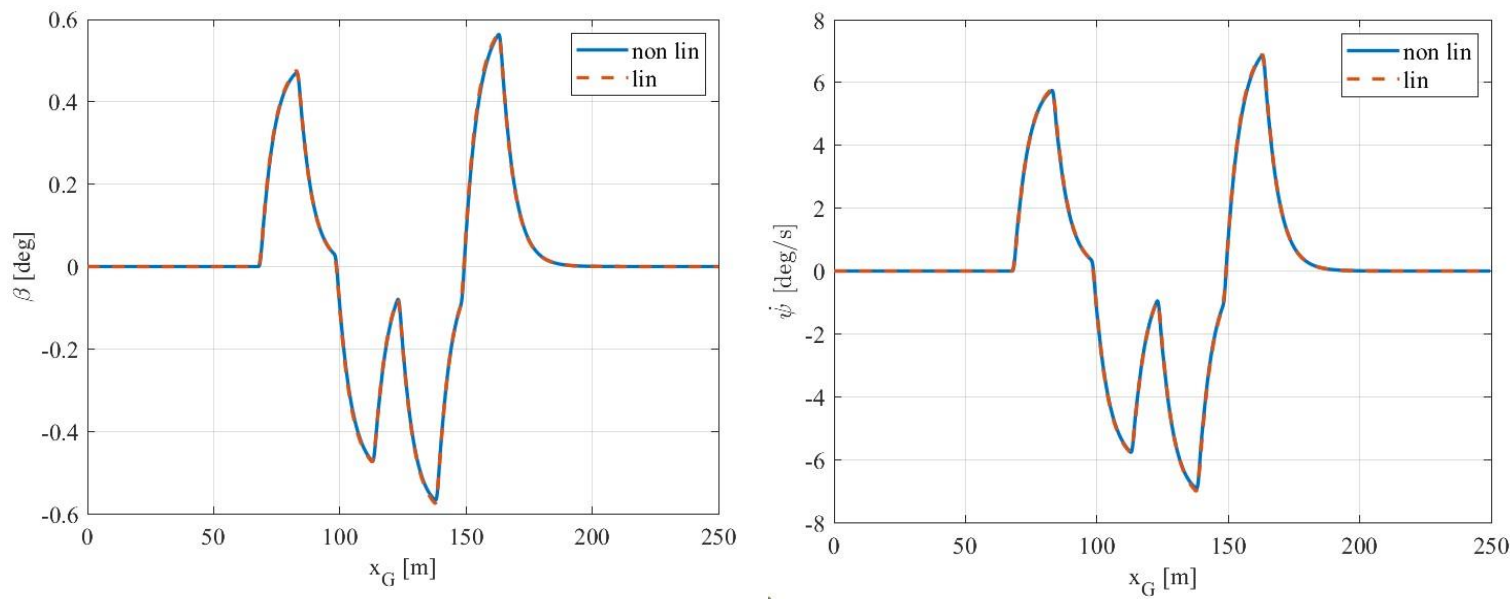
Path follower maneuvers:

Double lane change:

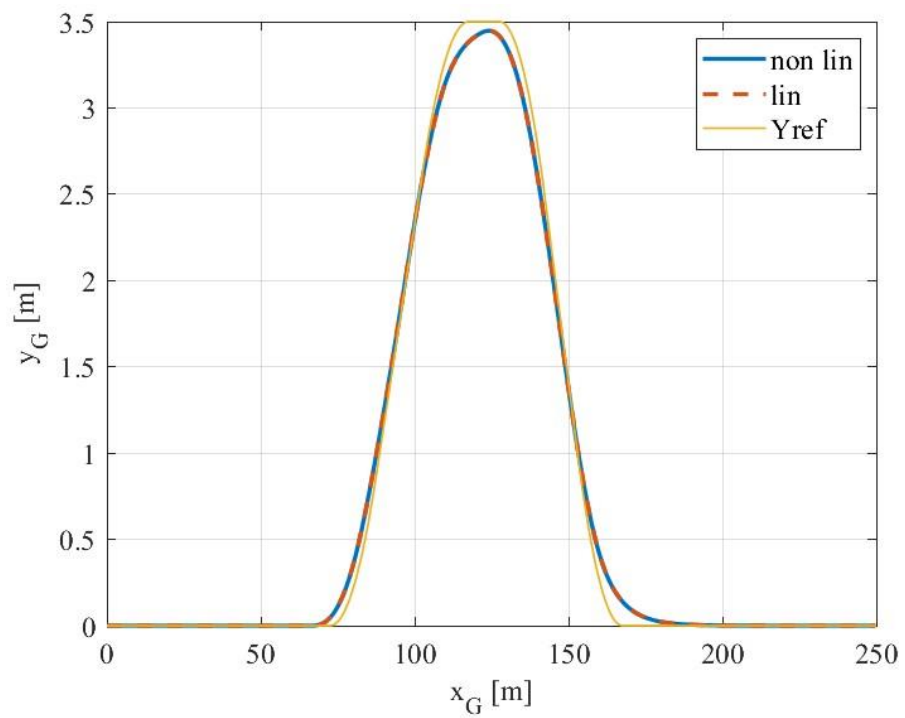
$K_{rol} = 0.5$ and $V = [50; 70; 90; 110]$ km/h

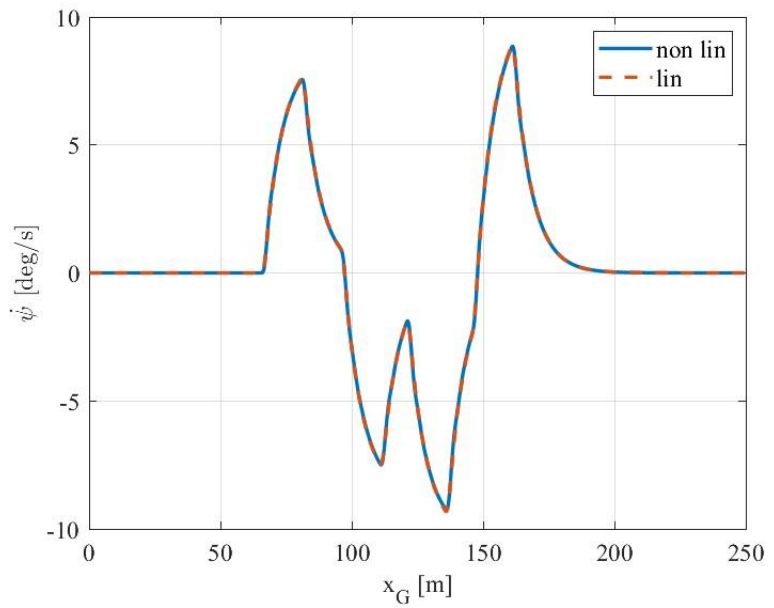
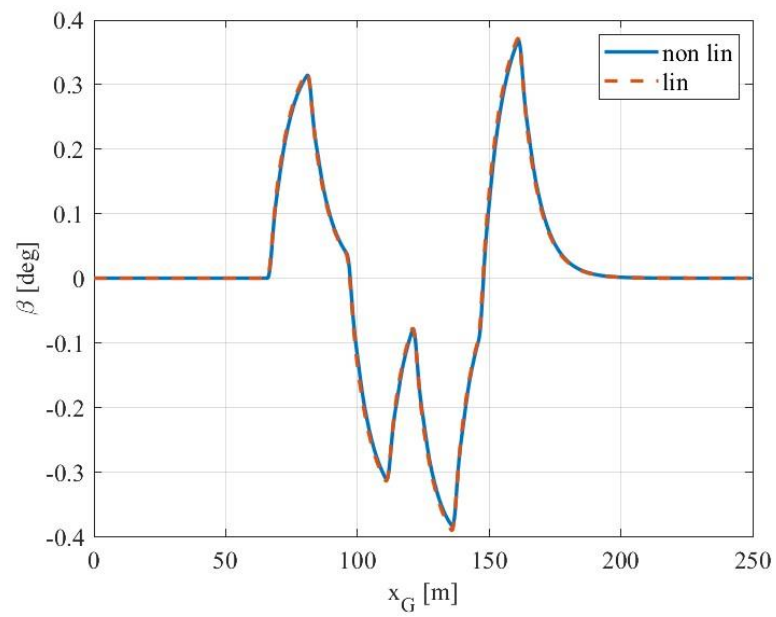
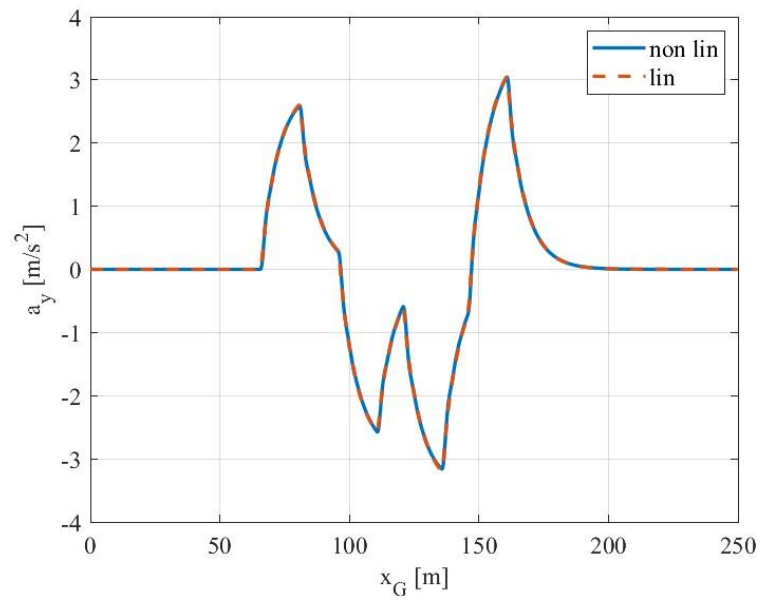
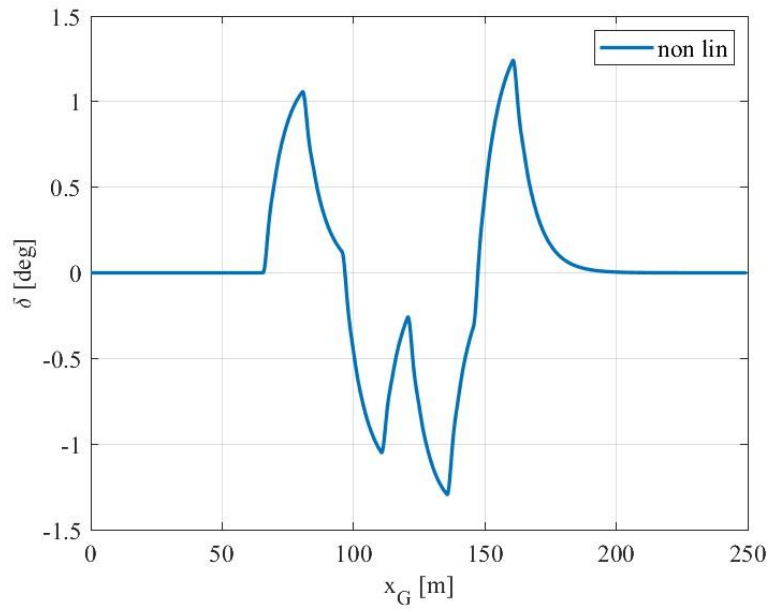
$V = 50$ km/h



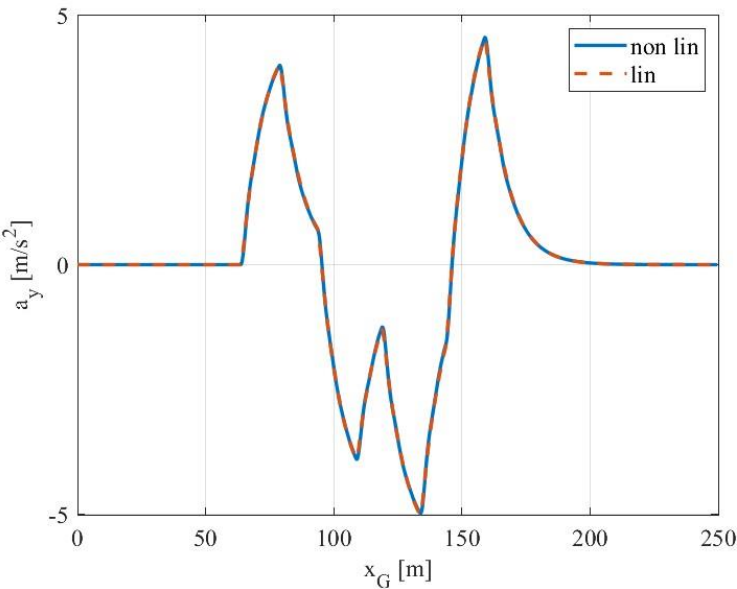
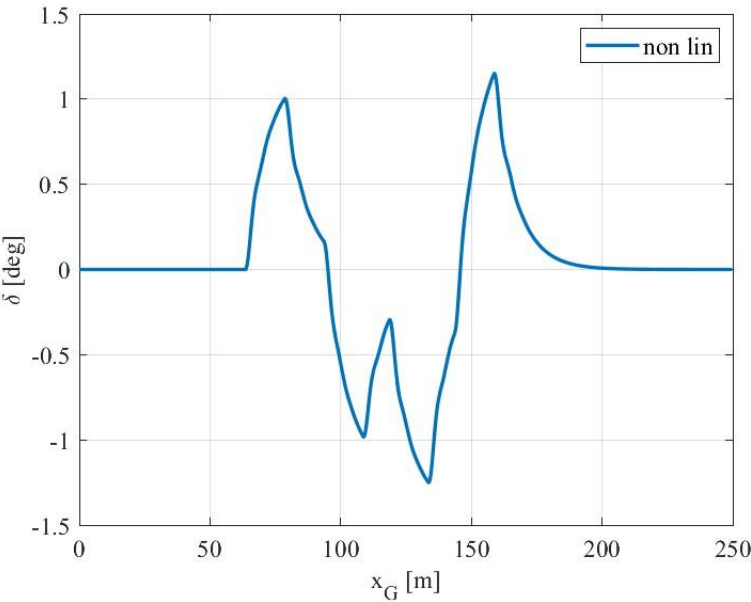
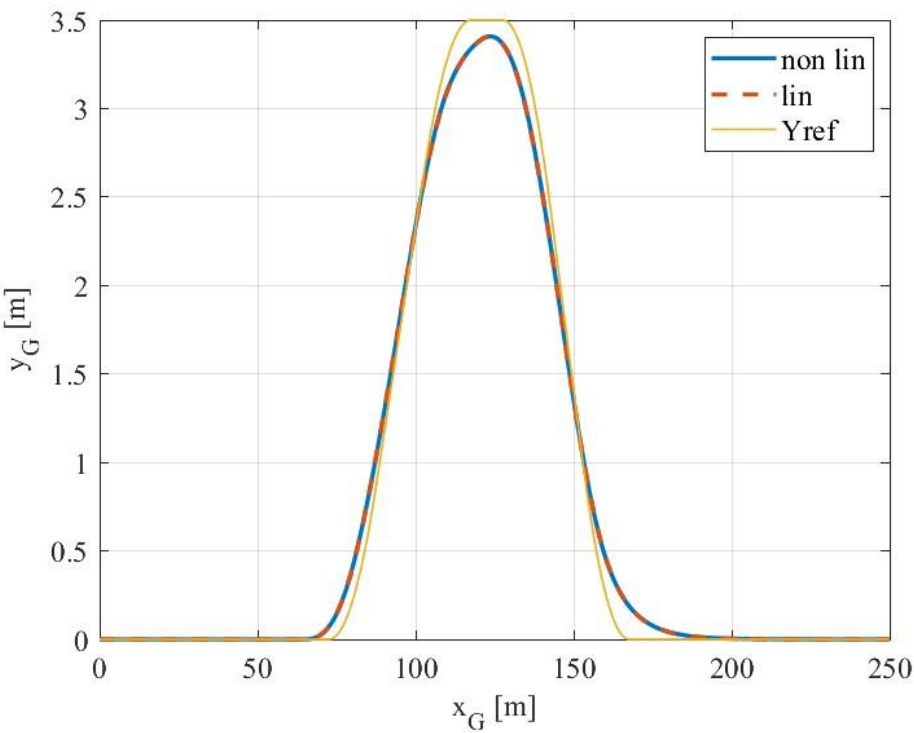


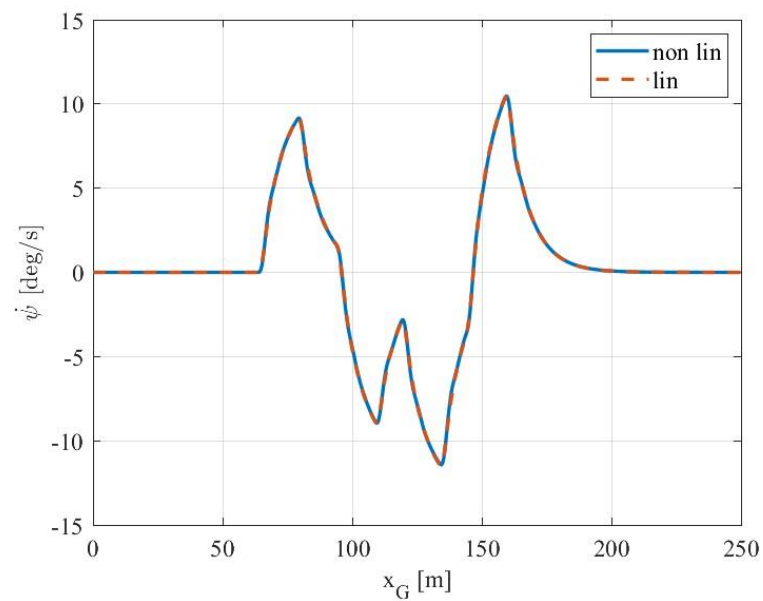
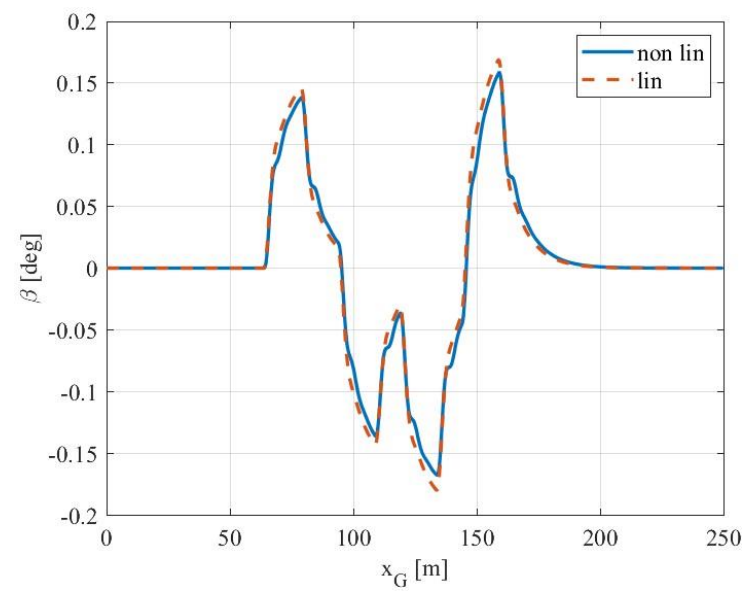
$V = 70 \text{ km/h}$



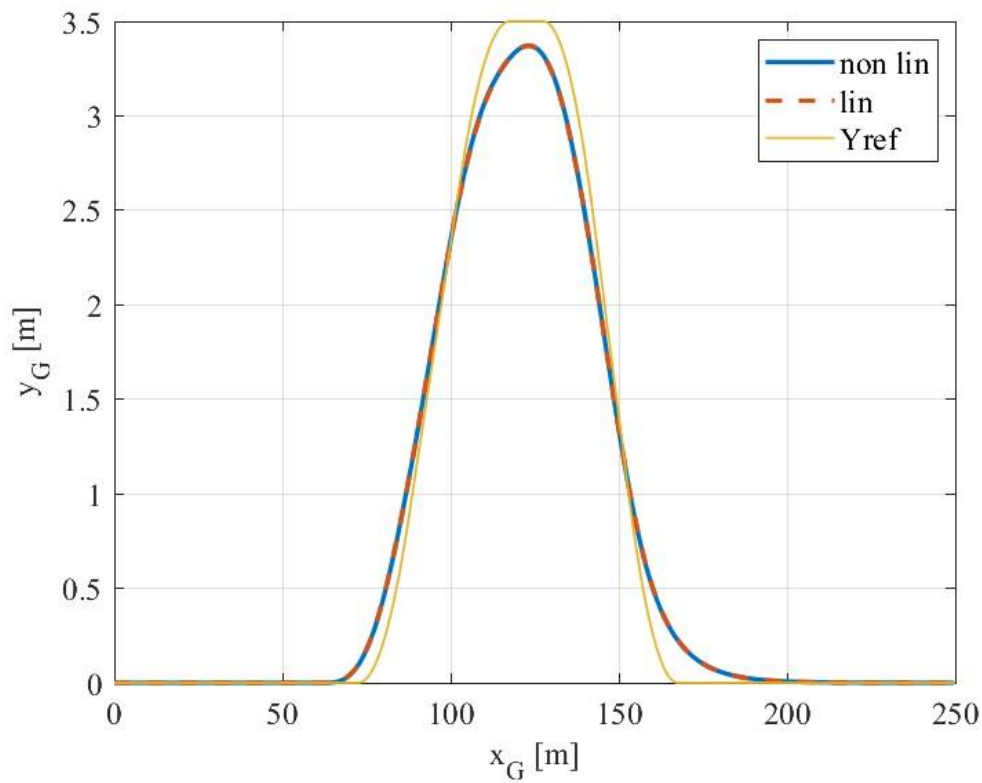


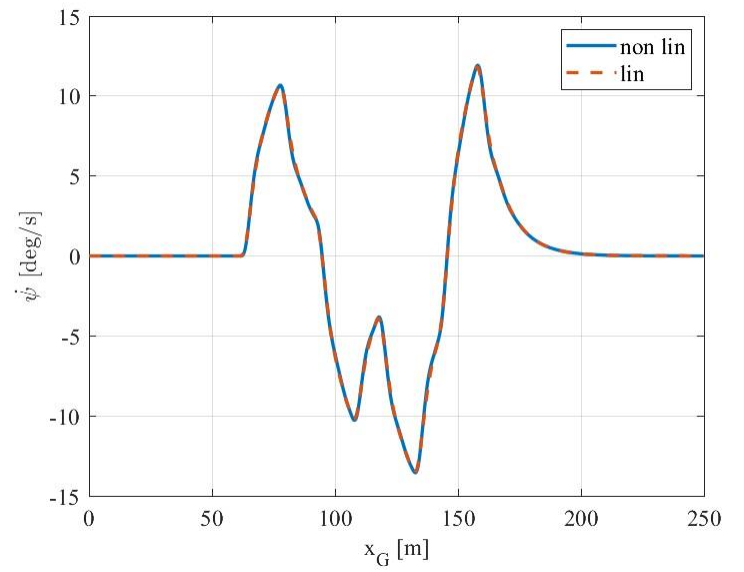
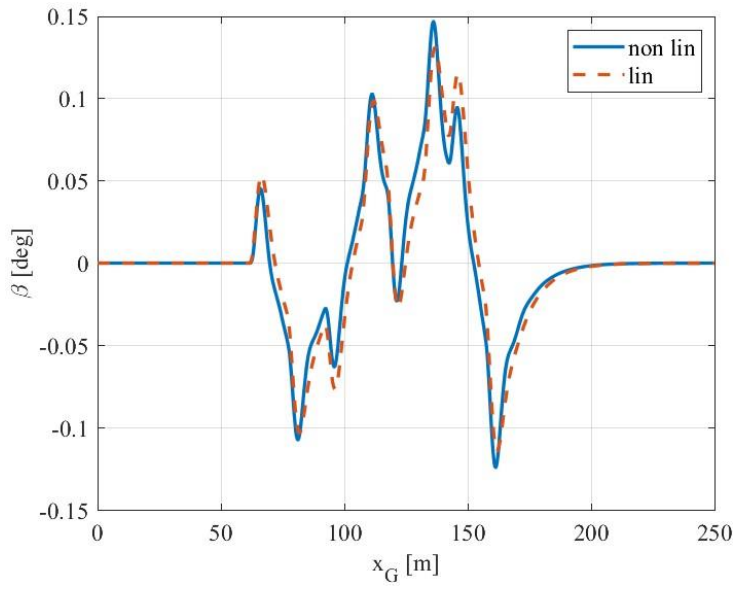
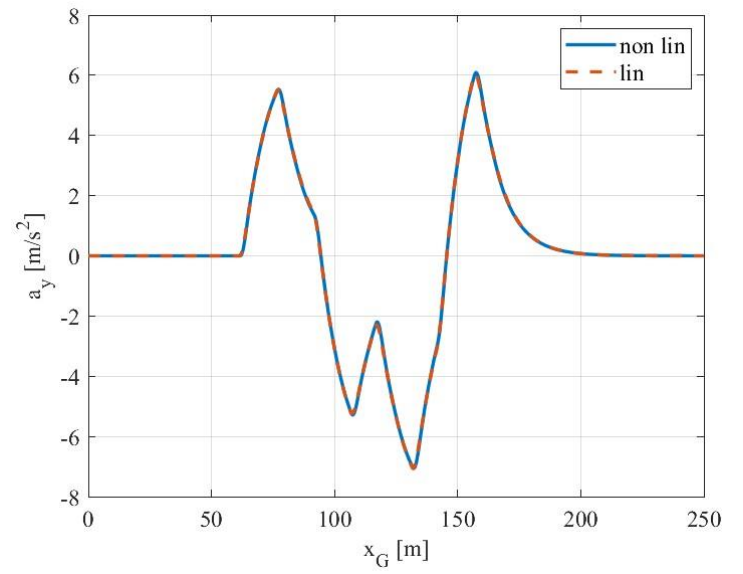
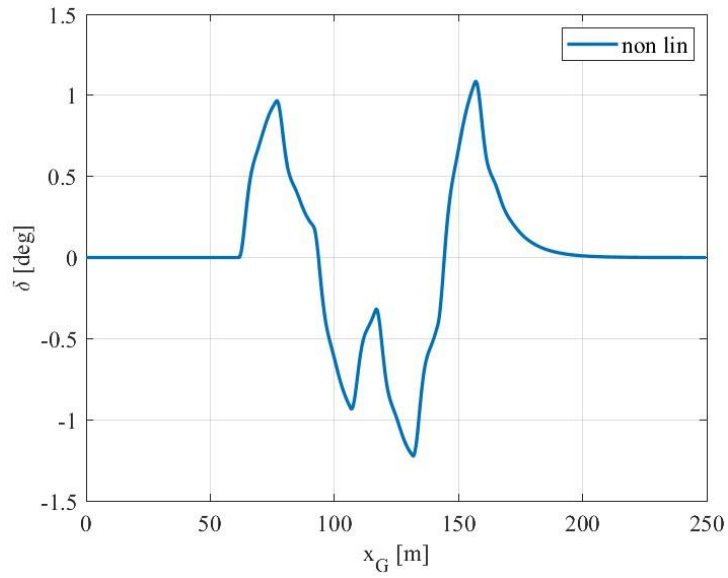
$V = 90 \text{ km/h}$





$V = 110 \text{ km/h}$



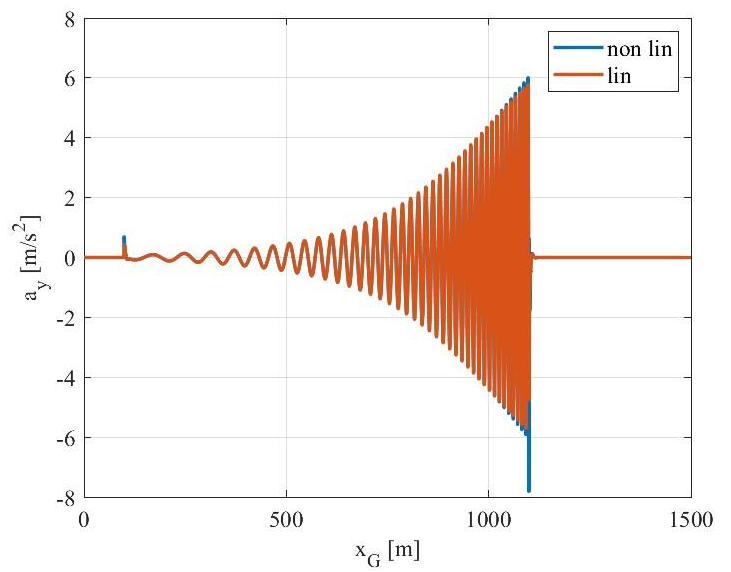
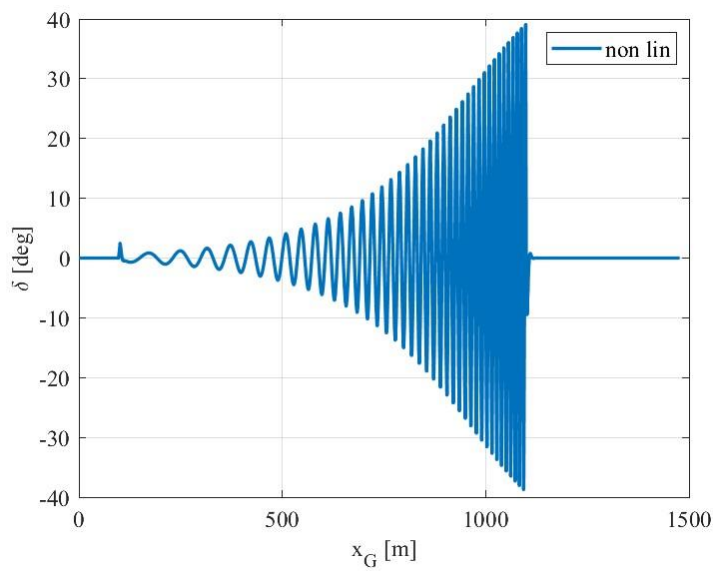
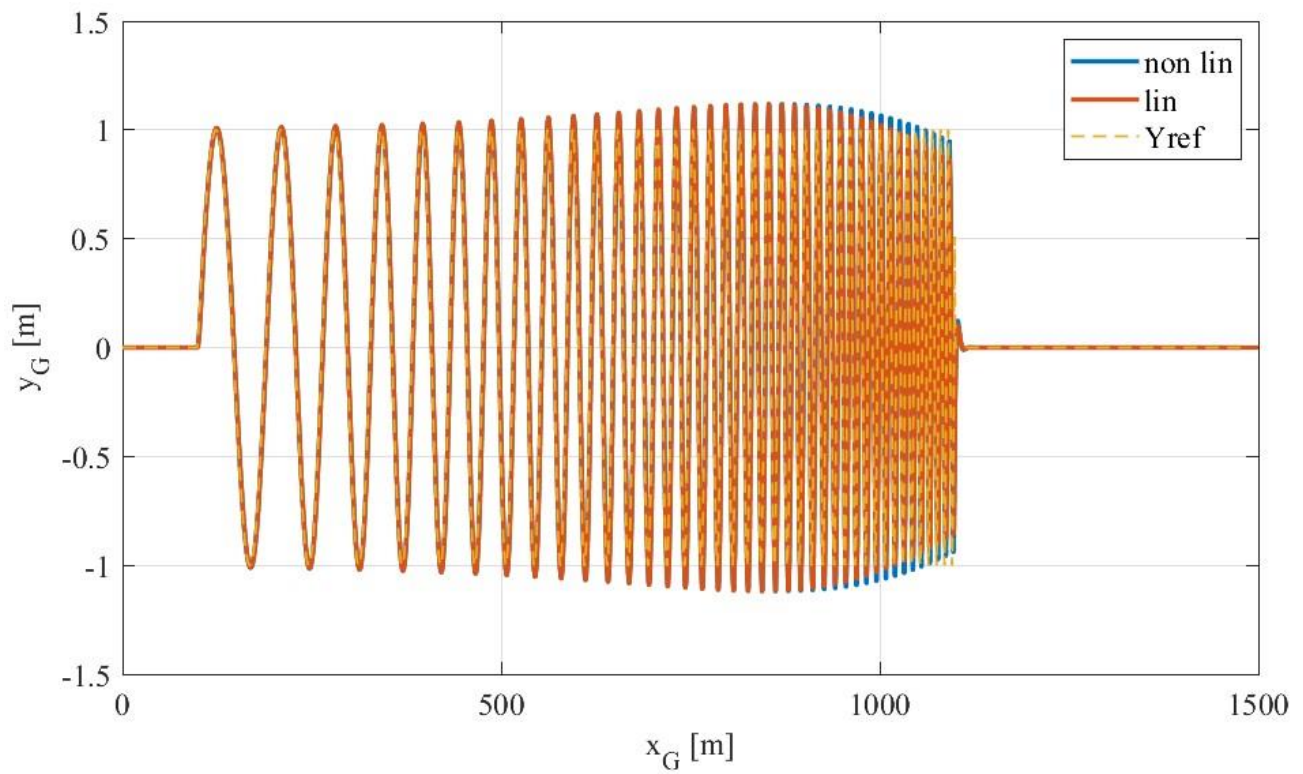


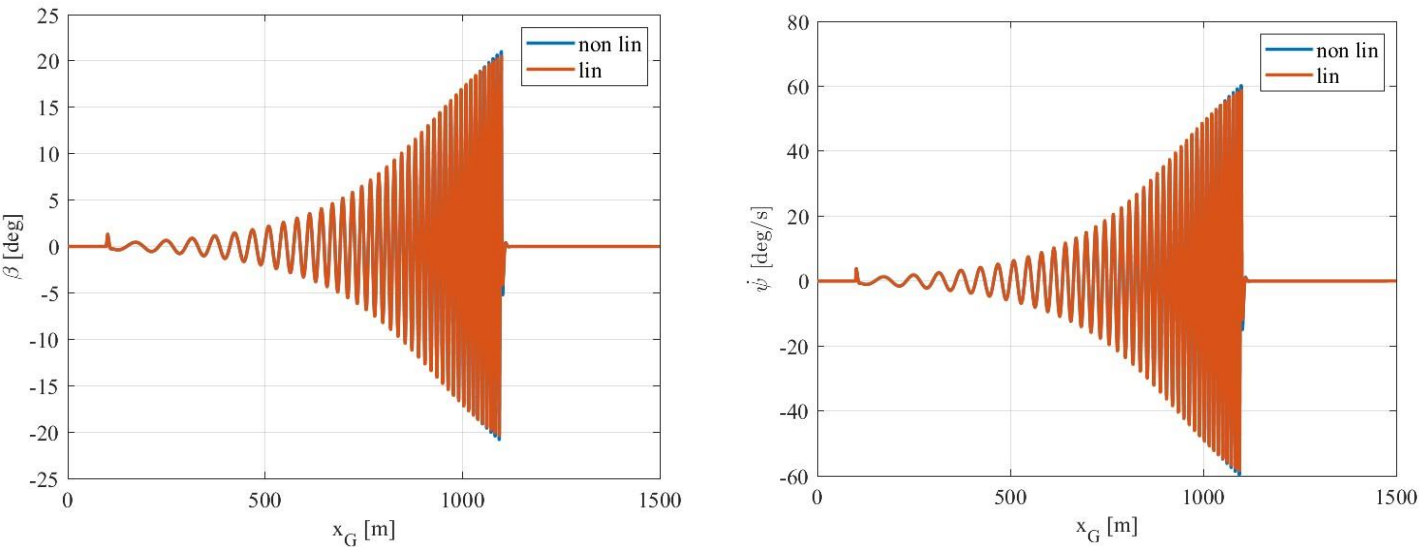
	driver.L [m]	driver.kp
V = 50 km/h	5	0.5
V = 70 km/h	7	0.4
V = 90 km/h	9	0.3
V = 110 km/h	11	0.2

Sine sweep:

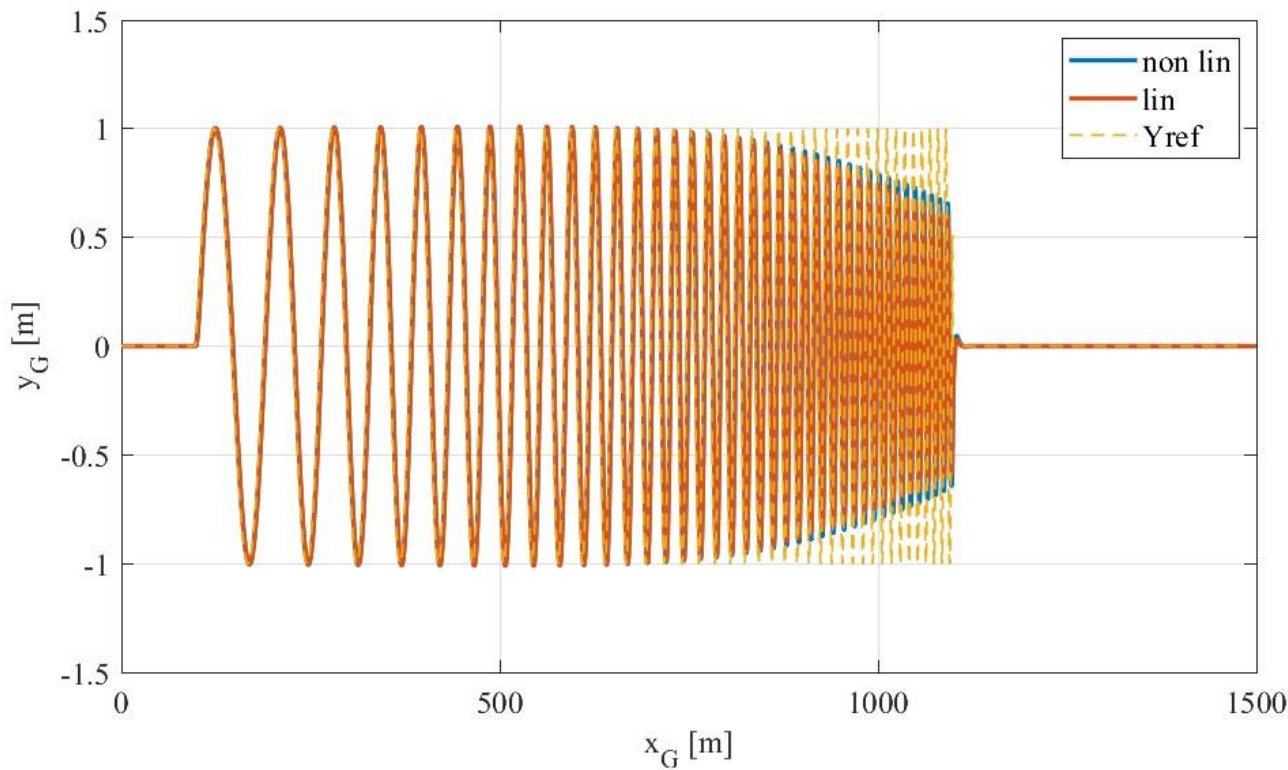
$K_{rol} = 0.5$ and $V = [15; 30; 50]$ km/h

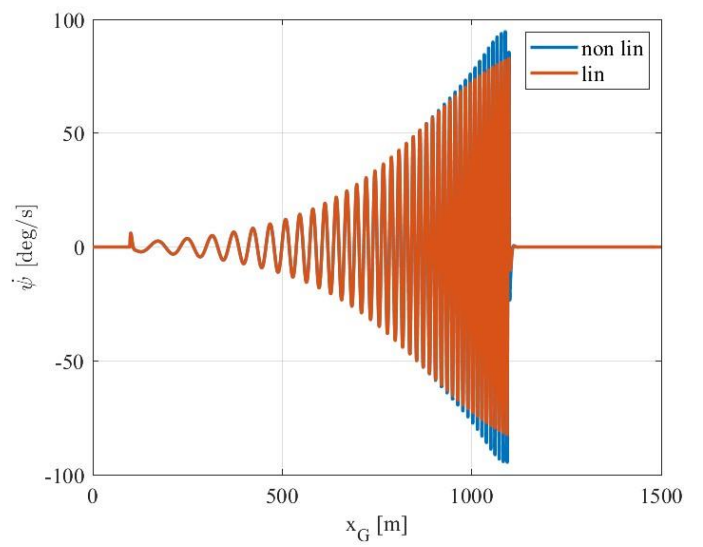
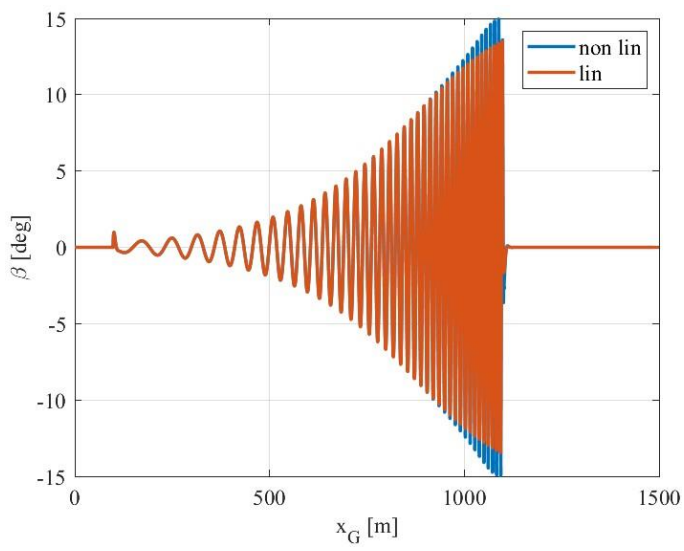
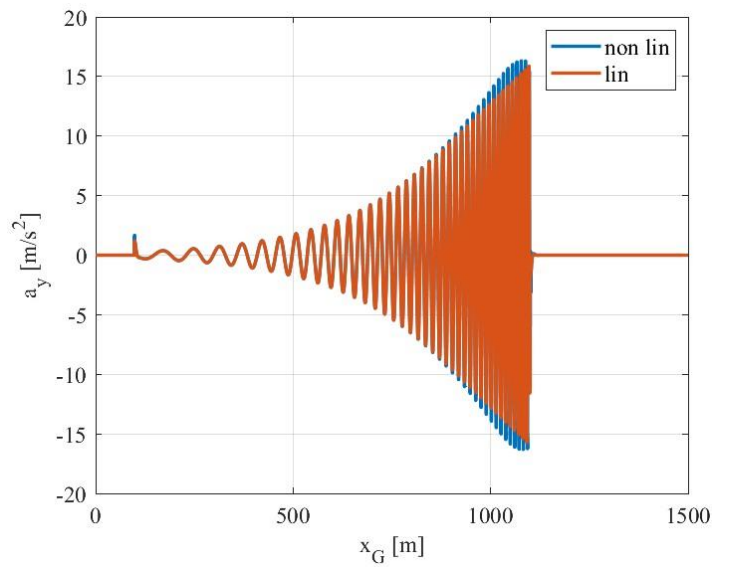
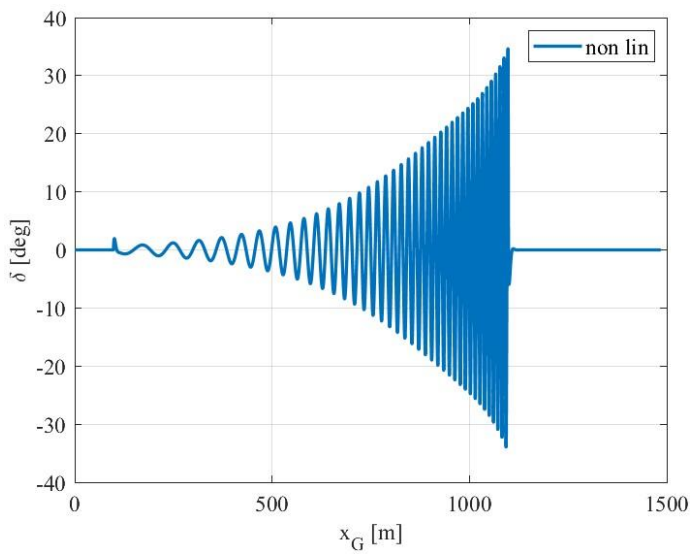
$V = 15$ km/h



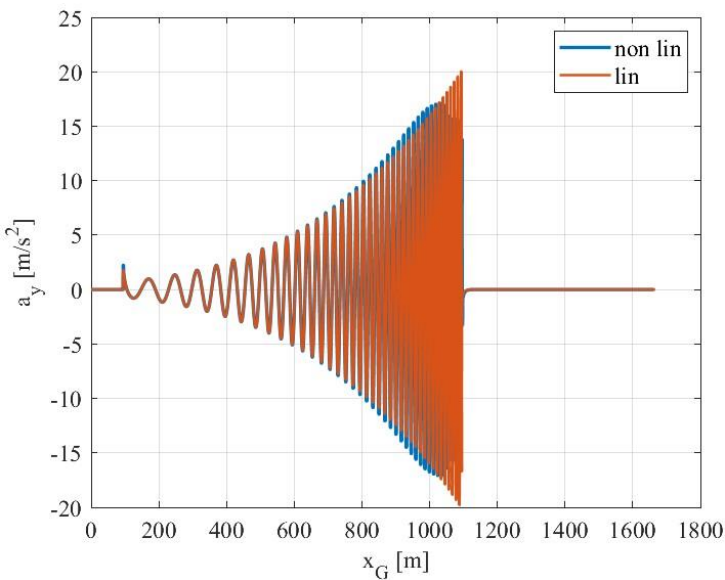
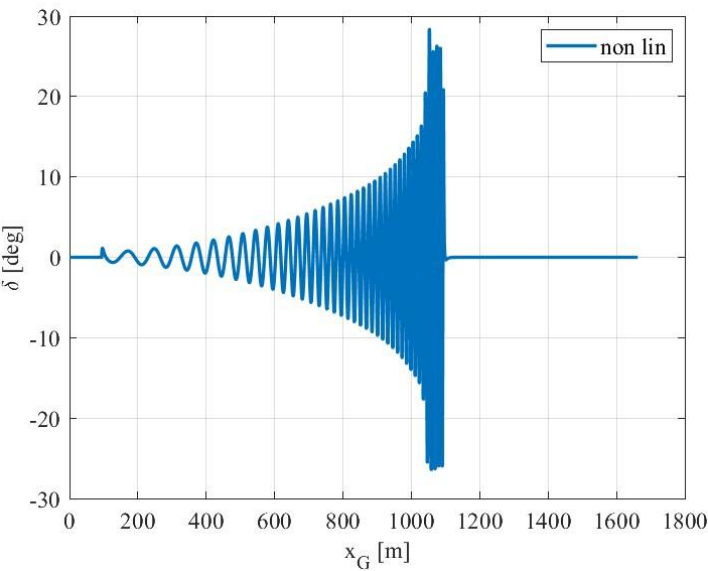
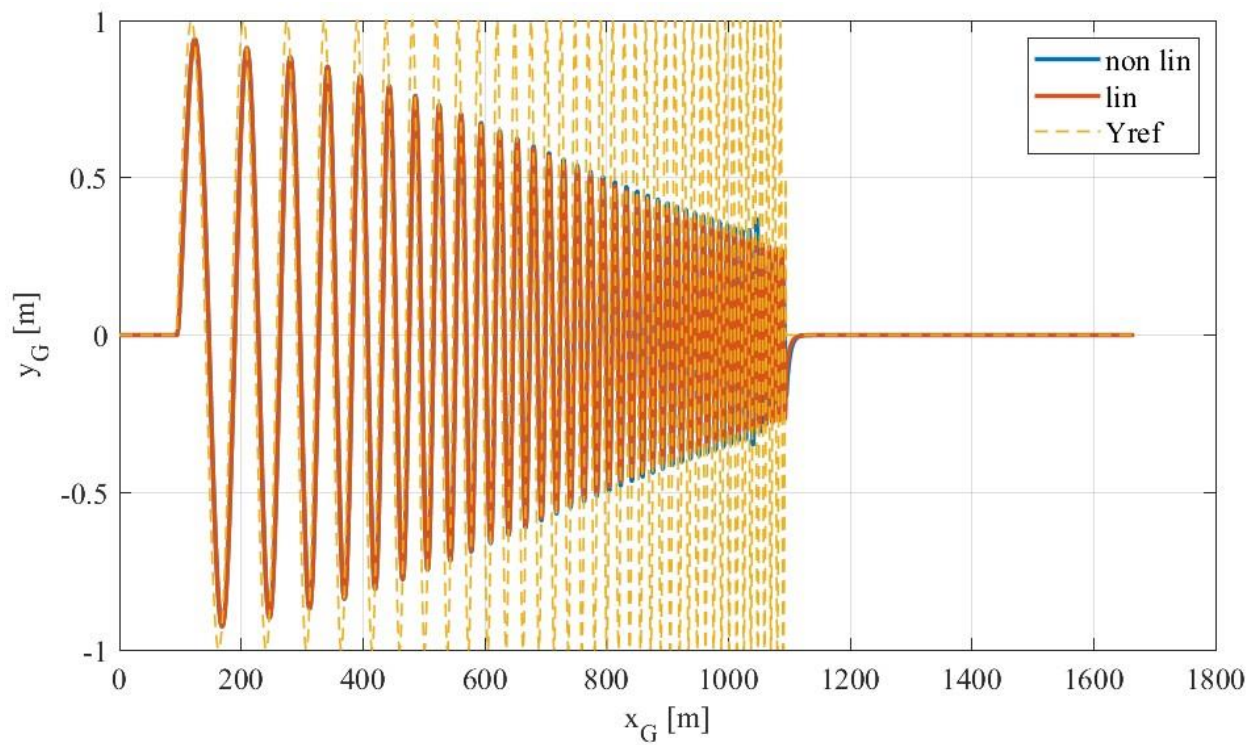


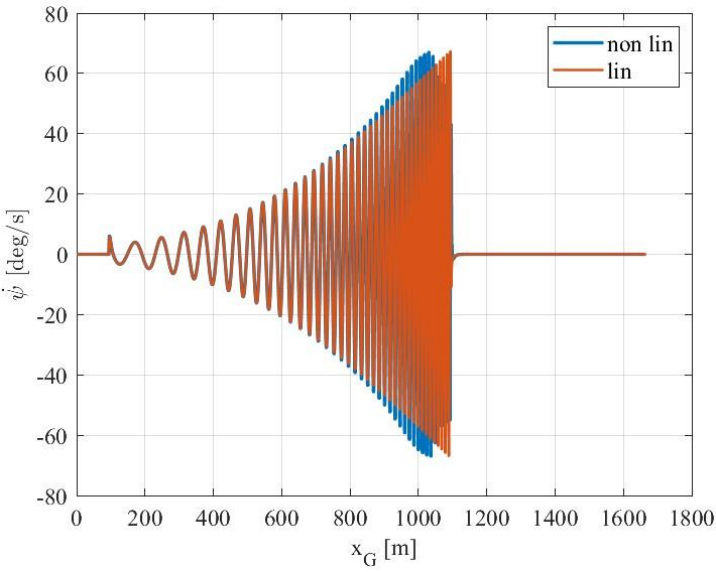
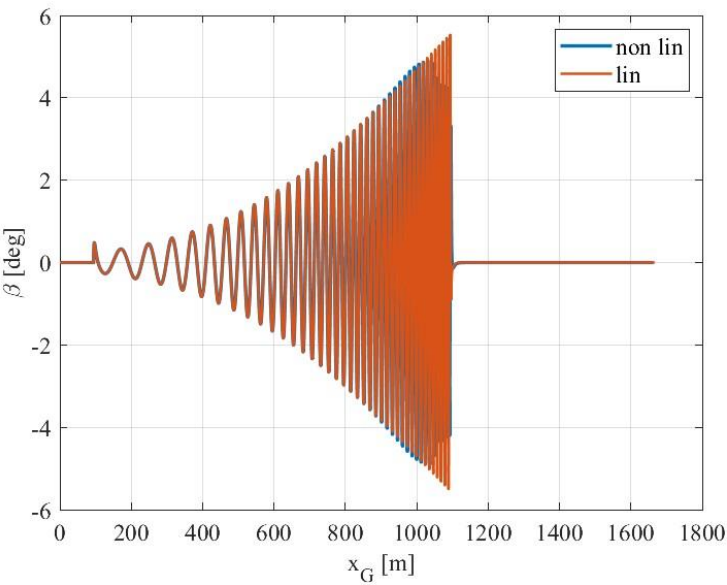
$V = 30 \text{ km/h}$





$V = 50 \text{ km/h}$



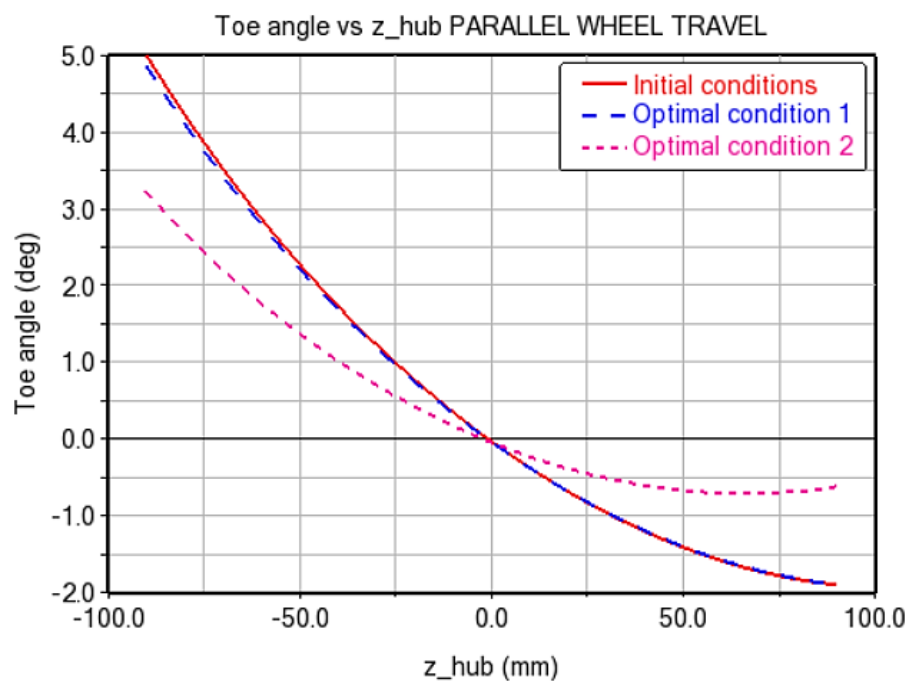
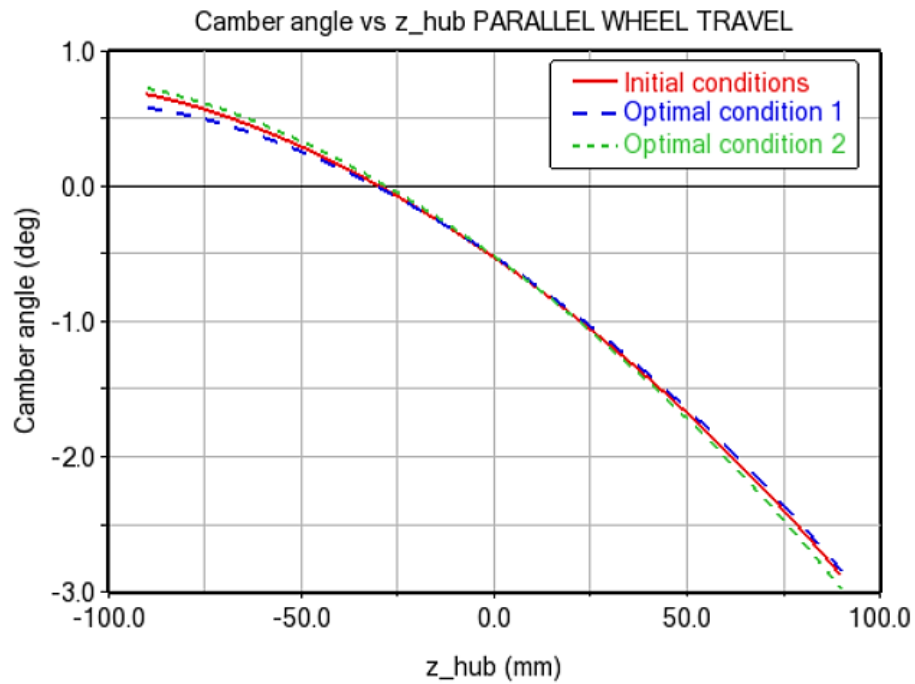


	driver.L [m]	driver.kp
V = 15 km/h	1	0.7
V = 30 km/h	2	0.5
V = 50 km/h	6	0.3

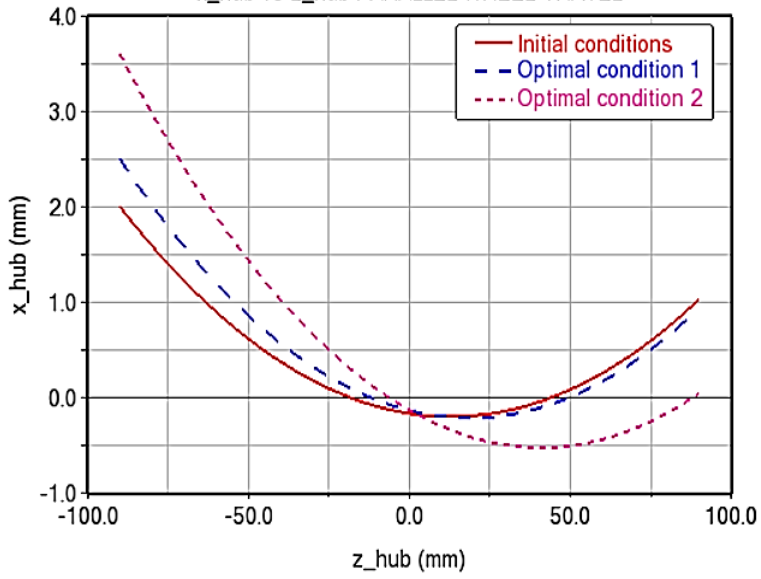
Vehicle Dynamics and Control: Adams Car Lab

Car: **Ferrari Testarossa**

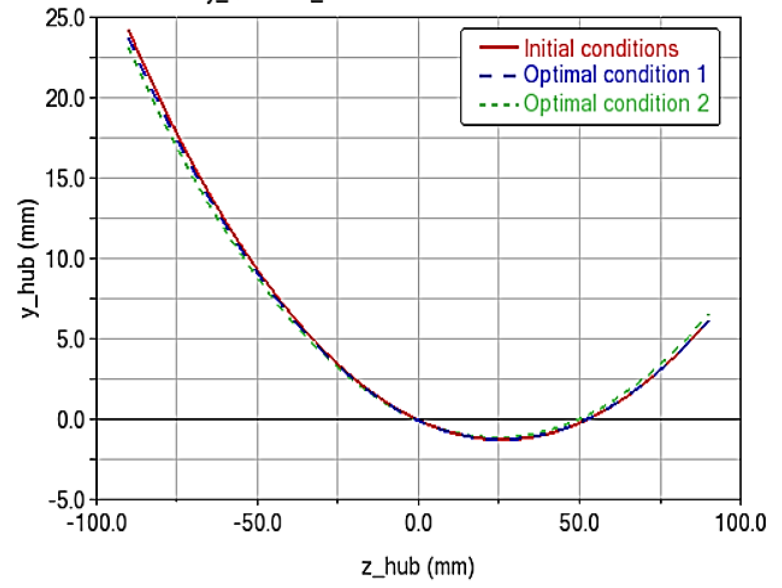
FRONT DOUBLE WISHBONE SUSPENSION



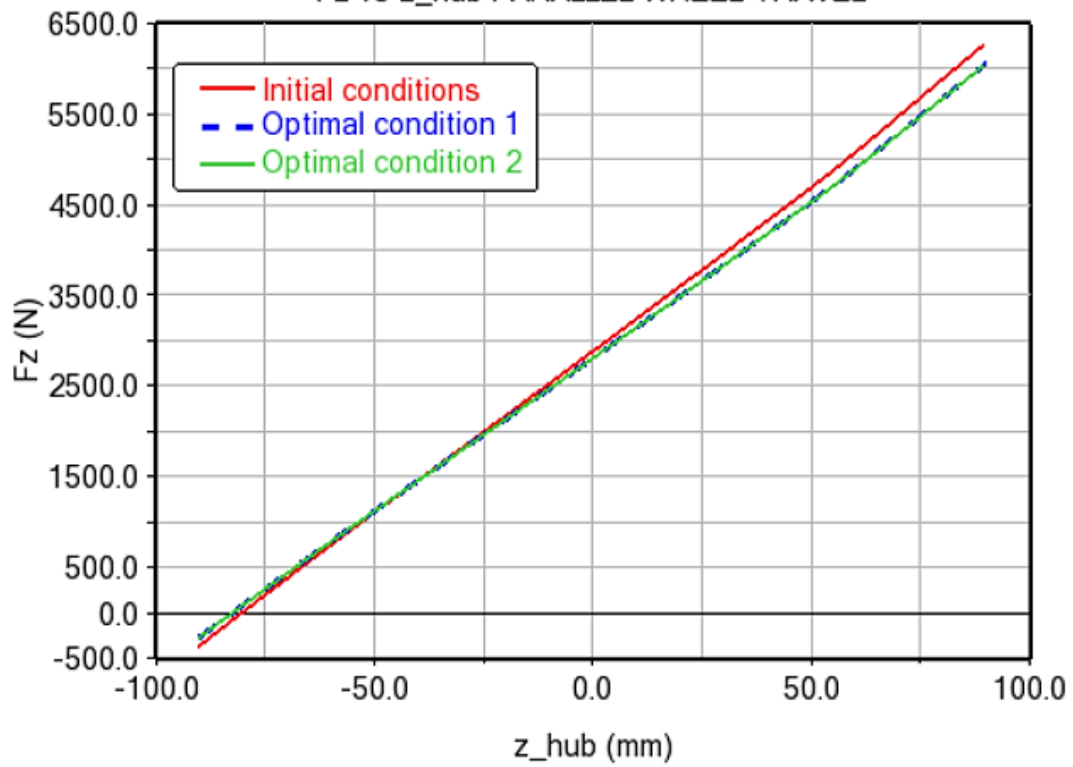
x_{hub} vs z_{hub} PARALLEL WHEEL TRAVEL



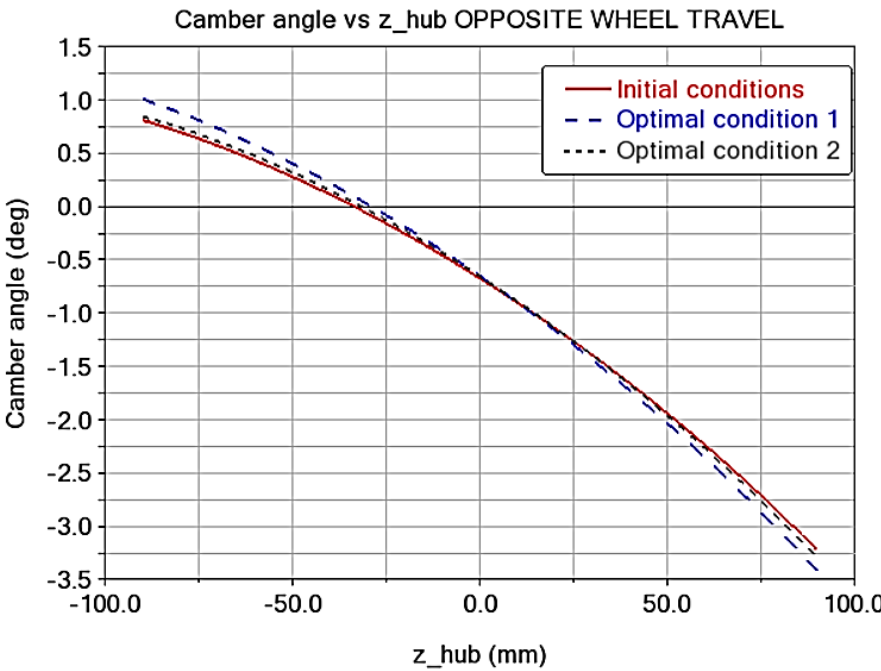
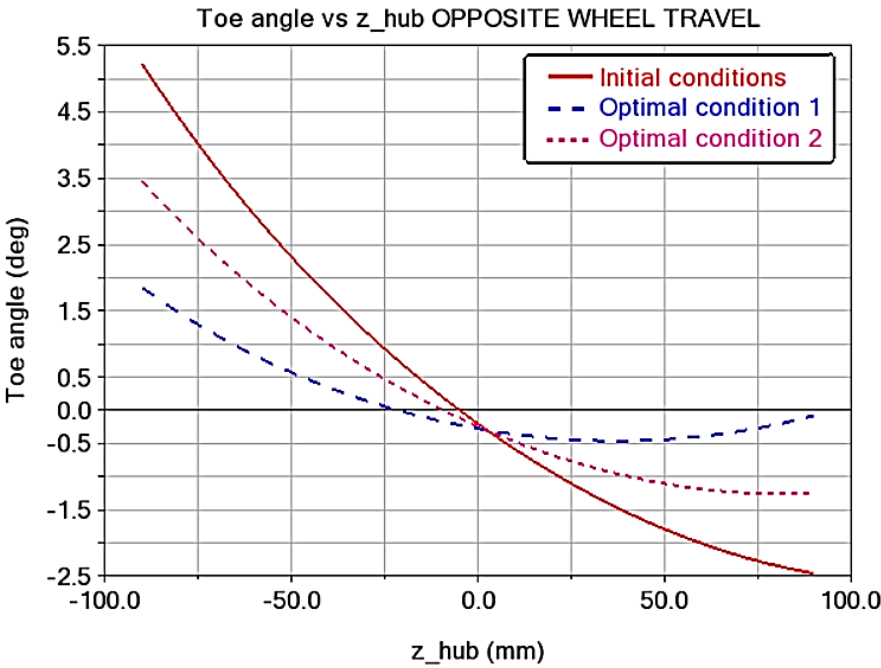
y_{hub} vs z_{hub} PARALLEL WHEEL TRAVEL



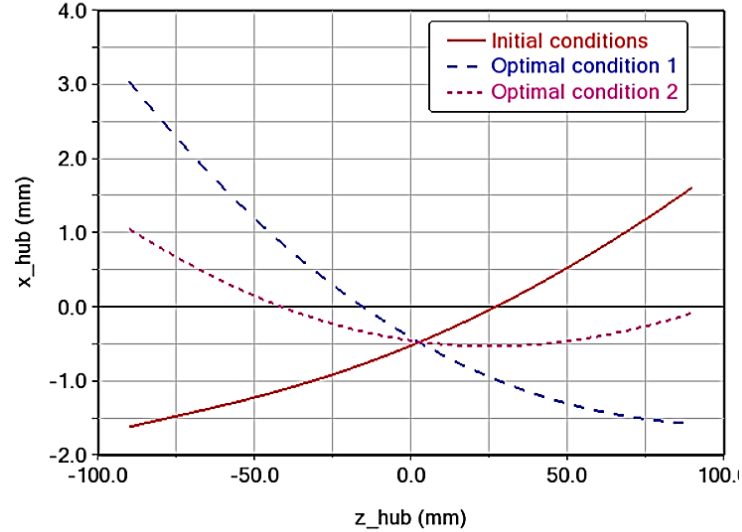
F_z vs z_{hub} PARALLEL WHEEL TRAVEL



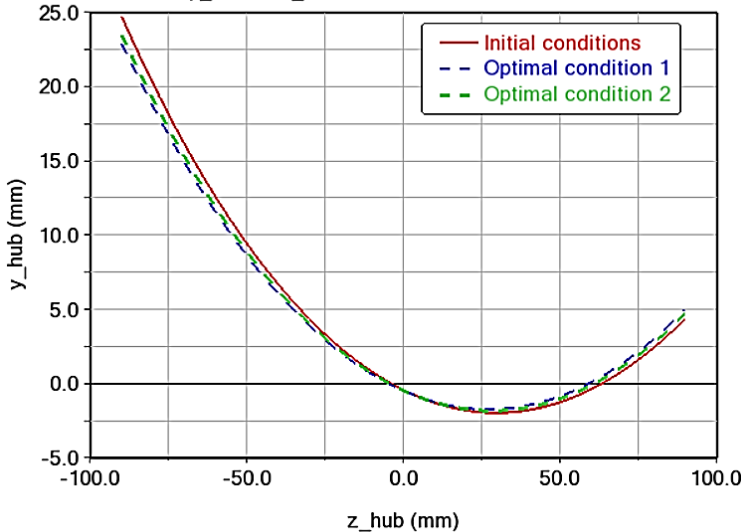
REAR DOUBLE WISHBONE SUSPENSION



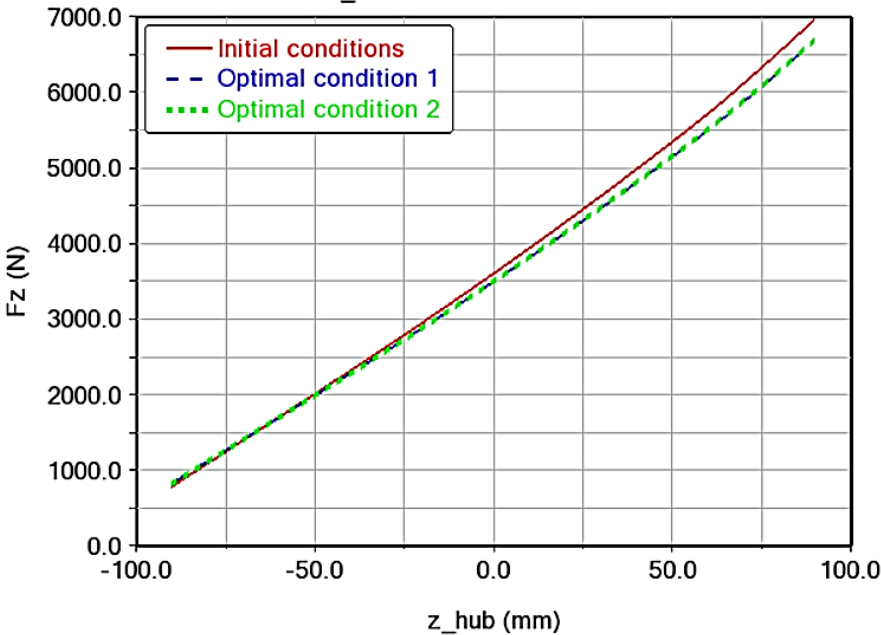
x_hub vs z_hub OPPOSITE WHEEL TRAVEL



y_hub vs z_hub OPPOSITE WHEEL TRAVEL

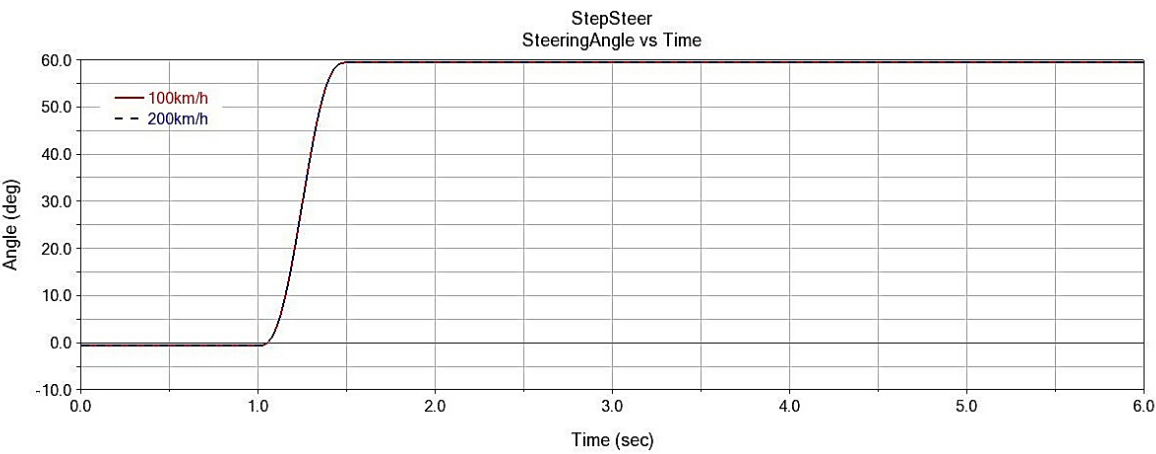
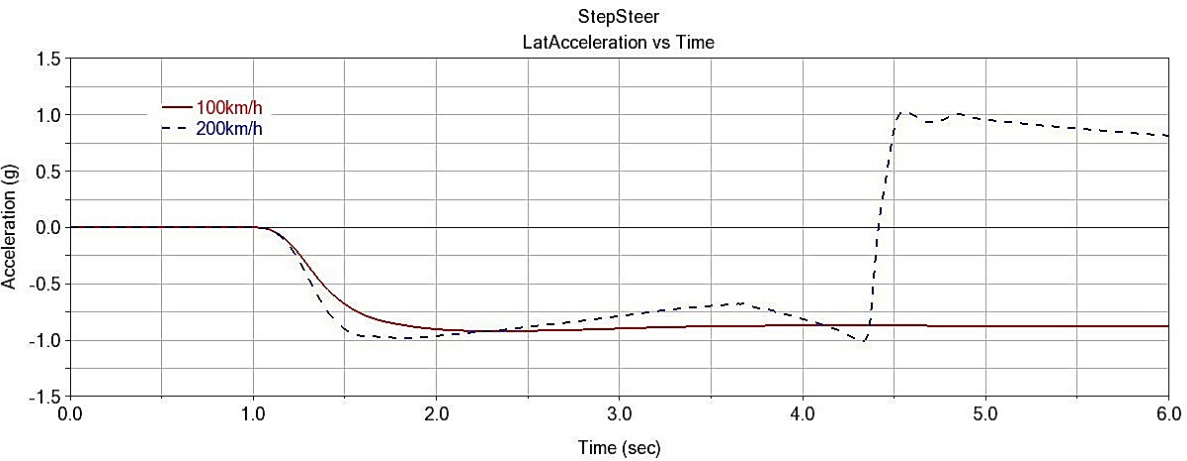
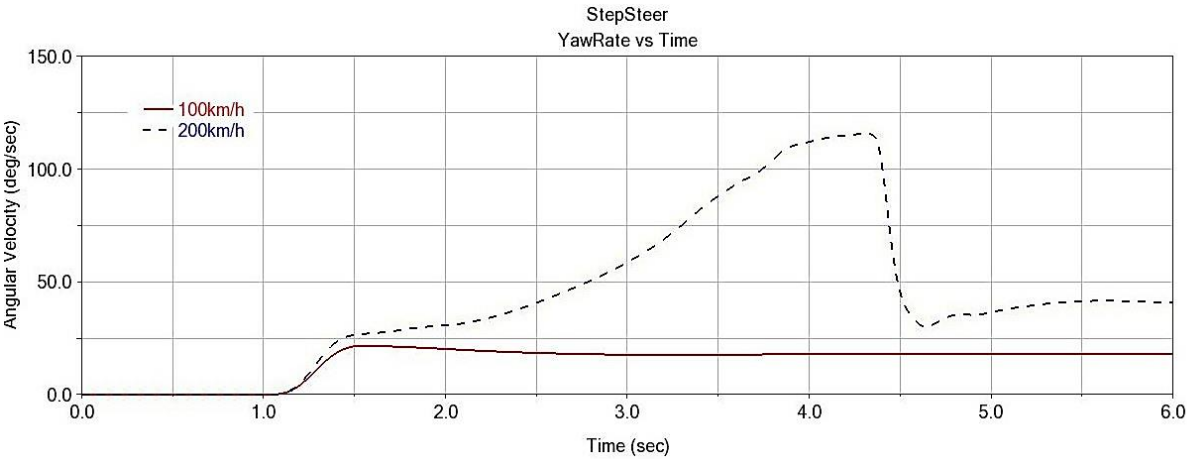


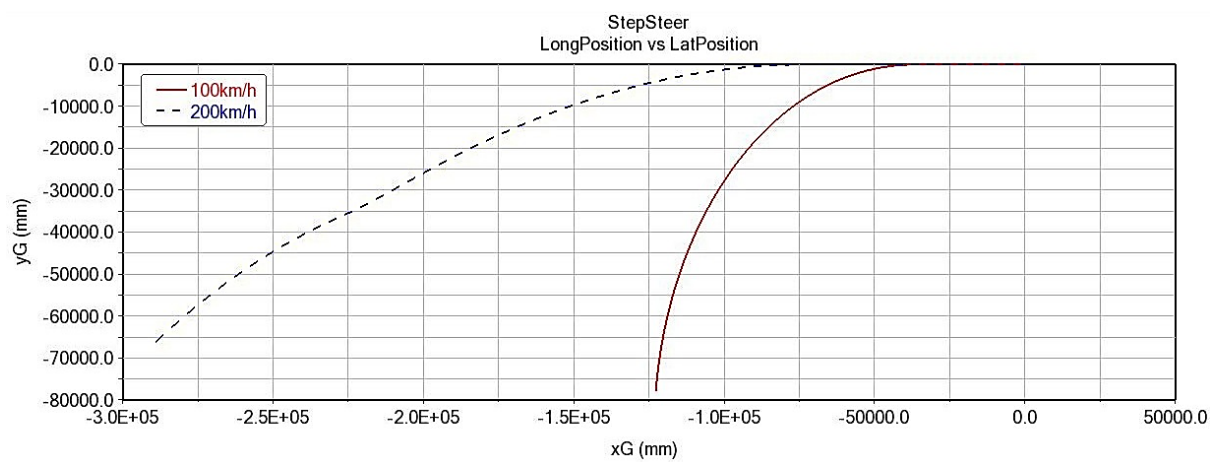
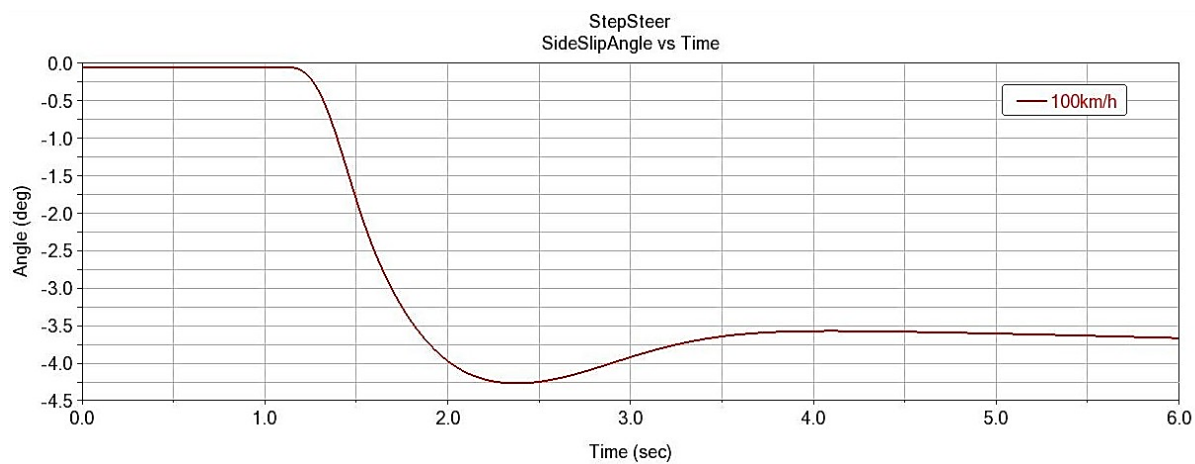
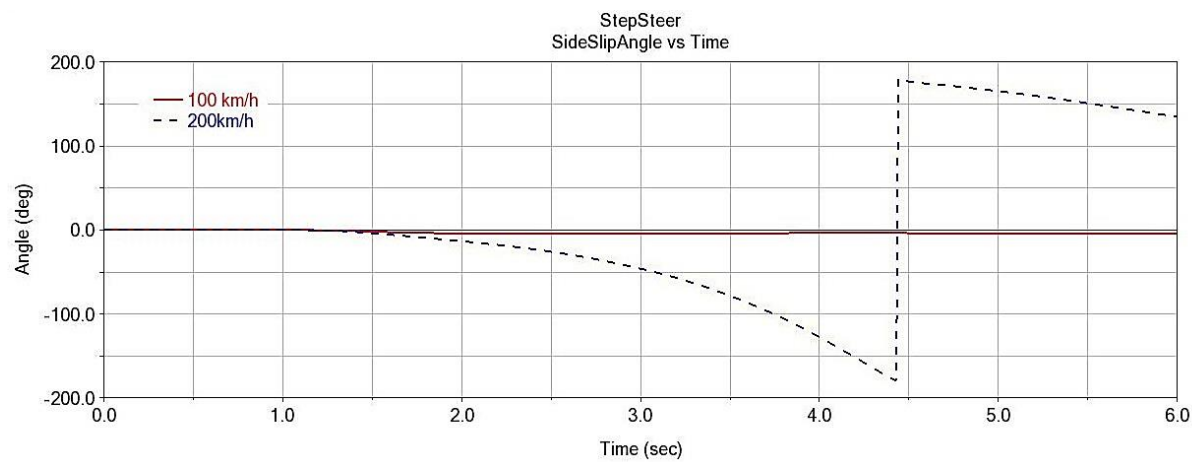
Fz vs z_hub OPPOSITE WHEEL TRAVEL



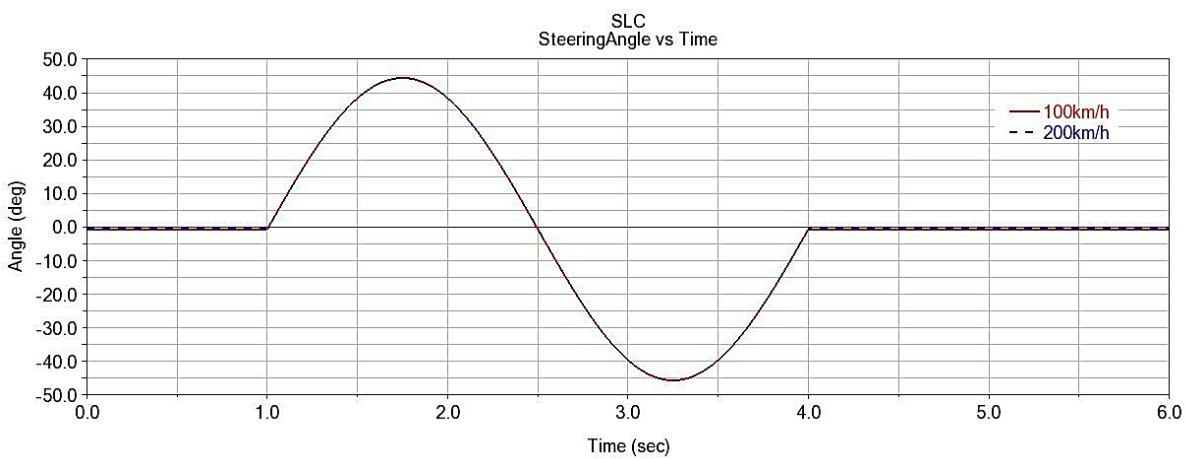
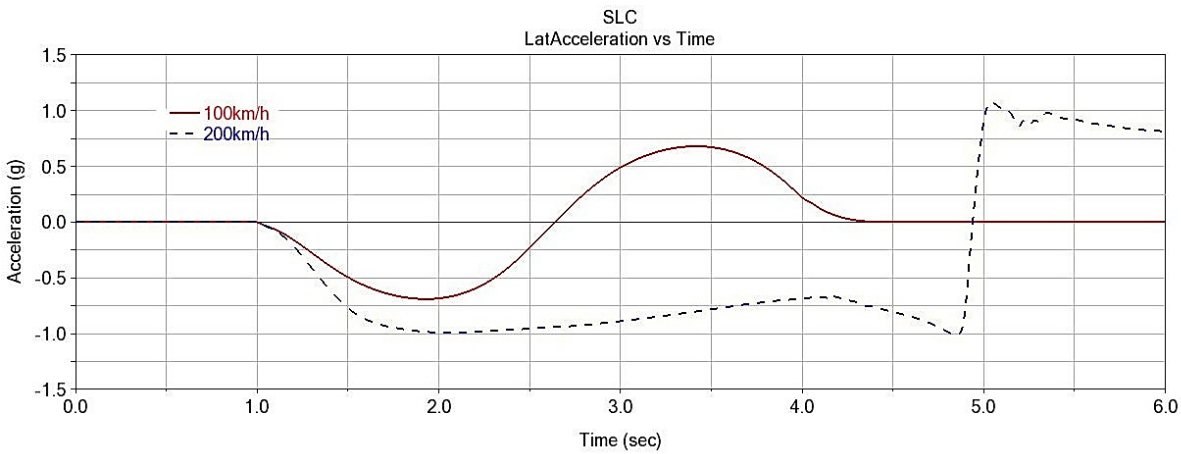
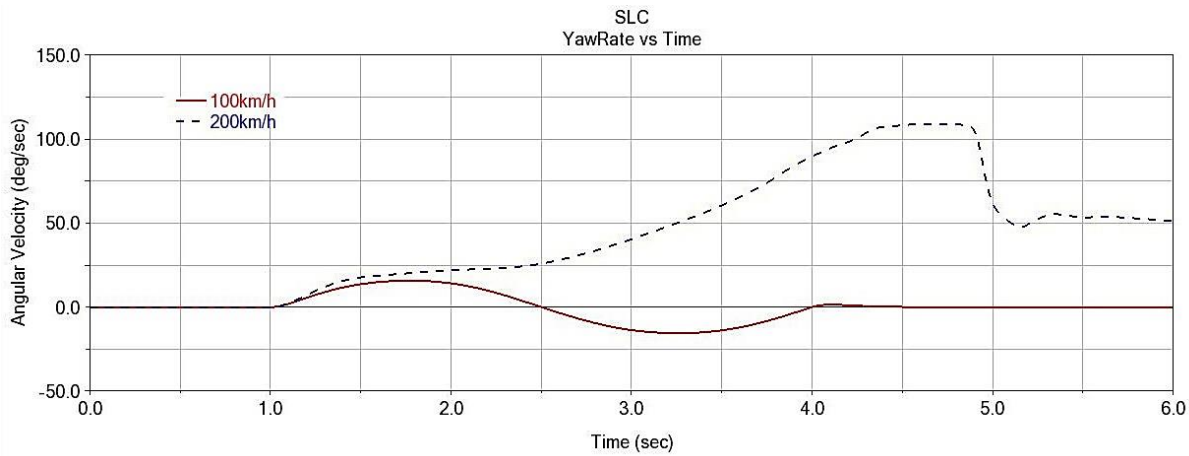
FULL VEHICLE MULTIBODY MODEL

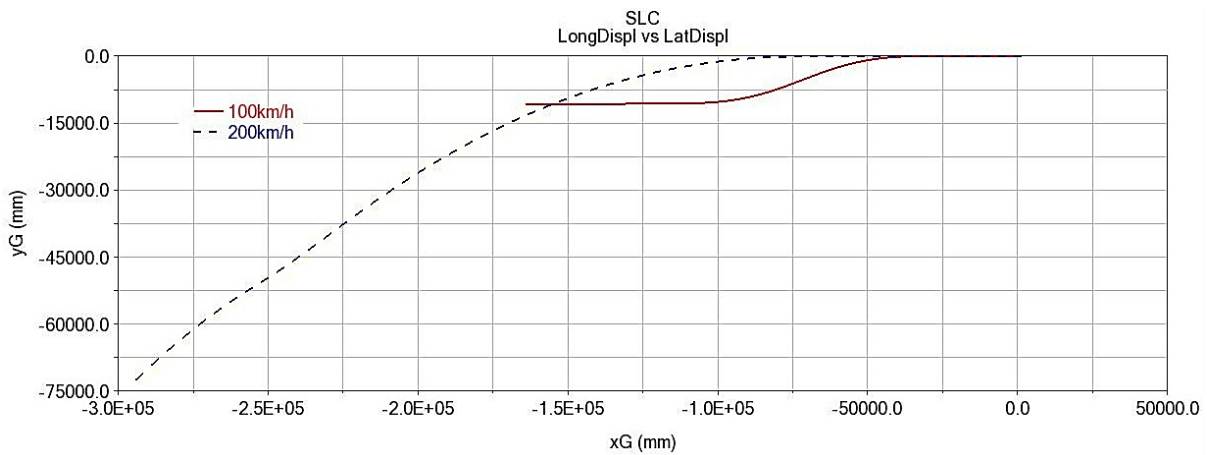
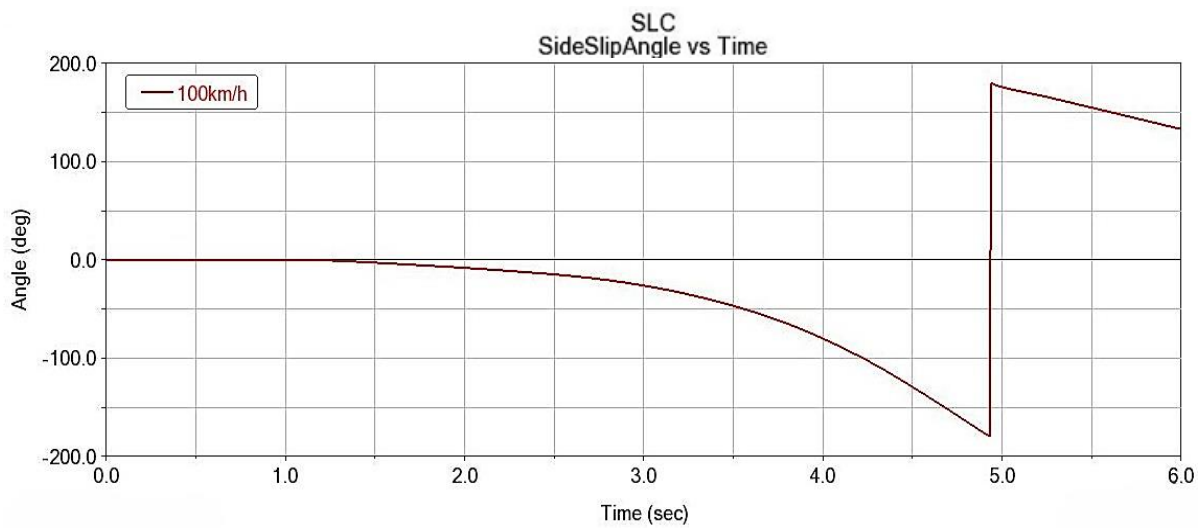
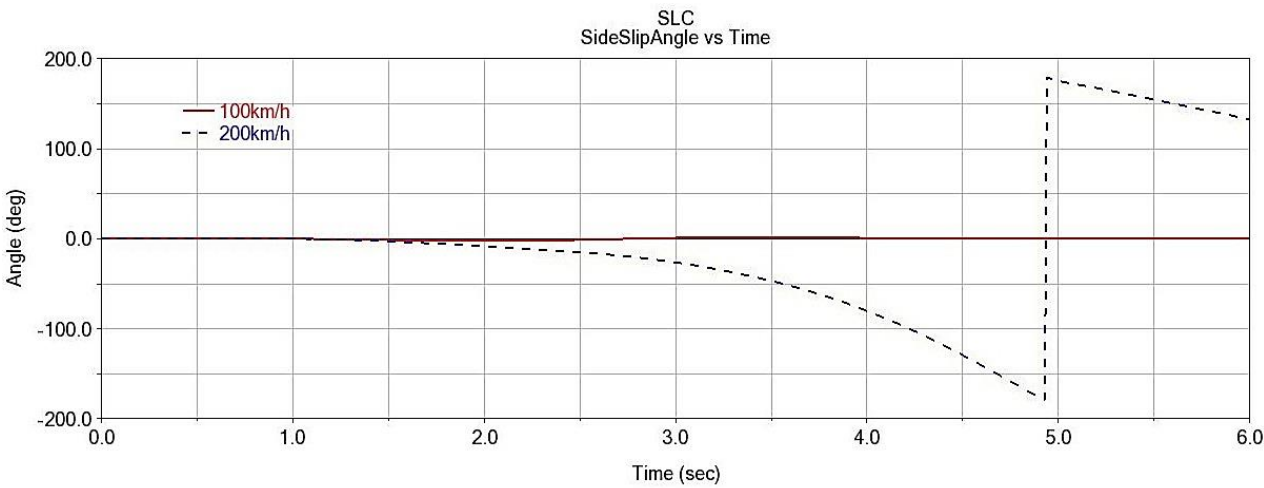
STEP STEER MANEUVER



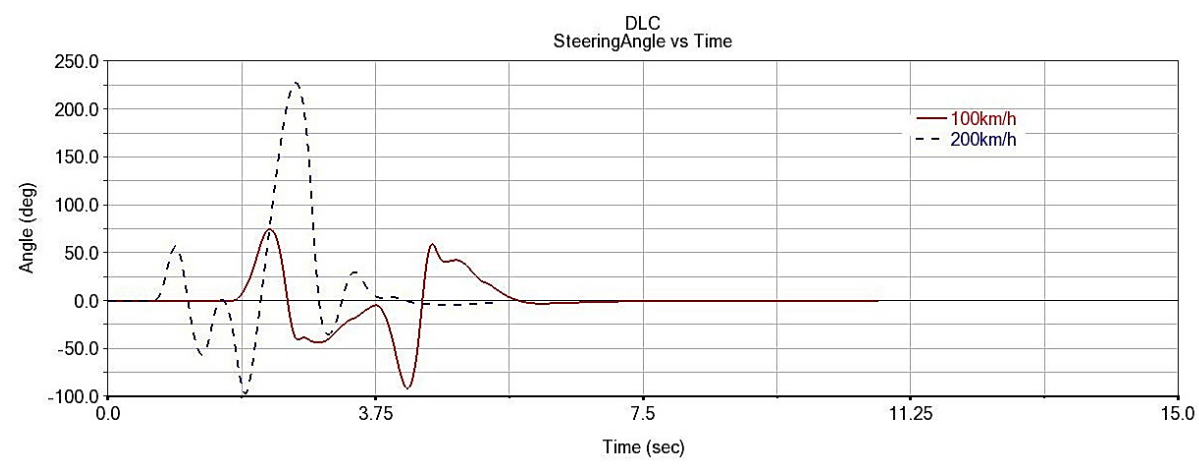
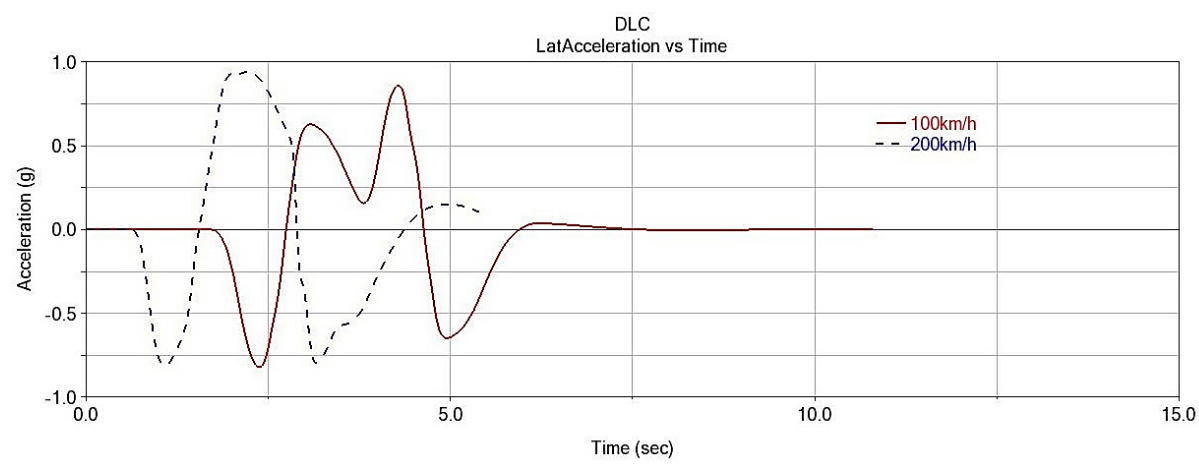
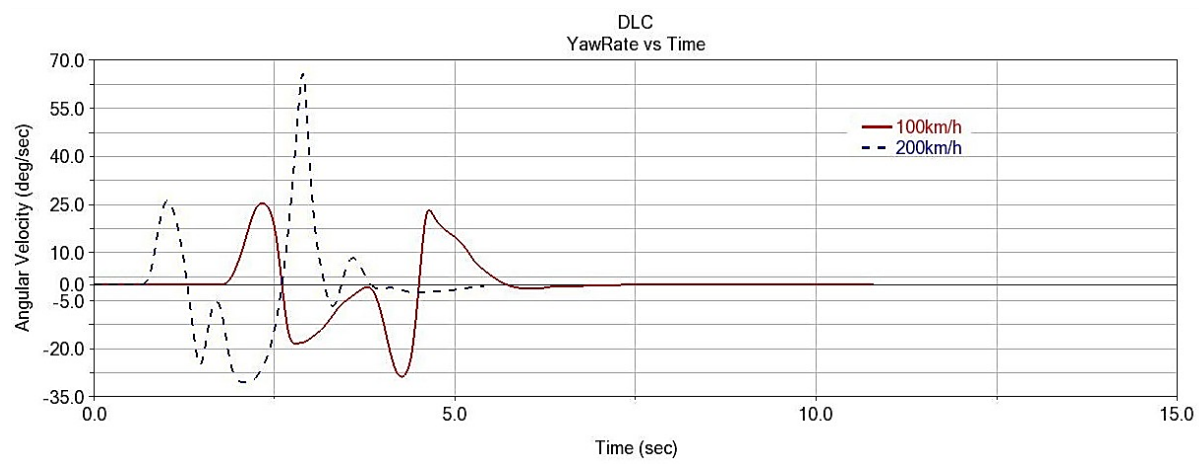


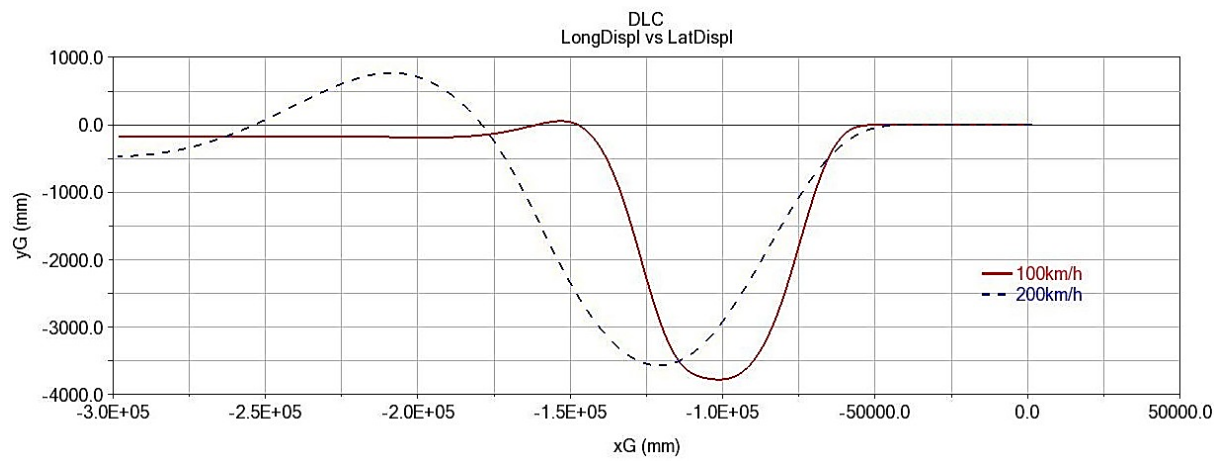
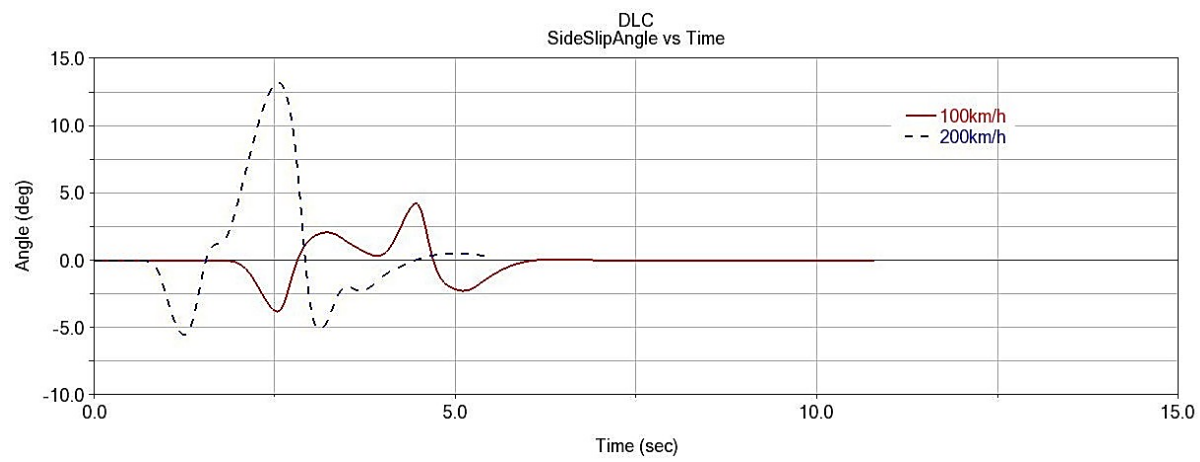
SINGLE LANE CHANGE



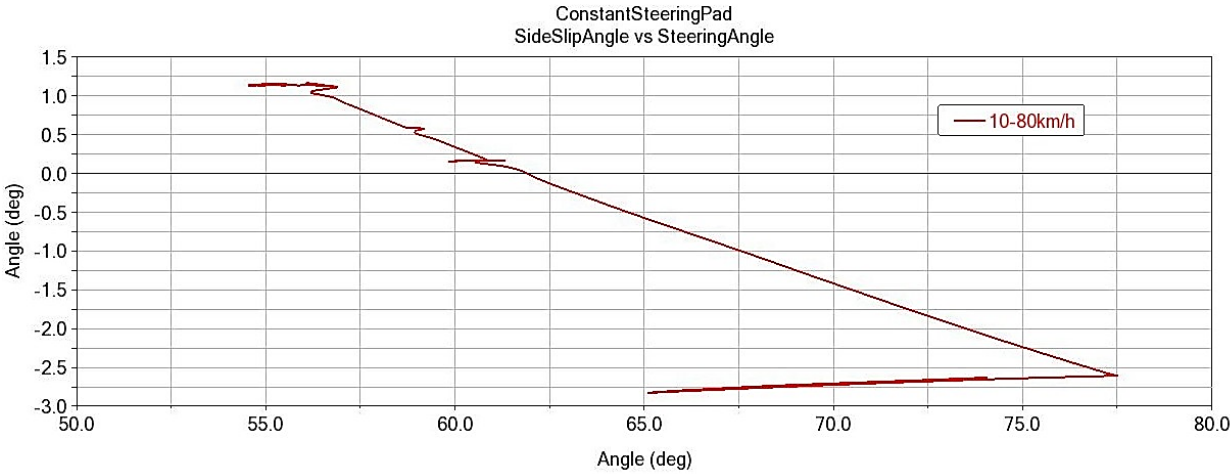
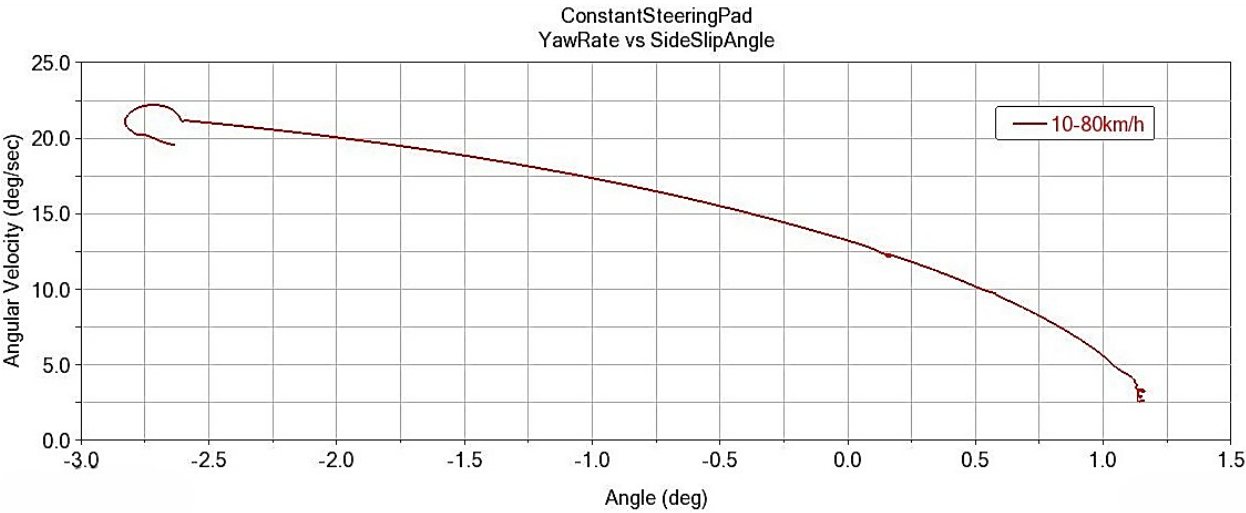
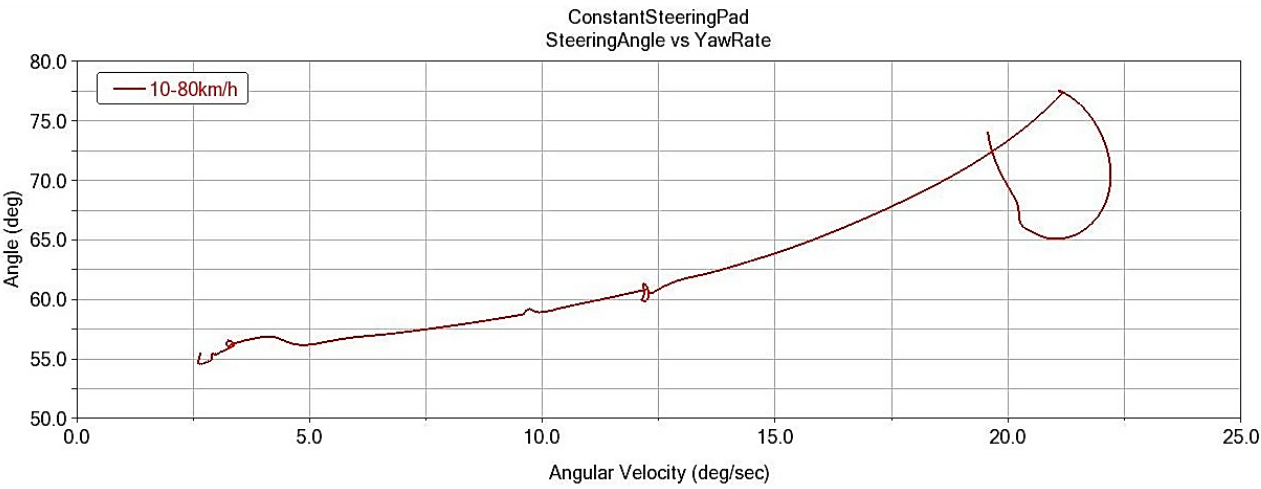


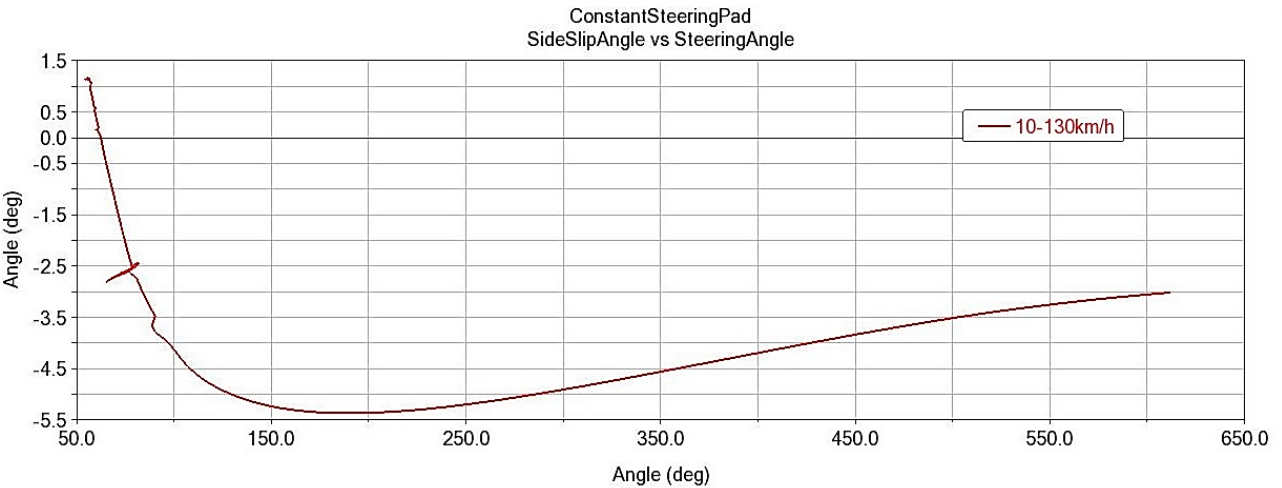
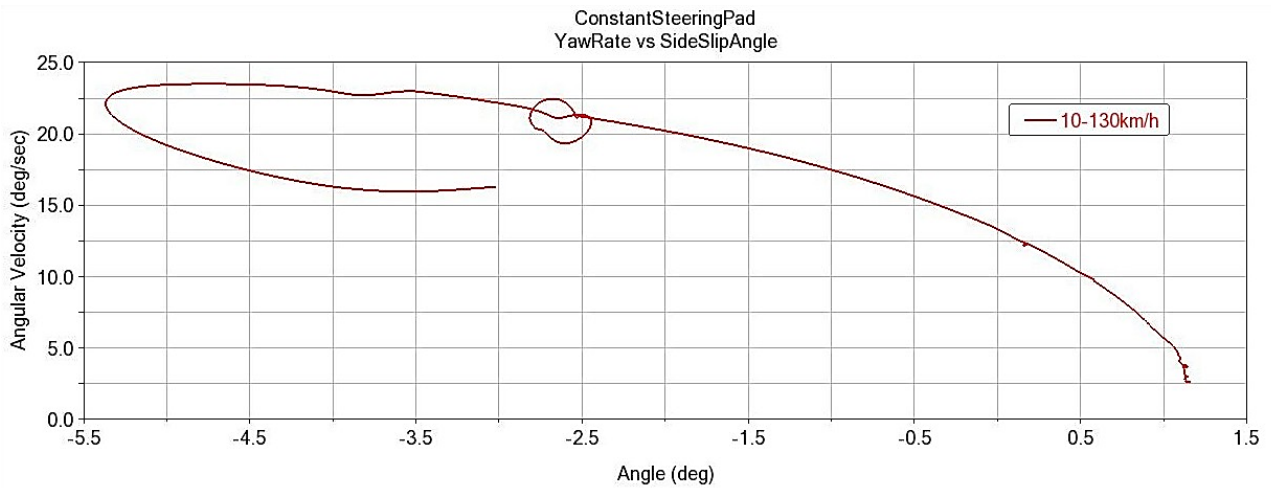
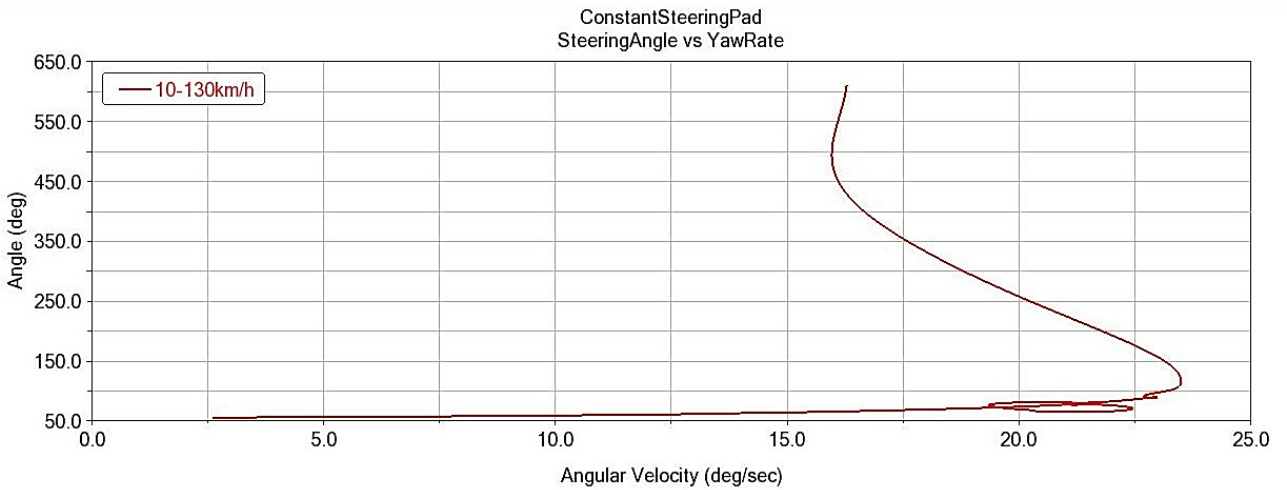
DOUBLE LANE CHANGE

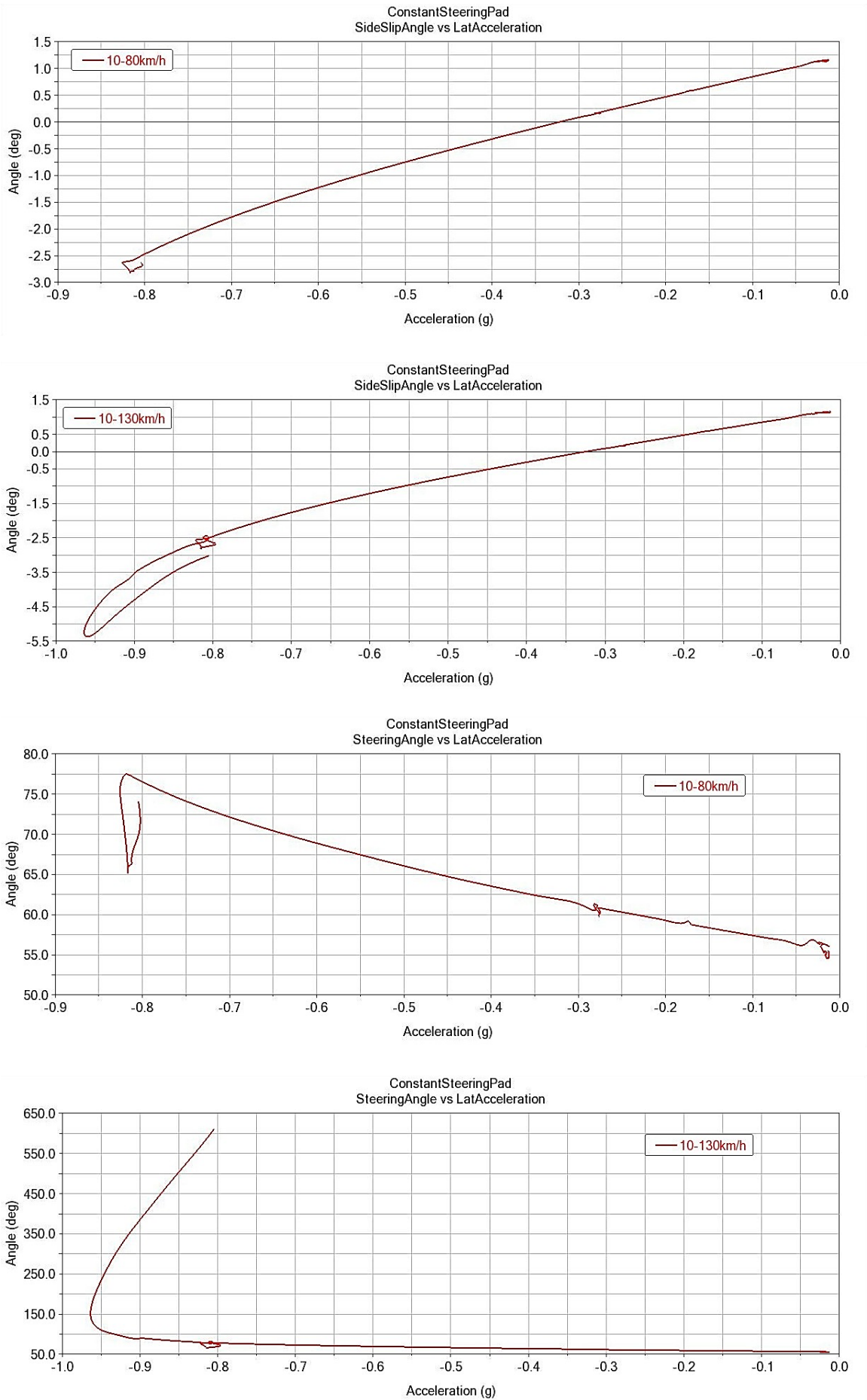




STEERING PAD: CONSTANT STEER







Vehicle Dynamics and Control: VI Grade Lab

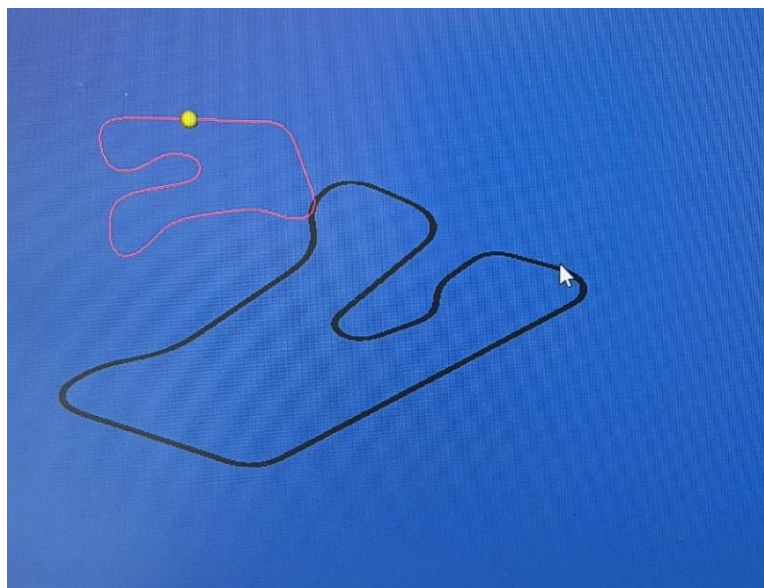
Car: Race car base implemented version

Working on:

- Different antiroll bars stiffness's coefficients
- Different aerodynamic set up of the car



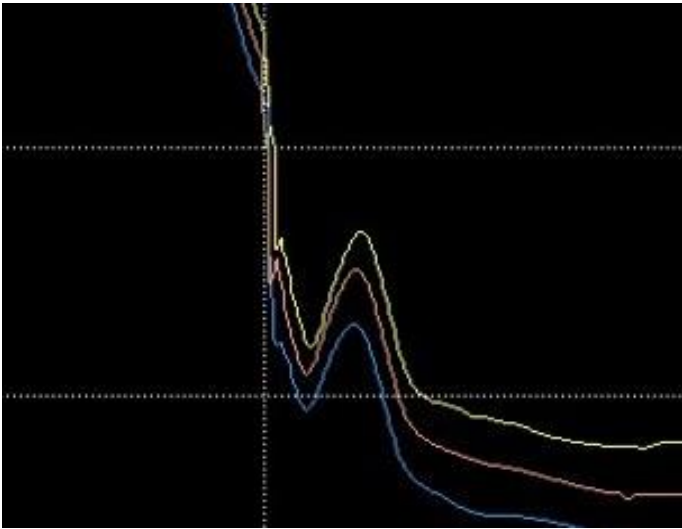
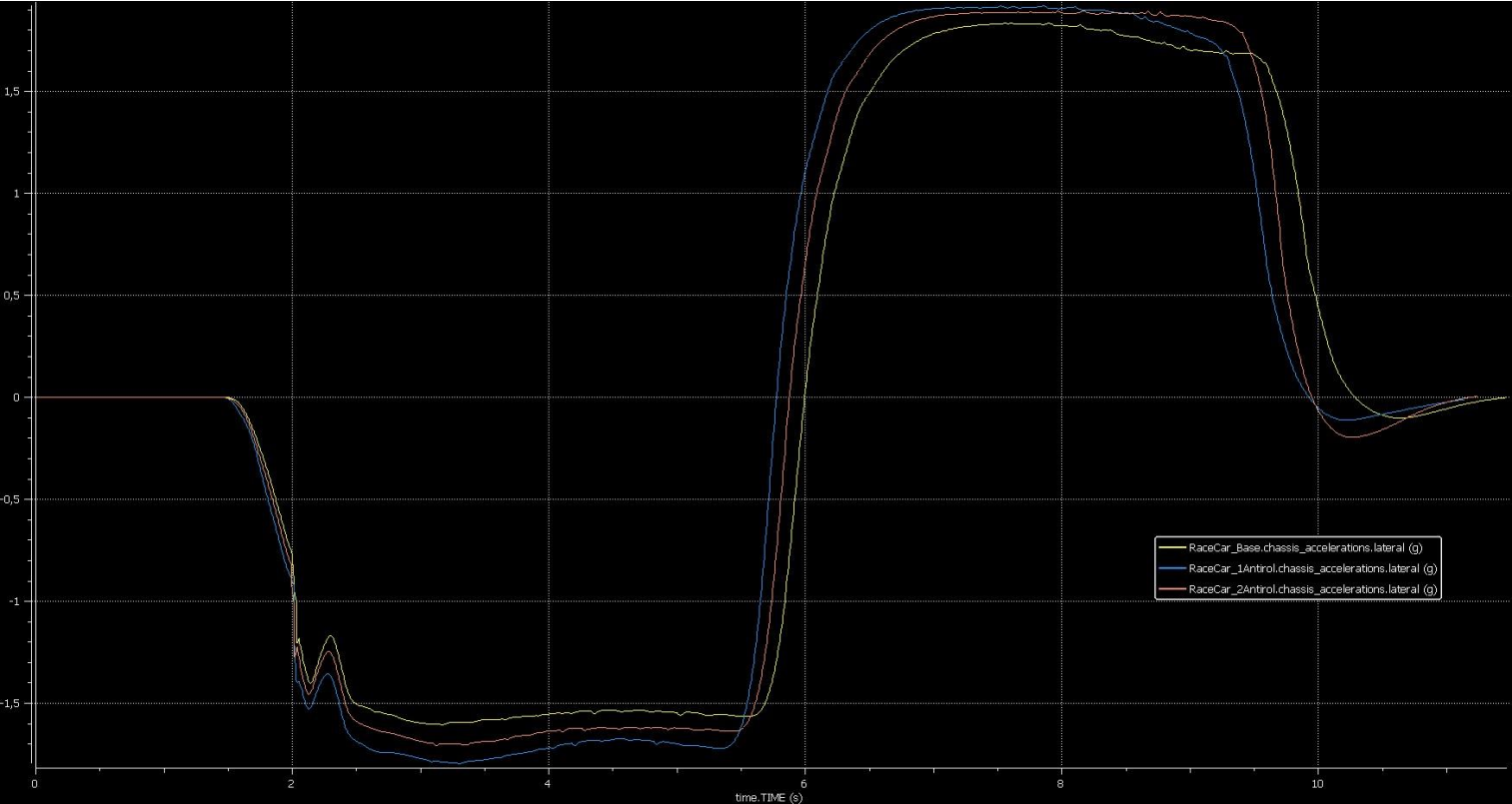
1. first simulation on a chicane of 100 meters radius
2. second simulation on base circuit already implemented



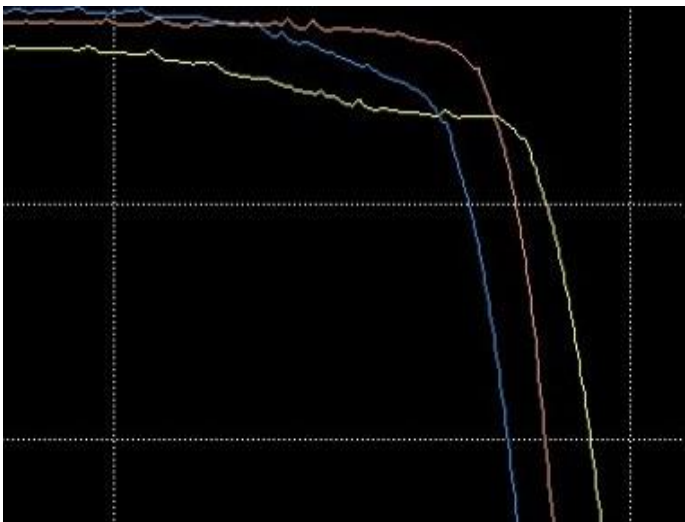
First simulation

Working on antiroll bar stiffness

Lateral Acceleration vs Time

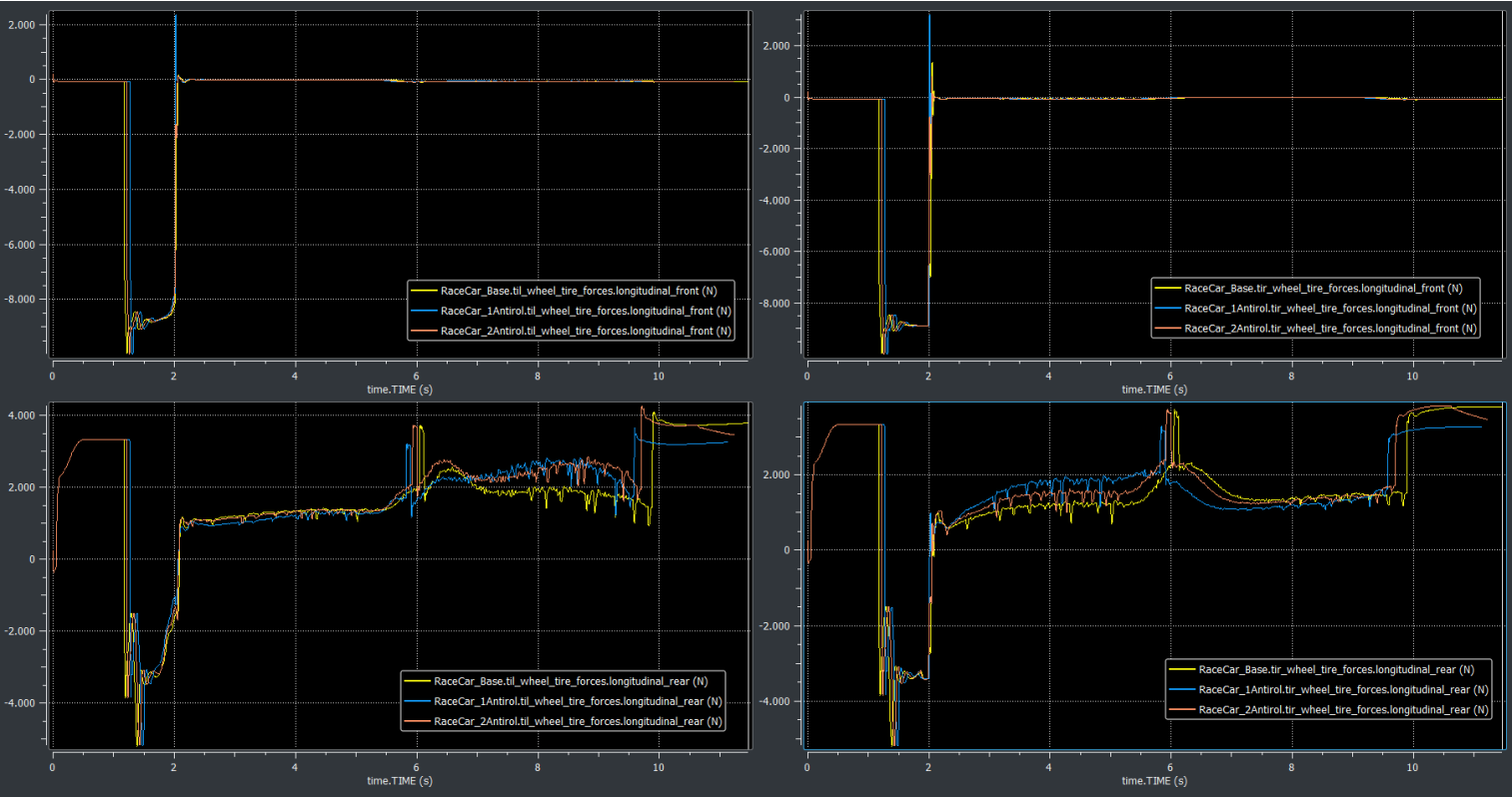


Braking while entering chicane

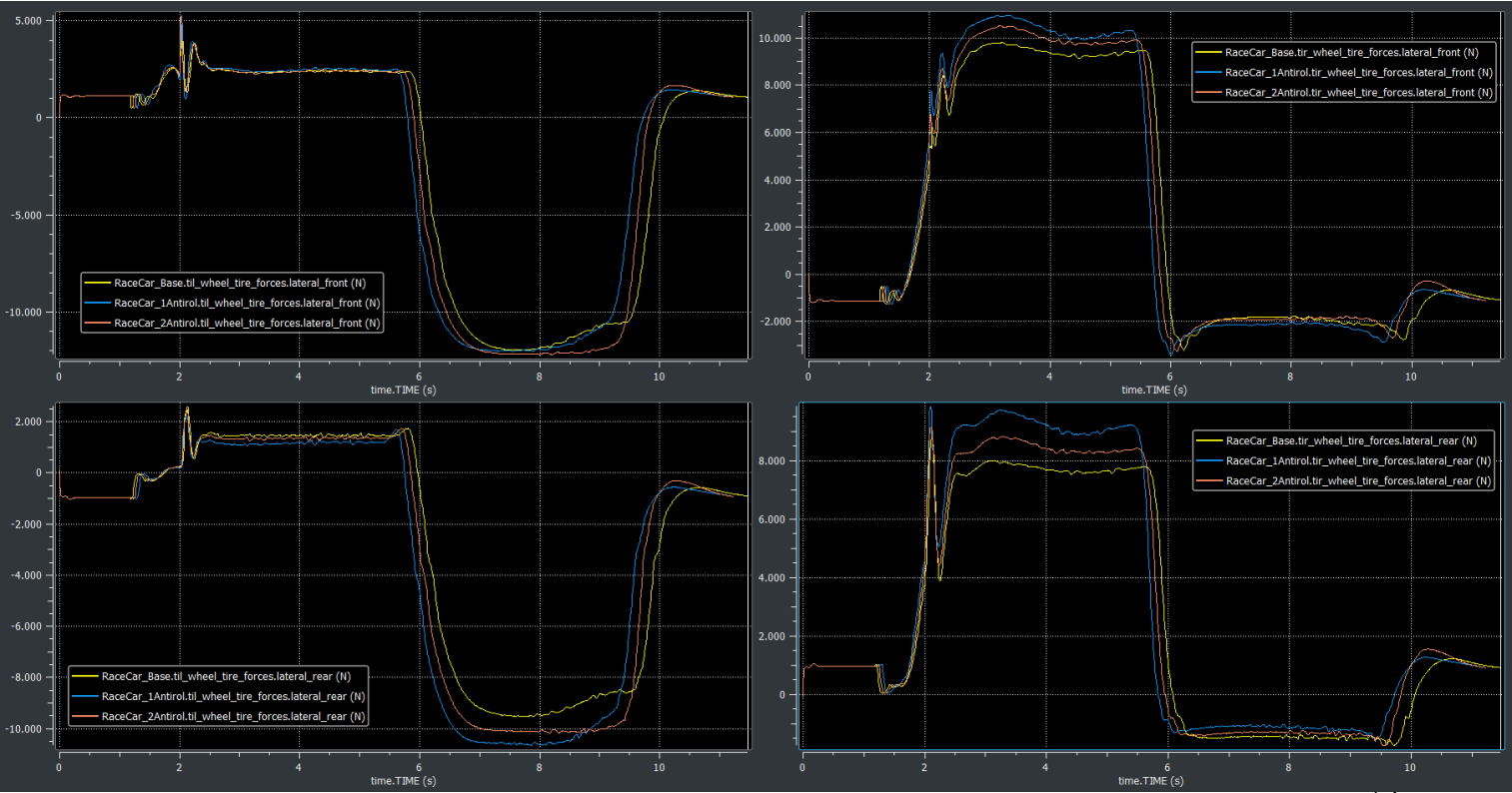


Acceleration while exiting chicane

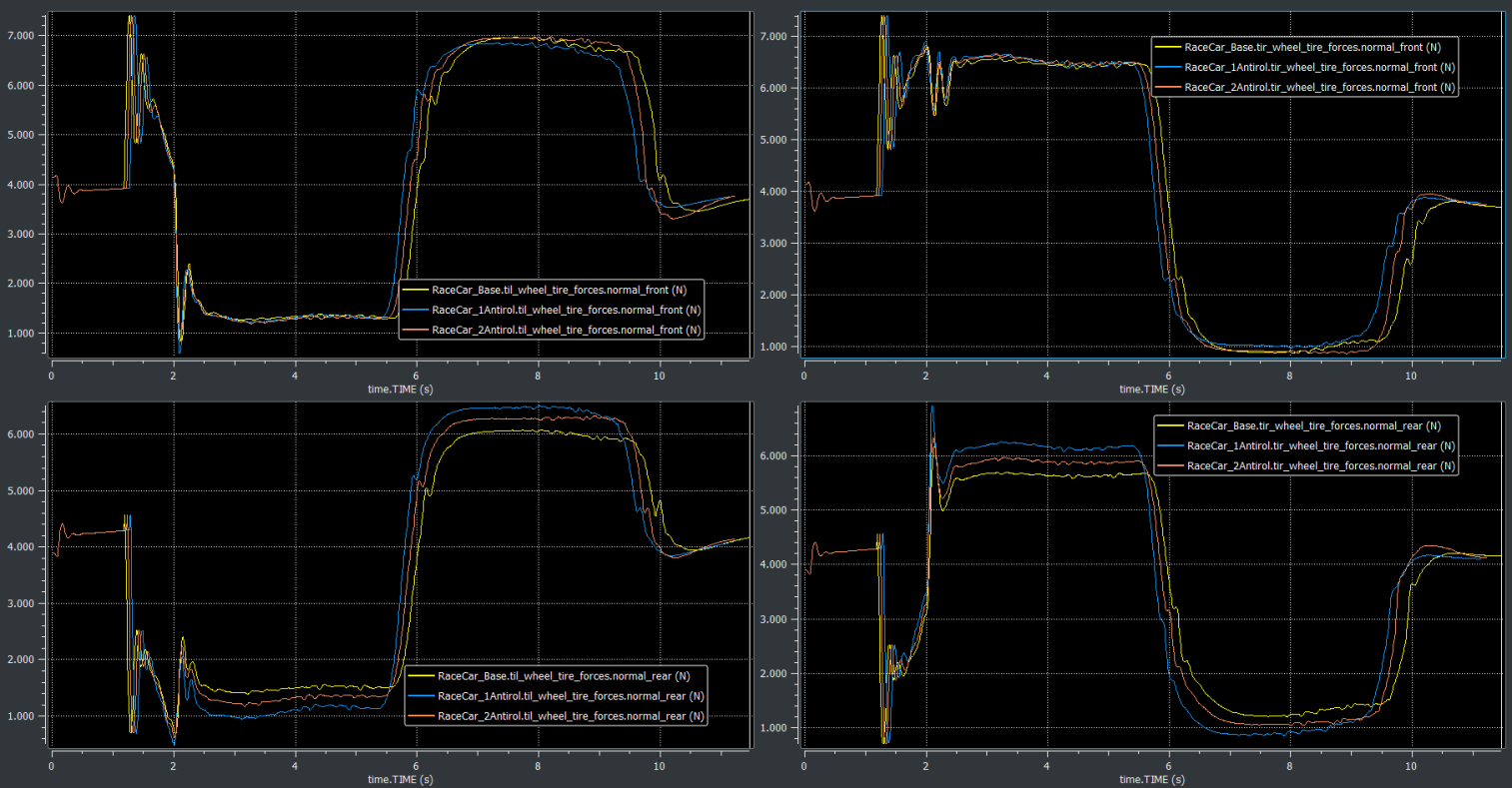
Longitudinal forces on tires vs Time



Lateral forces on tires vs Time

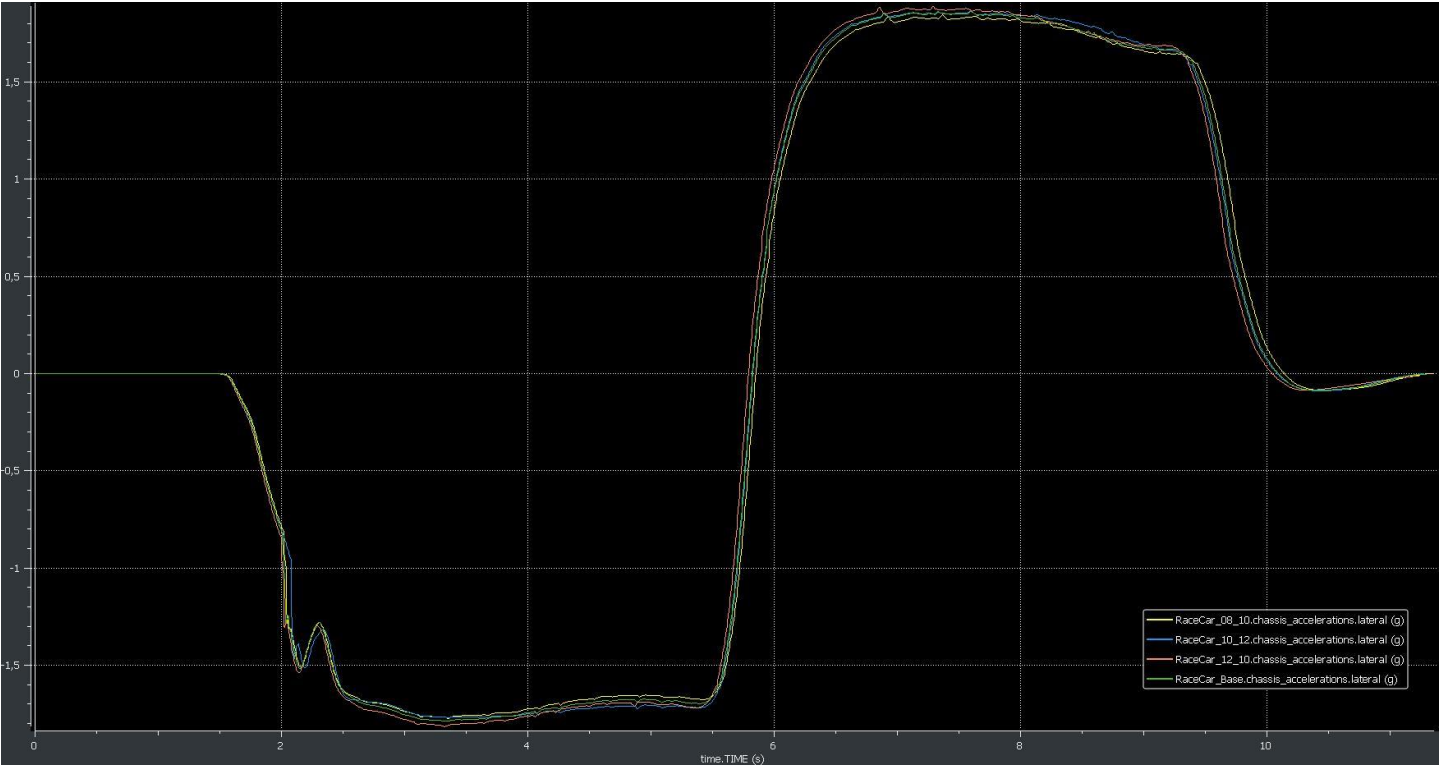


Vertical forces on tires vs Time

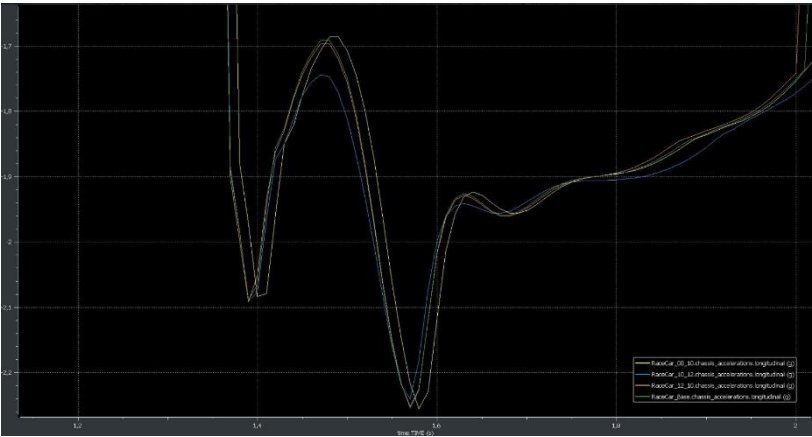
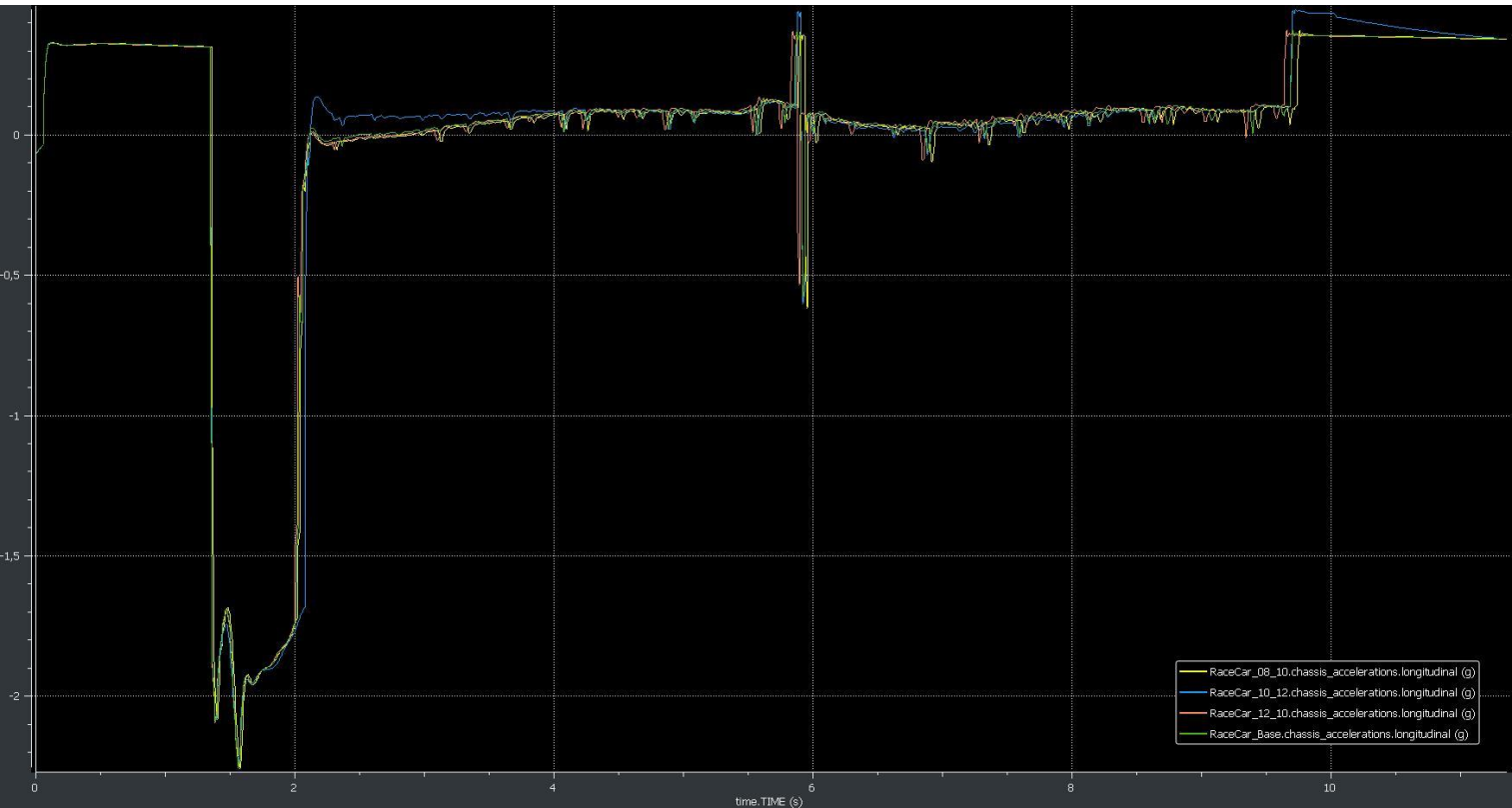


Working on aerodynamic set-up

Lateral acceleration vs Time

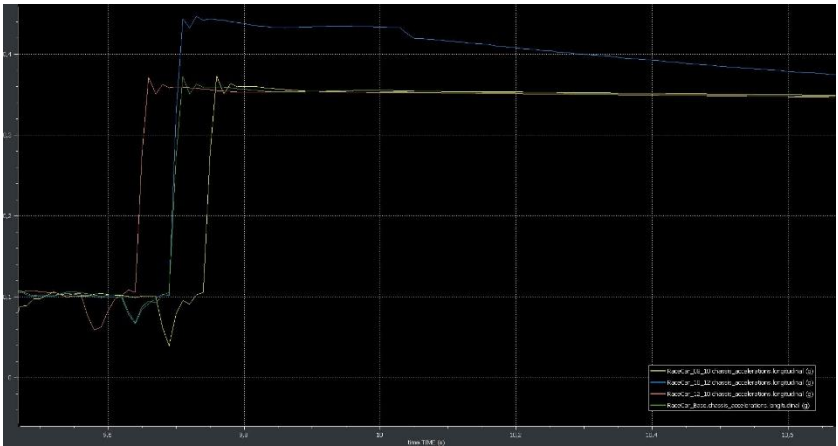


Longitudinal acceleration vs Time

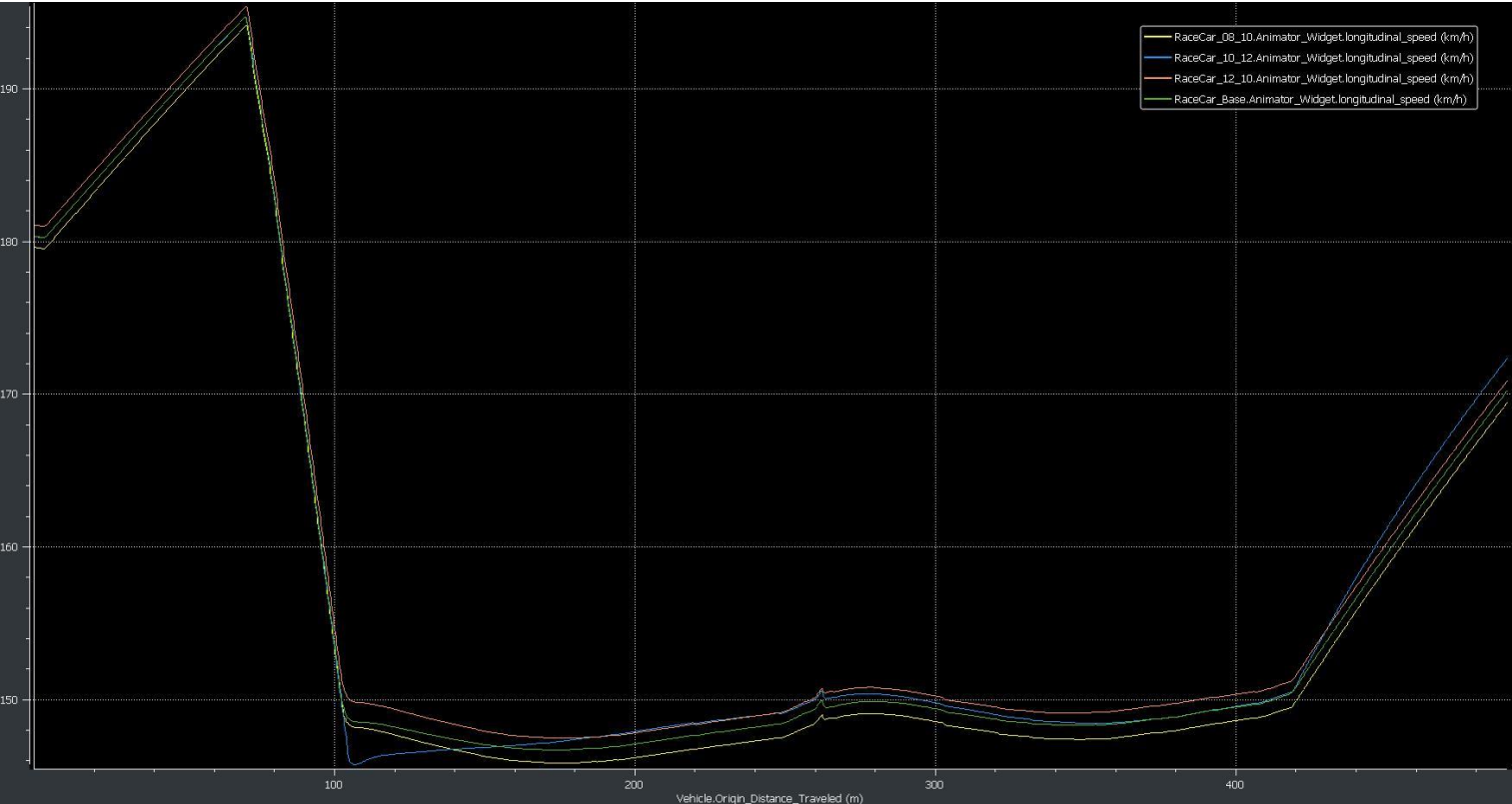


Braking while entering chicane

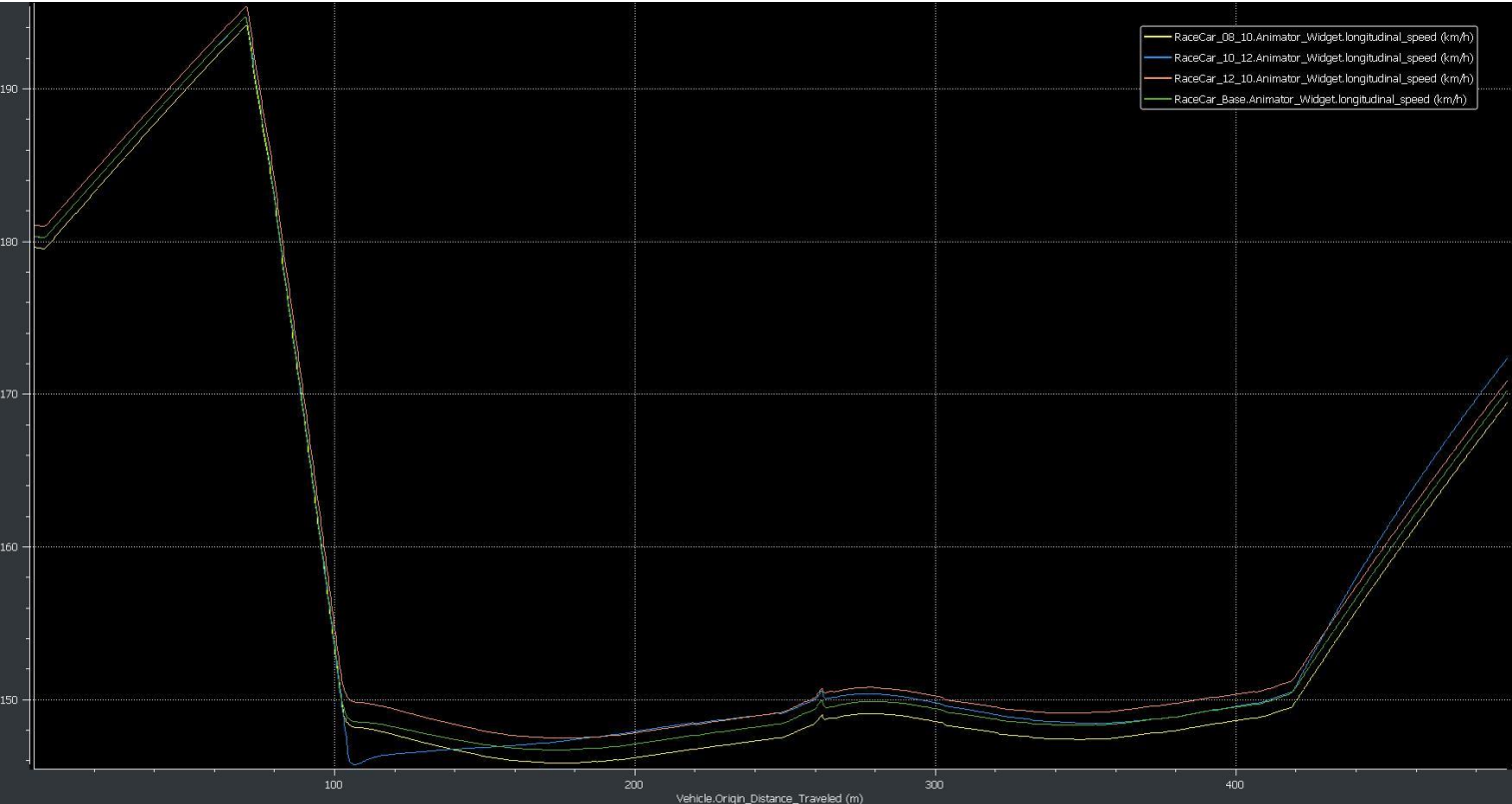
Accelerating while exiting chicane



Longitudinal speed vs Time



Longitudinal speed vs Travelled Distance



Second simulation

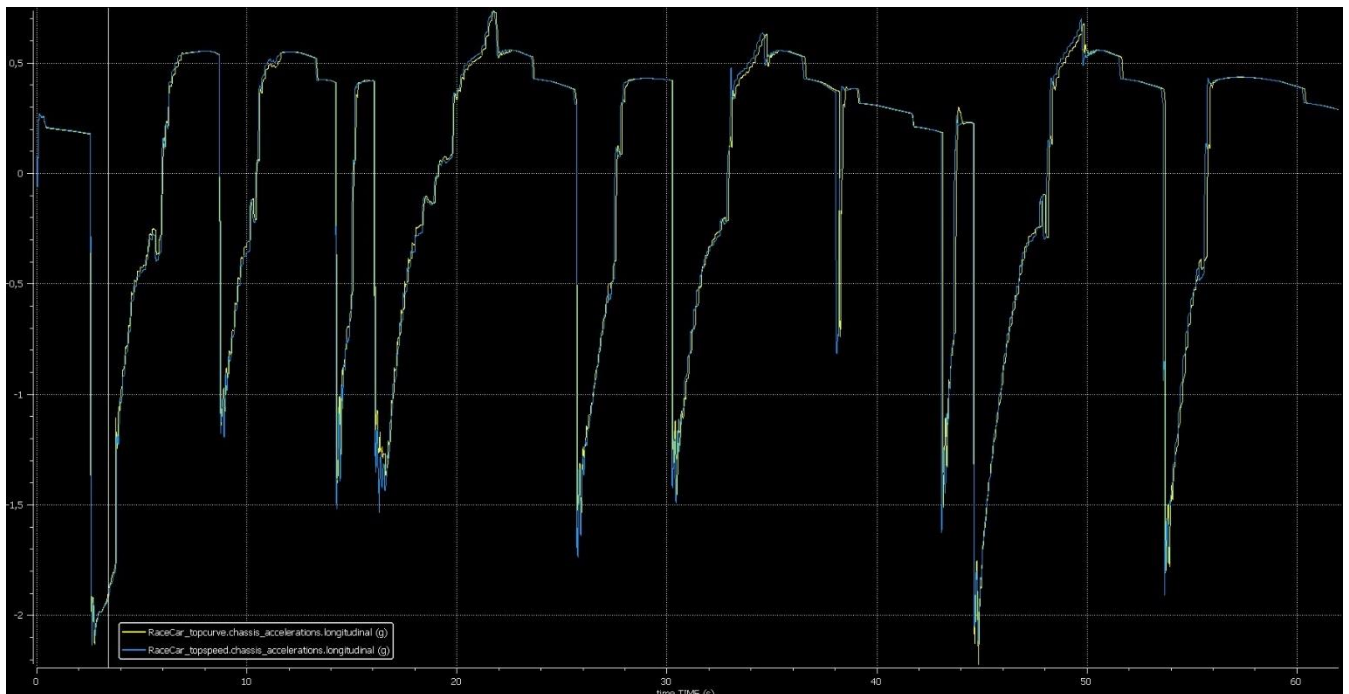
Here we worked out on 2 mixed configurations for the car: one is well prepared for top speed on straight line, the other for the best performance in curve.

Blu line: Top speed config. Yellow curve: Top curve config.

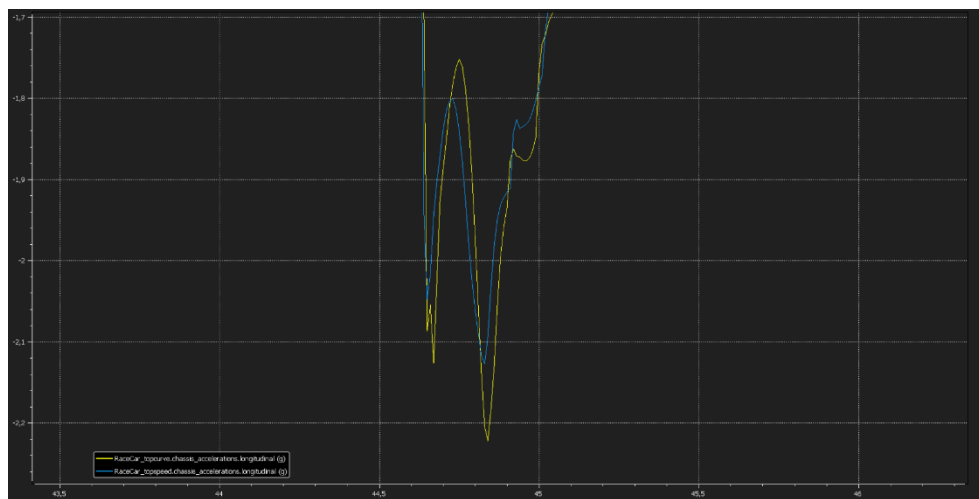
Parameters we have worked on:

- Aerodynamic balance
- Toe angle
- Front and rear anti-roll bar stiffness

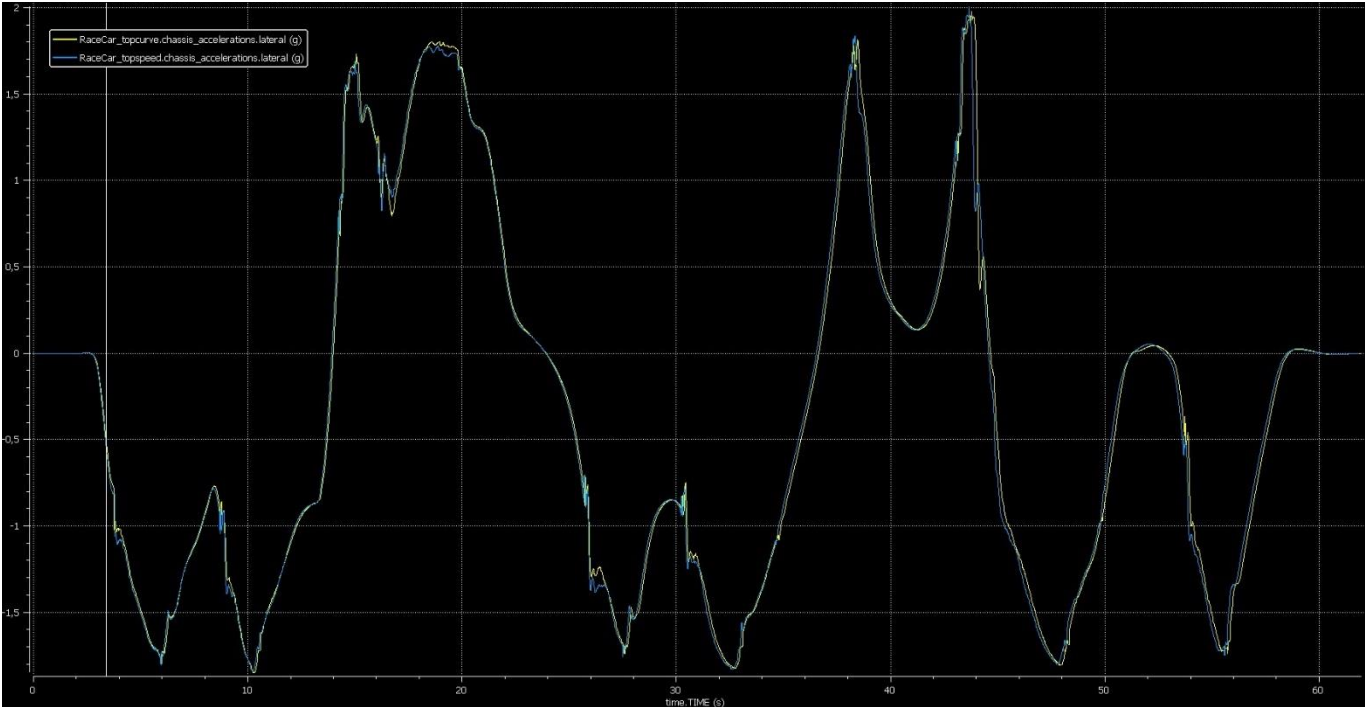
Longitudinal acceleration vs Time



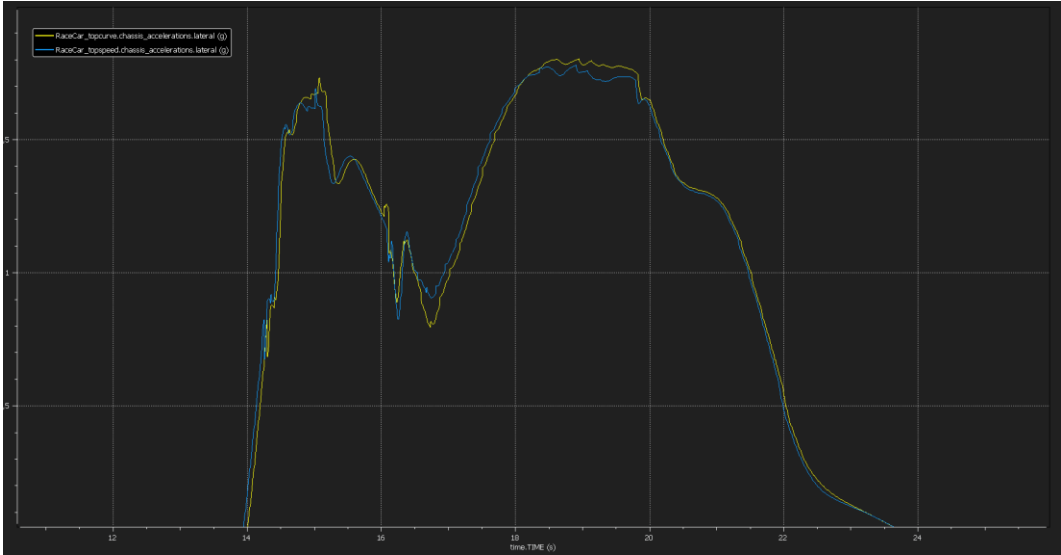
Braking point zoom



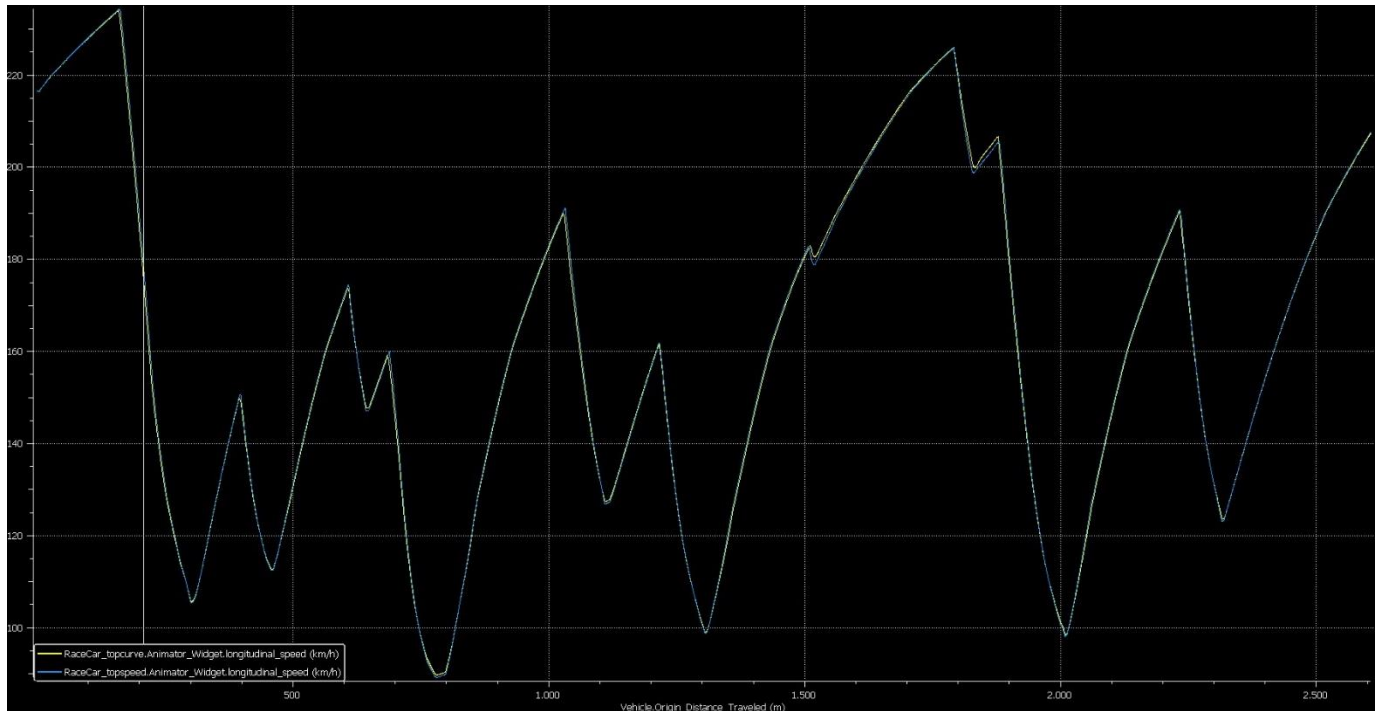
Lateral Acceleration vs Time



Curve 3 zoom



Longitudinal speed vs Time



For the TopSpeed vehicle, we set the front toe angle to 0° (low tire resistance in straights), increased the rear downforce (strong and stable rear in curve exits) and a stiffer antiroll bar at the front (less oversteering than the Base case).

Total downforce was decreased for the TopSpeed case with respect to the Base case, while increased for the TopCurve one.

For the TopCurve vehicle, we set the front toe angle to -0.4° (toe out, better curve entering), increased the front downforce (strong front in braking) and a softer antiroll bar at the front (more oversteering than the Base case).

Overall, the TopSpeed vehicle reaches better performances in terms of lap time, thanks to a better traction, stability and top speed.